

A LOCAL PROOF OF LANGLANDS'S FIRST MAIN LEMMA FOR LOCAL EPSILON FACTORS OVER NONARCHIMEDEAN LOCAL FIELDS

FUKUHIRO UEDA

ABSTRACT. We give a complete local proof of Langlands's First Main Lemma for local epsilon factors over nonarchimedean local fields. The proof builds on Dwork's p -adic calculations and Langlands's formulation of the local method. The present paper reformulates these calculations in a common normalization and extends the argument to equal characteristic. A more systematic treatment of stationary parameters and conductor cases is intended to make the calculations easier to follow. Norm–trace correction formulas yield the equal-characteristic extension.

The author is also completing a local proof of Langlands's Second Main Lemma and will verify it in Lean. Together with the present result and Langlands's remaining reductions, that work will give a complete proof of the local construction of local epsilon factors envisioned by Langlands.

OpenAI ChatGPT assisted with drafting and wording revisions of this manuscript; the author audited the manuscript, checked the proof, and assumes full responsibility.

Date: August 2026.

2020 Mathematics Subject Classification. 11S37, 11S40, 11R42.

Key words and phrases. local epsilon factors, Artin L -functions, local constants, Gauss sums, ramification, Lamprecht formula, cyclic extensions, equal characteristic.

CONTENTS

1. Introduction	3
Historical background	3
The Second Main Lemma and the local construction	3
Structure of the proof	3
Equal and mixed characteristic	4
Organization of the paper	4
2. Local constants and the elementary reduction	5
2.1. Local fields, characters, and exact conductors	5
2.2. The Haar-integral and finite realizations of the local constant	5
2.3. Norm and trace, elementary identities, and the error term	7
3. Ramification and norm filtration for cyclic prime extensions	8
4. Lamprecht's formula	14
5. Finite-field identities	20
6. Stationary parameters under norm	27
6.1. High-conductor parameters	31
6.2. Low-conductor parameters	35
7. Unramified and tame cases	37
8. Wild odd degree: stable range	41
9. The non-stable wild range	41
9.1. The endpoint conductors in wild prime degree	41
9.2. Odd residue characteristic	43
9.3. Wild quadratic extensions	56
9.4. Quadratic refinement in characteristic two	57
9.5. The four critical functions	58
9.6. Stationary-factor assembly	63
9.7. The norm-trace term and final cancellation	66
10. Completion of the First Main Lemma	67
References	68

1. INTRODUCTION

Let F be a nonarchimedean local field with Weil group W_F . In [3], Langlands constructed local epsilon factors for representations of W_F . Brauer induction reduces their construction to local constants of quasi-characters, but it also creates an independence problem: distinct Brauer decompositions must give the same result. Langlands reduced this problem to four local identities. The present paper proves the identity he called the First Main Lemma.

Let K/F be a cyclic extension of prime degree ℓ . For a (continuous) quasi-character $\chi_F : F^\times \rightarrow \mathbf{C}^\times$ and a nontrivial (continuous) additive character $\psi_F : F \rightarrow \mathbf{C}^\times$, put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \text{Tr}_{K/F}.$$

Let $S(K/F)$ be the group of continuous characters of $F^\times$ that are trivial on $N_{K/F}(K^\times)$.

The following is the identity quoted as Lemma 10.1 on [3, pp. 144].

Theorem 1.1 (First Main Lemma). *For every cyclic extension K/F of prime degree, every quasi-character χ_F of $F^\times$, and every nontrivial additive character ψ_F of F ,*

$$(1) \quad \Delta_K(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta_F(\mu \chi_F, \psi_F).$$

Here Δ is Langlands's normalized local constant, recalled in Section 2.

Historical background. Langlands attributes the First Main Lemma to Dwork [3, p. 6]. Dwork's 1954 thesis [2] contains the p -adic calculations for the cyclic-prime relation, including the endpoint and non-stable wild cases [2, Chapter III, pp. 72–95]. These calculations are carried out for finite extensions of $\mathbf{Q}_p$ and use a normalization different from Langlands's.

The surviving chapters of Langlands's manuscript already prove the unramified and tame cases and the stable odd-prime wild case of the First Main Lemma. Chapters 8 and 9 also develop the stationary parameters and finite-field identities used in the remaining wild cases. Chapter 10, entitled “The First Main Lemma”, is missing, and hence the manuscript does not contain the final assembly of these cases into a proof of the First Main Lemma. Langlands's editorial comments explain that Deligne's global argument provided a different proof of the existence of the local epsilon factors, after which Langlands did not complete the local manuscript [3].

The Second Main Lemma and the local construction. The present theorem settles the First Main Lemma, but Langlands's original local construction also requires the Second Main Lemma. The author is presently completing a companion local proof of the Second Main Lemma and will formalize and verify the finished proof in Lean 4. Once this is done, the Lean-verified First and Second Main Lemmas, together with Langlands's remaining reductions, will furnish a complete proof of the local construction of local epsilon factors envisioned in [3].

Structure of the proof. We compare the two sides of (1) using norm and trace identities for K/F . Neither local class field theory nor a global functional equation is used.

First, Proposition 2.4 converts Langlands's Haar-integral definition into a Gauss sum over a finite quotient of the unit group and proves this sum nonzero by Fourier duality, independently of Lamprecht's formula. This nonvanishing is used in the inverse-pairing identity of Lemma 2.11.

Second, Lemma 4.2 constructs the stationary parameter attached to a character as a quotient class. For conductor greater than one, Lamprecht's formula (Theorem 4.5) gives the factorization (46) into three factors: the multiplicative-character value at an admissible element, the stationary factor attached to a chosen representative, and a residue-field factor. The last factor is 1 for even conductor and the phase of the sum of the Hasse function over the residue field for odd conductor. Lemma 4.6 describes how changing the representative changes the stationary factor and the Hasse function; the product in the local constant is unchanged.

Third, Corollary 6.3 proves the transport of stationary classes under norm. Propositions 6.8, 6.9, and 6.10 then give stationary-class representatives over F and K , in the relevant conductor ranges.

Finally, Lemma 2.14 permits χ_F to be chosen with minimal conductor in its $S(K/F)$ -orbit. Write m for its conductor. For ramified K/F , let t be the lower ramification break and put $T = t + 1$, the conductor of every nontrivial character in $S(K/F)$. The proof is divided into the following cases.

Extension and conductor	Argument
Unramified	For $m = 0$, use the unramified-character formula. For $m = 1$, use the Hasse–Davenport lifting identity. For $m > 1$, use Lamprecht’s formula and compare sums over the residue fields (Proposition 7.1).
Tamely ramified, ℓ odd	For $m = 0, 1$, use the unramified-character and Hasse–Davenport product formulas, respectively. For $m > 1$, multiplication by a character in $S(K/F)$ preserves the stationary class of χ_F ; compare the remaining Gauss sums (Proposition 7.3).
Tamely ramified, $\ell = 2$	The cases $m = 0, 1$ use the same formulas as for tame odd degree. For $m > 1$, evaluate quadratic Gauss sums (Proposition 7.4).
Wildly ramified, $m \leq 1$ (<i>Endpoint cases</i>)	For $m = 0$, use the unramified-character formula. For $m = 1$, Frobenius identifies the Gauss sums for χ_F and χ_K ; norm identities compare the remaining character values (Subsection 9.1).
Wildly ramified, ℓ odd, $m \geq 2T$ (<i>stable range</i>)	Multiplication by a character in $S(K/F)$ preserves the stationary class of χ_F . Lamprecht’s formula reduces the comparison to sums over the residue field (Proposition 8.1).
Wildly ramified, ℓ odd, $1 < m < 2T$ (<i>non-stable range</i>)	Multiplication by a character in $S(K/F)$ can change the stationary class of χ_F . Compute the classes for χ_K and the twists of χ_F , then cancel the resulting character values and Gauss-sum factors (Corollary 9.16).
Wildly ramified, $\ell = 2$, $m > 1$	Compare the cases $m < T$, $m = T$, and $m > T$. When $m = T$ is odd, the additive-character values and Gauss-sum factors are cancelled together (Proposition 9.26).

Equal and mixed characteristic. The unramified, tame, endpoint conductors, and the stable wild odd-degree cases are proved by the same arguments in mixed and equal characteristic. Separate characteristic-dependent terms occur only in the remaining non-stable wild calculations.

In the odd-prime non-stable case, Proposition 9.13 reduces the remaining additive factor to a single norm–trace correction X . Proposition 9.14 gives a formula for X whose numerator contains the residue characteristic p , hence $X = 0$ in characteristic p . Proposition 9.15 shows that its contribution to the additive character is trivial in mixed characteristic, by bounding its valuation.

For a wild quadratic extension, Lemma 9.17 gives the analogous norm–trace correction; its numerator contains the factor 2, so the correction vanishes in characteristic two. In mixed characteristic, Proposition 9.25 shows that its additive phase is 1 except in the boundary case $m = T$ and T is odd. In that case, Proposition 9.26 shows that it cancels with the remaining quadratic factors, so the two sides of (1) are equal.

Organization of the paper. Section 2 establishes the two realizations of the local constant, proves nonvanishing, introduces the error term, and reduces the proof to characters of minimal conductor in their $S(K/F)$ -orbits. The next four sections develop the cyclic-prime ramification, stationary-phase, finite-field, and stationary-parameter tools. We then treat the unramified,

tame, endpoint, stable, odd non-stable, and quadratic cases. The last section combines these cases to prove Theorem 1.1.

2. LOCAL CONSTANTS AND THE ELEMENTARY REDUCTION

2.1. Local fields, characters, and exact conductors. Let E be a nonarchimedean local field. Write

$$\mathcal{O}_E = \{x \in E : v_E(x) \geq 0\}, \quad \mathfrak{p}_E = \{x \in E : v_E(x) \geq 1\},$$

and let $k_E = \mathcal{O}_E/\mathfrak{p}_E$ have cardinality q_E . We normalize $v_E(E^\times) = \mathbf{Z}$. Put

$$U_E^0 = \mathcal{O}_E^\times, \quad U_E^r = 1 + \mathfrak{p}_E^r \quad (r \geq 1).$$

All quasi-characters are continuous homomorphisms $E^\times \rightarrow \mathbf{C}^\times$, and all additive characters are continuous homomorphisms from the additive group of E to $\mathbf{C}^\times$.

Remark 2.1 (Characteristic convention). Write $p = \text{char } k_E$. Thus either $\text{char } E = 0$ with residue characteristic p , or $\text{char } E = p$. An integer occurring in an identity inside E denotes its image under $\mathbf{Z} \rightarrow E$; in particular $p = 0$ in equal characteristic p .

Definition 2.2 (Multiplicative conductor). The conductor exponent $m_E(\theta)$ is the least $m \geq 0$ for which θ is trivial on U_E^m . Thus $m_E(\theta) = 0$ precisely when θ is unramified.

Definition 2.3 (Additive conductor). For a nontrivial additive character ψ , the integer $n_E(\psi)$ is characterized by the condition that $\mathfrak{p}_E^{-n_E(\psi)}$ is the largest fractional ideal on which ψ is trivial.

2.2. The Haar-integral and finite realizations of the local constant. Define the total phase

$$\text{ph}(z) = \begin{cases} z/|z|, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Let $m = m_E(\theta)$ and $n = n_E(\psi)$, and choose $\gamma \in E^\times$ such that

$$(2) \quad \gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n}.$$

For a positive Haar measure du on U_E^0 , Langlands defines

$$(3) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \text{ph} \left(\int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du \right).$$

Put $Q_m = U_E^0/U_E^m$. For $\bar{u} \in Q_m$, represented by $u \in U_E^0$, define

$$g_{\theta, \psi, \gamma}(\bar{u}) = \psi(u/\gamma) \theta(u)^{-1}$$

and

$$(4) \quad G_E(\theta, \psi; \gamma) = \sum_{\bar{u} \in Q_m} g_{\theta, \psi, \gamma}(\bar{u}).$$

Proposition 2.4 (Finite realization and nonvanishing). *The summand just defined is well defined. There is a positive real number c_m , depending only on the chosen Haar measure and m , such that*

$$\int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du = c_m G_E(\theta, \psi; \gamma).$$

Moreover

$$G_E(\theta, \psi; \gamma) \neq 0.$$

Consequently

$$(5) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \text{ph}(G_E(\theta, \psi; \gamma)),$$

and the zero branch in the definition of ph is never used for a local constant.

Proof. If $u/v \in U_E^m$, then $\theta(u) = \theta(v)$ and $u - v \in \mathfrak{p}_E^m$. By (2), $(u - v)/\gamma \in \mathfrak{p}_E^{-n}$, so $\psi(u/\gamma) = \psi(v/\gamma)$. Thus the summand descends to Q_m . The integrand is constant on every coset of U_E^m in U_E^0 , and all such cosets have the same positive measure c_m ; this proves the stated integral formula.

It remains to prove nonvanishing. If $m = 0$, the quotient Q_0 is a singleton and $G_E(\theta, \psi; \gamma) = 1$. Assume $m > 0$ and put $A = \mathcal{O}_E/\mathfrak{p}_E^m$. The pairing

$$A \times A \longrightarrow \mathbf{C}^\times, \quad (\bar{x}, \bar{y}) \longmapsto \psi(xy/\gamma)$$

is perfect. Indeed, its annihilator in either variable is exactly $\mathfrak{p}_E^m$: this follows from (2) and the maximality of $\mathfrak{p}_E^{-n}$ in the definition of $n_E(\psi)$.

Extend the function $u \mapsto \theta(u)^{-1}$ on $A^\times$ by zero to a function $f : A \rightarrow \mathbf{C}$, and write

$$\hat{f}(x) = \sum_{u \in A^\times} \theta(u)^{-1} \psi(xu/\gamma).$$

Then $\hat{f}(1) = G_E(\theta, \psi; \gamma)$. If $x \in A^\times$, the substitution $u \mapsto x^{-1}u$ gives

$$\hat{f}(x) = \theta(x) \hat{f}(1).$$

If x is not a unit, exactness of the conductor provides $a \in U_E^{m-1}$ with $\theta(a) \neq 1$. Since $x(a-1)u/\gamma \in \mathfrak{p}_E^{-n}$ for every $u \in U_E^0$, multiplication by a permutes $A^\times$ and gives

$$\hat{f}(x) = \theta(a)^{-1} \hat{f}(x),$$

whence $\hat{f}(x) = 0$. If $\hat{f}(1)$ were zero, the formula for the Fourier coefficients at units and the preceding vanishing would make every Fourier coefficient of f zero. Perfect finite Fourier duality would then give $f = 0$, contradicting $f(1) = 1$. Thus $G_E(\theta, \psi; \gamma) \neq 0$. Finally the positive factor c_m does not change phase, proving (5). $\square$

Corollary 2.5 (Nonvanishing of the local Gauss integral). *For every quasi-character θ and every nontrivial additive character ψ , the integral in (3) is nonzero.*

Lemma 2.6 (Independence of the admissible denominator). *The right side of (3) is independent of the choice of γ satisfying (2).*

Proof. Let $\gamma' = a\gamma$ with $a \in U_E^0$. Multiplication by a permutes Q_m , and (4) gives

$$G_E(\theta, \psi; \gamma') = \theta(a)^{-1} G_E(\theta, \psi; \gamma).$$

The restriction of θ to U_E^0 factors through the finite group Q_m , so $|\theta(a)| = 1$. By nonvanishing,

$$\text{ph}(G_E(\theta, \psi; \gamma')) = \theta(a)^{-1} \text{ph}(G_E(\theta, \psi; \gamma)).$$

Since $\theta(\gamma') = \theta(a)\theta(\gamma)$, the two factors cancel. $\square$

The same comparison shows immediately that the value is independent of the positive normalization of Haar measure.

Lemma 2.7 (Normalization of the residual additive character). *Let ψ_F have conductor exponent n , and fix a uniformizer ϖ_F . The formula*

$$(6) \quad \bar{\psi}(\bar{x}) = \psi_F(x\varpi_F^{-n-1}), \quad x \in \mathcal{O}_F \text{ a lift of } \bar{x} \in k_F,$$

defines a nontrivial additive character of k_F , independently of the choice of the lift x . Define the canonical additive character $\psi_{\text{can}} : k_F \rightarrow \mathbf{C}^\times$ by

$$(7) \quad x \longmapsto \exp\left(\frac{2\pi i}{p} \text{Tr}_{k_F/\mathbf{F}_p}(x)\right).$$

Then there is a unit $a \in \mathcal{O}_F^\times$ such that the residual character attached to $\psi_F^a(x) = \psi_F(ax)$ is ψ_{can} .

Proof. Well-definedness follows because two lifts differ by an element of $\mathfrak{p}_F$, and therefore their arguments differ by an element of $\mathfrak{p}_F^{-n}$, on which ψ_F is trivial. Nontriviality follows from the maximality of $\mathfrak{p}_F^{-n}$. Since the trace pairing $k_F \times k_F \rightarrow \mathbf{F}_p$ is nondegenerate, there is a unique $c \in k_F^\times$ such that $\bar{\psi}(x) = \psi_{\text{can}}(cx)$. Choose a unit $a \in \mathcal{O}_F^\times$ with residue c^{-1} ; then the residual character attached to ψ_F^a is ψ_{can} . $\square$

Let k be a finite field, let $\varphi : k \rightarrow \mathbf{C}^\times$ be a nontrivial additive character, and let $\eta : k^\times \rightarrow \mathbf{C}^\times$ be a multiplicative character. Define

$$(8) \quad \tau_k(\eta, \varphi) = - \sum_{x \in k^\times} \eta(x)^{-1} \varphi(x).$$

Thus $\tau_k(1, \varphi) = 1$.

If $m_E(\theta) = 0$ and ϖ_E is a uniformizer, then $\gamma = \varpi_E^{n_E(\psi)}$ is admissible and

$$(9) \quad \Delta_E(\theta, \psi) = \theta(\varpi_E)^{n_E(\psi)}.$$

If $m_E(\theta) = 1$, for $\bar{u} \in k_E^\times$ and $\bar{x} \in k_E$ choose lifts $u \in \mathcal{O}_E^\times$ and $x \in \mathcal{O}_E$, and define

$$\bar{\theta}(\bar{u}) = \theta(u), \quad \bar{\psi}(\bar{x}) = \psi(x/\varpi_E^{n_E(\psi)+1}).$$

Then

$$(10) \quad \Delta_E(\theta, \psi) = -\theta(\varpi_E)^{n_E(\psi)+1} \text{ph}(\tau_{k_E}(\bar{\theta}, \bar{\psi})).$$

These are the local-to-residue-field formulas used in the unramified, tame, and endpoint cases.

2.3. Norm and trace, elementary identities, and the error term. Let K/F be a finite extension. Write

$$N_{K/F} : K^\times \rightarrow F^\times, \quad \text{Tr}_{K/F} : K \rightarrow F.$$

For a nontrivial additive character ψ_F and a quasi-character χ_F , put

$$\psi_K = \psi_F \circ \text{Tr}_{K/F}, \quad \chi_K = \chi_F \circ N_{K/F}.$$

If K/F is abelian, set

$$S(K/F) = \{\mu : F^\times \rightarrow \mathbf{C}^\times : \mu \text{ continuous and } \mu \circ N_{K/F} = 1\}.$$

By the universal property of the quotient, this is the group of continuous characters of the topological quotient $F^\times/N_{K/F}(K^\times)$. For cyclic K/F of prime degree, the quotient and $S(K/F)$ are proved directly to have order ℓ in Corollary 3.11 below. That calculation uses the norm filtration and not local class field theory.

Lemma 2.8 (Change of additive character). *For $a \in E^\times$, put $\psi^a(x) = \psi(ax)$. Then*

$$\Delta_E(\theta, \psi^a) = \theta(a) \Delta_E(\theta, \psi).$$

Proof. If $v_E(a) = r$, then $n_E(\psi^a) = n_E(\psi) + r$. Hence $a\gamma$ is admissible for (θ, ψ^a) whenever γ is admissible for (θ, ψ) , and $\psi^a(u/(a\gamma)) = \psi(u/\gamma)$. $\square$

Corollary 2.9 (Scaling invariance of the First Main identity). *For every $a \in F^\times$, the First Main Lemma for ψ_F is equivalent to the First Main Lemma for ψ_F^a .*

Proof. By Corollary 3.11, the group $S(K/F)$ has ℓ elements. By Lemma 2.8, the two sides of (1) acquire the respective factors

$$\chi_K(a) \prod_{\mu \in S(K/F)} \mu(a) \quad \text{and} \quad \chi_F(a)^\ell \prod_{\mu \in S(K/F)} \mu(a).$$

Since $N_{K/F}(a) = a^\ell$ for $a \in F^\times$, these factors agree. $\square$

Lemma 2.10 (The trivial character). *For every nontrivial additive character ψ ,*

$$(11) \quad \Delta_E(1, \psi) = 1.$$

Proof. Here $m_E(1) = 0$. Formula (9) gives the result. $\square$

Lemma 2.11 (Inverse pairing). *Let μ be a finite-order character of $E^\times$. Then*

$$(12) \quad \Delta_E(\mu, \psi) \Delta_E(\mu^{-1}, \psi) = \mu(-1).$$

Proof. Choose a common admissible γ and write

$$I_\mu = \int_{U_E^0} \psi(u/\gamma) \mu(u)^{-1} du.$$

Both I_μ and $I_{\mu^{-1}}$ are nonzero by Corollary 2.5. Since μ and ψ are unitary,

$$\overline{I_\mu} = \int_{U_E^0} \psi(-u/\gamma) \mu(u) du = \mu(-1) I_{\mu^{-1}}.$$

The product of the two phases is therefore $\mu(-1)^{-1} = \mu(-1)$, and the factors $\mu(\gamma)$ and $\mu^{-1}(\gamma)$ cancel. $\square$

Corollary 2.12. *If K/F is cyclic of odd prime degree, then*

$$(13) \quad \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F) = 1.$$

Proof. The finiteness of $S(K/F)$ follows from Corollary 3.11. For $x \in F^\times$, viewed as an element of $K^\times$, one has $N_{K/F}(x) = x^\ell$. Hence, for every $\mu \in S(K/F)$,

$$\mu(x)^\ell = \mu(N_{K/F}(x)) = 1.$$

Thus every nontrivial μ has order ℓ , which is odd, and in particular $\mu(-1) = 1$. Apply (12) to each pair μ, μ^{-1} and (11) to the trivial character. $\square$

Definition 2.13 (Error term). Define

$$(14) \quad \mathcal{E}_{K/F}(\chi_F) = \frac{\Delta_K(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F)}{\prod_{\mu \in S(K/F)} \Delta_F(\mu \chi_F, \psi_F)}.$$

The denominator is nonzero by Corollary 2.5, and the First Main Lemma is the assertion $\mathcal{E}_{K/F}(\chi_F) = 1$.

Lemma 2.14 (Twist invariance). *For every $\nu \in S(K/F)$,*

$$(15) \quad \mathcal{E}_{K/F}(\nu \chi_F) = \mathcal{E}_{K/F}(\chi_F).$$

Proof. Since $\nu \circ N_{K/F} = 1$, the character on $K^\times$ is unchanged. Multiplication by ν permutes the group $S(K/F)$, including its trivial character, so the denominator is only reindexed. $\square$

Proposition 2.15 (Minimal-orbit reduction). *To prove Theorem 1.1, it is enough, for each $S(K/F)$ -orbit of quasi-characters, to prove $\mathcal{E}_{K/F}(\chi_F) = 1$ for one member of minimal multiplicative conductor.*

Proof. The conductors in an orbit are nonnegative integers, so a minimum exists. Equation (15) shows that $\mathcal{E}_{K/F}(\chi_F)$ is constant on each orbit. $\square$

3. RAMIFICATION AND NORM FILTRATION FOR CYCLIC PRIME EXTENSIONS

Let K/F be a finite Galois extension of nonarchimedean local fields, with Galois group G . For an integer $i \geq -1$, the lower ramification group is

$$G_i = \{\sigma \in G : \sigma(x) \equiv x \pmod{\mathfrak{p}_K^{i+1}} \text{ for every } x \in \mathcal{O}_K\}.$$

For a real $u \geq -1$, put $G_u = G_{[u]}$. Define the Herbrand function and its inverse by

$$\varphi_{K/F}^H(u) = \int_0^u \frac{ds}{[G_0 : G_s]}, \quad \psi_{K/F}^H = (\varphi_{K/F}^H)^{-1}.$$

The superscript H is used throughout to prevent confusion with additive characters. This notation avoids the conflict noted by Langlands on pp. 28 and 62.

When K/F is cyclic of prime degree ℓ and ramified, it is totally ramified and there is a unique integer $t \geq 0$ such that

$$G = G_0 = \cdots = G_t, \quad G_{t+1} = 1.$$

Consequently

$$(16) \quad \psi_{K/F}^H(u) = \begin{cases} u, & -1 \leq u \leq t, \\ t + \ell(u - t), & u \geq t. \end{cases}$$

For a finite separable extension E/F , let $\mathfrak{D}_{E/F} \subset \mathcal{O}_E$ denote the different ideal, and write

$$D_{E/F} = v_E(\mathfrak{D}_{E/F})$$

for its exponent.

Theorem 3.1 (Ramification, trace, and norm filtration for a cyclic prime extension). *Let K/F be ramified cyclic of prime degree ℓ , with lower break t . Then:*

(a) *The different exponent is*

$$D_{K/F} = (\ell - 1)(t + 1),$$

and for every integer a ,

$$(17) \quad \mathrm{Tr}_{K/F}(\mathfrak{p}_K^a) = \mathfrak{p}_F^{\lfloor (a + D_{K/F})/\ell \rfloor}.$$

(b) *For $0 \leq i \leq t$, let*

$$\overline{N}_i : \mathrm{gr}^i U_K \longrightarrow \mathrm{gr}^i U_F, \quad \mathrm{gr}^i U_E = U_E^i / U_E^{i+1},$$

be the map induced by the norm. For $0 \leq i < t$ it is an isomorphism. At $i = t$, its kernel and cokernel both have order ℓ , and its kernel is the image of G_t/G_{t+1} under the ramification map

$$\sigma \longmapsto \begin{cases} \overline{\sigma(\varpi_K) / \varpi_K}, & t = 0, \\ \overline{(\sigma(\varpi_K) - \varpi_K) / \varpi_K^{t+1}}, & t > 0. \end{cases}$$

(c) *Splicing the graded maps below the break gives*

$$(18) \quad K^\times / U_K^t \xrightarrow{\sim} F^\times / U_F^t, \quad U_K^r / U_K^t \xrightarrow{\sim} U_F^r / U_F^t \quad (0 \leq r \leq t).$$

(d) *For every integer $r > t$, put $a = \psi_{K/F}^H(r)$. The norm induces an isomorphism*

$$U_K^a / U_K^{a+1} \longrightarrow U_F^r / U_F^{r+1}.$$

Moreover, for every $a \geq 0$, the subgroup $N_{K/F}(U_K^a)$ is closed in U_F^0 , and consequently

$$(19) \quad N_{K/F}(U_K^{\psi_{K/F}^H(r)}) = U_F^r, \quad N_{K/F}(U_K^{\psi_{K/F}^H(r)+1}) = U_F^{r+1}.$$

Proof. Choose a uniformizer ϖ_K . For every $\sigma \neq 1$ one has $v_K(\sigma(\varpi_K) - \varpi_K) = t + 1$. The different formula

$$v_K(\mathfrak{D}_{K/F}) = \sum_{\sigma \neq 1} v_K(\sigma(\varpi_K) - \varpi_K),$$

which is [4, Chapter IV, §1, Proposition 4], gives the first assertion in (a).

For the trace formula, the trace-dual of $\mathfrak{p}_K^a$ under $(x, y) \mapsto \mathrm{Tr}_{K/F}(xy)$ is $\mathfrak{p}_K^{-a - D_{K/F}}$ by [4, Chapter III, §3, Proposition 7]. Write $\mathrm{Tr}(\mathfrak{p}_K^a) = \mathfrak{p}_F^c$. An element $y \in F$ belongs to the dual of this trace ideal if and only if

$$y\mathfrak{p}_K^a \subseteq \mathfrak{p}_K^{-D_{K/F}}, \quad \text{equivalently} \quad \ell v_F(y) + a \geq -D_{K/F}.$$

Thus its dual is $\mathfrak{p}_F^{\lfloor (-a - D_{K/F})/\ell \rfloor}$, whereas the dual of $\mathfrak{p}_F^c$ is $\mathfrak{p}_F^{-c}$. Hence

$$c = \left\lfloor \frac{a + D_{K/F}}{\ell} \right\rfloor,$$

which proves (17).

The graded maps in (b), including the critical kernel, are the cyclic-prime specialization of [4, Chapter V, §3, Propositions 4–5]. For (c), splice the graded isomorphisms for $0 \leq i < t$. The valuation quotient is preserved because the residue degree is one: $v_F(N_{K/F}x) = v_K(x)$. If $t = 0$, then $K^\times/U_K^0 \simeq \mathbf{Z}$ and norm is the identity on the valuation quotient. If $t > 0$, the residue norm $U_K^0/U_K^1 \rightarrow U_F^0/U_F^1$ is $x \mapsto x^\ell = x^p$, hence is Frobenius, and the remaining graded pieces below t are isomorphisms. This proves (18).

For (d), the above-break graded isomorphism is [4, Chapter V, §3, Corollaries 1–3]. Each U_K^a is compact, so $N_{K/F}(U_K^a)$ is compact and hence closed in U_F^0 . Successive lifting through the graded isomorphisms therefore converges to a norm and gives (19). $\square$

Lemma 3.2 (Frobenius differences and trace-zero hyperplanes). *Let $k/\mathbf{F}_p$ be a finite field extension and let $\lambda \in k^\times$. Then the $\mathbf{F}_p$ -linear maps*

$$\wp(z) = z^p - z, \quad \wp_\lambda(z) = z^p - \lambda^{p-1}z$$

satisfy

$$\ker(\wp) = \mathbf{F}_p, \quad \text{im}(\wp) = \ker(\text{Tr}_{k/\mathbf{F}_p}),$$

and

$$(20) \quad \ker(\wp_\lambda) = \lambda \mathbf{F}_p, \quad \text{im}(\wp_\lambda) = \lambda^p \ker(\text{Tr}_{k/\mathbf{F}_p}).$$

Under the perfect trace pairing $(a, x) \mapsto \text{Tr}_{k/\mathbf{F}_p}(ax)$, the annihilator of the last image is $\lambda^{-p}\mathbf{F}_p$.

Proof. The kernel of $\wp$ is the prime field. Hence its image has $\mathbf{F}_p$ -dimension $[k : \mathbf{F}_p] - 1$. Frobenius invariance of the trace gives $\text{Tr}(z^p - z) = 0$, so the image is contained in the trace-zero hyperplane; the two spaces have the same dimension and therefore coincide. Since

$$\wp_\lambda(\lambda y) = \lambda^p(y^p - y),$$

the scaled kernel and image follow. The orthogonal complement of $\ker(\text{Tr}_{k/\mathbf{F}_p})$ is $\mathbf{F}_p$: it contains $\mathbf{F}_p$, and both have dimension one. Scaling the second variable by λ^p gives the annihilator $\lambda^{-p}\mathbf{F}_p$. $\square$

Proposition 3.3 (Critical norm polynomial at a wild break). *Let K/F be totally ramified cyclic of prime degree $p = \text{char } k_F$, with positive lower break t , and identify the residue fields with k . Choose a uniformizer ϖ_K , put $\varpi_F = N_{K/F}(\varpi_K)$, choose a generator σ of $\text{Gal}(K/F)$, and set*

$$(21) \quad \lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

For $z, w \in k$, let $\tilde{z}, \tilde{w} \in \mathcal{O}_F \subset \mathcal{O}_K$ be any lifts. We use the standard identifications

$$\text{gr}^t U_K = U_K^t/U_K^{t+1} \simeq (k, +), \quad (1 + \varpi_K^t \tilde{z})U_K^{t+1} \longleftrightarrow z,$$

and

$$\text{gr}^t U_F = U_F^t/U_F^{t+1} \simeq (k, +), \quad (1 + \varpi_F^t \tilde{w})U_F^{t+1} \longleftrightarrow w.$$

In these source and target coordinates, the critical graded norm is the additive polynomial

$$(22) \quad P_\lambda(Z) = Z^p - \lambda^{p-1}Z.$$

Consequently

$$(23) \quad \ker P_\lambda = \lambda \mathbf{F}_p, \quad \text{im } P_\lambda = \lambda^p \ker(\text{Tr}_{k/\mathbf{F}_p}),$$

and its annihilator under the trace pairing is $\lambda^{-p}\mathbf{F}_p$. The linear coefficient is the normalized trace:

$$(24) \quad \frac{\overline{\text{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t} = -\lambda^{p-1}.$$

Proof. For $2 \leq j \leq p-1$, let s_j be the j th elementary symmetric function of the conjugates of ϖ_K^t . Lemma 3.4 gives

$$v_F(s_j) \geq \left\lfloor \frac{jt + (p-1)(t+1)}{p} \right\rfloor \geq t+1,$$

since $(j-1)t \geq 1$. There are no such terms when $p=2$. The trace-ideal formula (17) gives $\text{Tr}_{K/F}(\varpi_K^t) \in \mathfrak{p}_F^t$, while

$$N_{K/F}(\varpi_K^t \tilde{z}) = \varpi_F^t \tilde{z}^p.$$

Thus the norm expansion gives

$$N_{K/F}(1 + \varpi_K^t \tilde{z}) \equiv 1 + \text{Tr}_{K/F}(\varpi_K^t) \tilde{z} + \varpi_F^t \tilde{z}^p \pmod{\mathfrak{p}_F^{t+1}}.$$

The critical norm map is therefore represented by $Z^p + aZ$, where

$$a = \frac{\overline{\text{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t}.$$

By Theorem 3.1(b), its kernel is $\lambda \mathbf{F}_p$. In particular $\lambda^p + a\lambda = 0$; since $\lambda \neq 0$, this gives $a = -\lambda^{p-1}$. This proves (22) and (24). The kernel, image, and annihilator assertions follow from Lemma 3.2. $\square$

Lemma 3.4 (Strong wild symmetric-coefficient estimate). *Assume that K/F is wildly ramified cyclic of prime degree $\ell = p = \text{char } k_F$, with lower break t , and put $T = t+1$. For $x \in K$ let $s_j(x)$ be the j th elementary symmetric function of its p conjugates. If $v_K(x) \geq a$, then, for $1 \leq j \leq p-1$,*

$$(25) \quad v_F(s_j(x)) \geq \left\lfloor \frac{ja + (p-1)T}{p} \right\rfloor.$$

For the last coefficient,

$$(26) \quad v_F(s_p(x)) = v_F(N_{K/F}x) = v_K(x) \geq a.$$

Proof. Put $D = D_{K/F} = v_K(\mathfrak{D}_{K/F})$. By Theorem 3.1(a), $D = (p-1)T$. Put $P_i(x) = \text{Tr}_{K/F}(x^i)$. Since $x^i \in \mathfrak{p}_K^{ia}$, equation (17) gives

$$v_F(P_i(x)) \geq \left\lfloor \frac{ia + D}{p} \right\rfloor \quad (i \geq 1).$$

We now use Newton's identities

$$js_j(x) = \sum_{i=1}^j (-1)^{i-1} s_{j-i}(x) P_i(x), \quad s_0(x) = 1.$$

Because $j < p$, the integer j is a unit of $\mathcal{O}_F$. We prove (25) by induction on j . For $j=1$ it is equation (17), applied to x . Suppose it is known below j . The term $i=j$ in Newton's identity has the required valuation by (17), applied to x^j . For $i < j$, the induction hypothesis and (17), applied to x^i , give

$$\left\lfloor \frac{(j-i)a + D}{p} \right\rfloor + \left\lfloor \frac{ia + D}{p} \right\rfloor.$$

For real numbers A, B one has $\lfloor A \rfloor + \lfloor B \rfloor \geq \lfloor A+B \rfloor - 1$. Here the sum before taking floors is

$$\frac{ja + D}{p} + \frac{D}{p}.$$

The extension is wild, so $T \geq 2$ and $D/p = (p-1)T/p \geq 1$. Hence the last displayed integral lower bound is at least $\lfloor (ja + D)/p \rfloor$. Every term in Newton's identity therefore has the required valuation. Finally $s_p(x) = N_{K/F}(x)$, and total ramification gives $v_F(N_{K/F}x) = v_K(x)$. $\square$

Corollary 3.5 (Norm truncation). *Let $q, r \geq 0$ and assume*

$$2q + (p-1)T \geq pr.$$

Then, for every $y \in \mathfrak{p}_K^q$,

$$(27) \quad N_{K/F}(1+y) \equiv 1 + \text{Tr}_{K/F}(y) + N_{K/F}(y) \pmod{\mathfrak{p}_F^r}.$$

Proof. For $2 \leq j \leq p-1$, Equation (25) gives

$$v_F(s_j(y)) \geq \left\lfloor \frac{jq + (p-1)T}{p} \right\rfloor \geq \left\lfloor \frac{2q + (p-1)T}{p} \right\rfloor \geq r.$$

The norm expansion then gives (27). $\square$

Proposition 3.6 (Stable-to-critical normalized trace). *Let K/F be totally ramified cyclic of odd prime degree p , with positive lower break t . Choose a uniformizer ϖ_K and a generator σ of $\text{Gal}(K/F)$, put $\varpi_F = N_{K/F}(\varpi_K)$, and define λ by (21). Suppose $d \geq t+1$ and put*

$$d' = pd - \frac{p-1}{2}t.$$

Then

$$\frac{\varpi_F}{\varpi_K^p} \equiv 1 \pmod{\mathfrak{p}_K^{t+1}},$$

$$\varpi_K^{2d'} \equiv \varpi_F^{2d-t} \varpi_K^t \pmod{\mathfrak{p}_K^{2d'+1}},$$

and the normalized traces are units with common residue

$$(28) \quad \frac{\text{Tr}_{K/F}(\varpi_K^{2d'})}{\varpi_F^{2d}} = \frac{\text{Tr}_{K/F}(\varpi_K^t)}{\varpi_F^t} = -\lambda^{p-1}.$$

Proof. For $j \in \mathbf{F}_p$, put

$$a_j = \frac{\sigma^j(\varpi_K)}{\varpi_K} - 1.$$

The ramification homomorphism gives $a_j \in \mathfrak{p}_K^t$ and $\overline{a_j/\varpi_K^t} = j\lambda$. Products of at least two of the a_j lie in $\mathfrak{p}_K^{t+1}$, and $\sum_{j \in \mathbf{F}_p} j = 0$. Hence

$$\frac{\varpi_F}{\varpi_K^p} = \prod_{j \in \mathbf{F}_p} (1 + a_j) \equiv 1 \pmod{\mathfrak{p}_K^{t+1}},$$

which proves the first congruence. Since $2d' = p(2d-t) + t$, raising the inverse ratio to the power $2d-t$ and multiplying by ϖ_K^t gives the second congruence.

The trace-ideal formula (17) sends the difference in the second congruence into $\mathfrak{p}_F^{2d+1}$. After division by ϖ_F^{2d} , the normalized traces have equal residues. In the critical source coordinate the linear coefficient of the graded norm is $\overline{\text{Tr}(\varpi_K^t)/\varpi_F^t}$. Proposition 3.3 identifies it with $-\lambda^{p-1}$, which is nonzero. This proves (28) and the unit assertions. $\square$

Corollary 3.7 (Different and additive conductors). *Let $\psi_K = \psi_F \circ \text{Tr}_{K/F}$. If K/F is totally ramified of degree ℓ , then*

$$(29) \quad n_K(\psi_K) = \ell n_F(\psi_F) + D_{K/F}.$$

Proof. The largest fractional ideal I on which $\psi_F \circ \text{Tr}$ is trivial is characterized by $\text{Tr}(I) \subseteq \mathfrak{p}_F^{-n_F(\psi_F)}$. By the trace duality used in the proof of (17), this ideal is $\mathfrak{p}_K^{-\ell n_F(\psi_F) - D_{K/F}}$. $\square$

Corollary 3.8 (Ramified norm quotient and norm characters). *Let K/F be ramified cyclic of prime degree ℓ . Then $F^\times/N_{K/F}(K^\times)$ is cyclic of order ℓ , the group $S(K/F)$ has ℓ elements, and every nontrivial $\mu \in S(K/F)$ has conductor exponent $t+1$.*

Proof. Write $N = N_{K/F}$. The first isomorphism in (18) shows that

$$F^\times = N(K^\times)U_F^t.$$

Its injectivity also shows that

$$N(K^\times) \cap U_F^t = N(U_K^t).$$

Consequently

$$F^\times/N(K^\times) \simeq U_F^t/N(U_K^t).$$

Taking $r = t+1$ in (19) gives

$$U_F^{t+1} = N(U_K^{\psi_{K/F}^H(t+1)}) \subseteq N(U_K^t),$$

because $\psi_{K/F}^H(t+1) > t$. Hence

$$U_F^t/N(U_K^t) \simeq \text{coker} \left(U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1} \right).$$

By Theorem 3.1(b), this cokernel has order ℓ . Thus $F^\times/N(K^\times)$ has order ℓ , and is cyclic because ℓ is prime.

By (19), $U_F^{t+1} \subseteq N(K^\times)$, so $N(K^\times)$ is open and the quotient is discrete. The elements of $S(K/F)$ are therefore exactly the complex characters of this cyclic quotient. A character is determined by the image of a generator, which may be any ℓ th root of unity. Hence $S(K/F)$ has ℓ elements.

The identification $F^\times/N(K^\times) \simeq U_F^t/N(U_K^t)$ shows that U_F^t maps onto the norm quotient, whereas U_F^{t+1} maps to the identity. A nontrivial norm character is therefore nontrivial on U_F^t and trivial on U_F^{t+1} . Its conductor exponent is exactly $t+1$. $\square$

Proposition 3.9 (Conductor of norm pullback). *Let K/F be ramified cyclic of prime degree ℓ , with break t . Let $m = m_F(\chi_F)$ and $m' = m_K(\chi_F \circ N_{K/F})$.*

(i) *If $m > t+1$, then*

$$(30) \quad m' - 1 = \psi_{K/F}^H(m - 1).$$

(ii) *If $m \leq t$, then $m' = m$.*

(iii) *If $m = t+1$, then $m' \leq m$; moreover $m' < m$ if and only if there is a $\mu \in S(K/F)$ for which $m_F(\mu\chi_F) < t+1$.*

Proof. For (i), put $a = \psi_{K/F}^H(m - 1)$. By (19), $N_{K/F}(U_K^{a+1}) = U_F^m$ and $N_{K/F}(U_K^a) = U_F^{m-1}$. Since m is the exact conductor of χ_F , the pullback is trivial on U_K^{a+1} and nontrivial on U_K^a . Thus $m' = a + 1$, which is (30).

For (ii), first separate $m = 0$: an unramified character is trivial on U_F^0 , hence its norm pullback is unramified and $m' = 0$. Now assume $1 \leq m \leq t$. The character χ_F is nontrivial on U_F^{m-1}/U_F^m . The graded norm map is an isomorphism at every level $< t$, so its pullback is nontrivial on U_K^{m-1}/U_K^m and trivial on U_K^m . Hence $m' = m$.

For (iii), the pullback is always trivial on U_K^{t+1} , so $m' \leq t+1 = m$. It has smaller conductor precisely when the restriction of χ_F to the image

$$I = \text{im}(U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1})$$

is trivial. By Theorem 3.1(b), the quotient $(U_F^t/U_F^{t+1})/I$ has order ℓ . The proof of Corollary 3.8 identifies this quotient with $F^\times/N_{K/F}(K^\times)$, so its character group is exactly the restriction of $S(K/F)$ to the last unit layer. Therefore $\chi_F|_I = 1$ if and only if there is a unique restriction $\mu|_{U_F^t/U_F^{t+1}}$ with $(\mu\chi_F)|_{U_F^t/U_F^{t+1}} = 1$, equivalently $m_F(\mu\chi_F) < t+1$. $\square$

This is [3, Lemmas 8.8 and 8.12, pp. 73 and 78–79].

Lemma 3.10 (Unramified norm, trace, and conductor compatibility). *Let K/F be an unramified extension of degree f , and let ϖ be a uniformizer of F . Then ϖ is also a uniformizer of K , $[k_K : k_F] = f$, and the following statements hold.*

(a) *Reduction commutes with trace and norm:*

$$(31) \quad \overline{\text{Tr}_{K/F}(x)} = \text{Tr}_{k_K/k_F}(\bar{x}) \quad (x \in \mathcal{O}_K), \quad \overline{N_{K/F}(u)} = N_{k_K/k_F}(\bar{u}) \quad (u \in \mathcal{O}_K^\times).$$

(b) *For every integer a ,*

$$(32) \quad \text{Tr}_{K/F}(\mathfrak{p}_K^a) = \mathfrak{p}_F^a.$$

(c) *For every $r \geq 0$,*

$$(33) \quad N_{K/F}(U_K^r) = U_F^r, \quad N_{K/F}(\varpi) = \varpi^f.$$

(d) *If $\psi_K = \psi_F \circ \text{Tr}_{K/F}$ and $\chi_K = \chi_F \circ N_{K/F}$, then*

$$n_K(\psi_K) = n_F(\psi_F), \quad m_K(\chi_K) = m_F(\chi_F).$$

(e) The quotient $F^\times/N_{K/F}(K^\times)$ is cyclic of order f , generated by the class of ϖ .

Proof. Write $N = N_{K/F}$. Since K/F is unramified, its ramification index is 1. Thus $\mathfrak{p}_K = \varpi \mathcal{O}_K$, and the residue extension has degree f . The conjugates of an integral element reduce to the conjugates of its residue class. Taking the first and last coefficients of the characteristic polynomial therefore gives (31).

We use two elementary finite-field facts. The map Tr_{k_K/k_F} is onto: if $q = |k_F|$, its polynomial

$$X + X^q + \cdots + X^{q^{f-1}}$$

is nonzero and has degree $q^{f-1} < q^f = |k_K|$, so it cannot vanish on all of k_K ; a nonzero k_F -linear map to the one-dimensional space k_F is surjective. The map $N_{k_K/k_F} : k_K^\times \rightarrow k_F^\times$ is onto because $k_K^\times$ is cyclic and

$$N_{k_K/k_F}(x) = x^{(q^f-1)/(q-1)}$$

has image of order $q - 1$.

Choose $x \in \mathcal{O}_K$ whose residue has trace 1. Then $\mathrm{Tr}_{K/F}(x)$ is a unit of $\mathcal{O}_F$; after multiplying x by its inverse, we obtain an integral element of trace 1. Hence $\mathrm{Tr}_{K/F}(\mathcal{O}_K) = \mathcal{O}_F$. Since $\mathfrak{p}_K^a = \varpi^a \mathcal{O}_K$ for every $a \in \mathbf{Z}$, this proves (32).

For $r \geq 1$ and $x \in \mathcal{O}_K$, multiplication of the conjugates gives

$$N(1 + \varpi^r x) \equiv 1 + \varpi^r \mathrm{Tr}_{K/F}(x) \pmod{\mathfrak{p}_F^{r+1}}.$$

Thus the map induced by norm on $U_K^r/U_K^{r+1} \rightarrow U_F^r/U_F^{r+1}$ is the finite-field trace and is surjective. Starting with $v \in U_F^r$, choose a lift of its class, divide v by the norm of that lift, and repeat at levels $r+1, r+2, \dots$. Completeness makes the product of the successive lifts converge to an element of U_K^r with norm v . This proves $N(U_K^r) = U_F^r$ for $r \geq 1$. Surjectivity of the finite-field norm, followed by the case $r = 1$, gives $N(U_K^0) = U_F^0$. Finally, because $\varpi \in F$, its norm is ϖ^f . This proves (33).

Put $n = n_F(\psi_F)$. By (32), $\psi_F \circ \mathrm{Tr}$ is trivial on $\mathfrak{p}_K^{-n}$ and is not trivial on $\mathfrak{p}_K^{-n-1}$, so $n_K(\psi_K) = n$. By (33), for every $r \geq 0$ the character $\chi_F \circ N$ is trivial on U_K^r if and only if χ_F is trivial on U_F^r . Hence the multiplicative conductors are equal. This proves the conductor equalities in part (d).

Every $y \in K^\times$ has the form $y = \varpi^a u$ with $u \in U_K^0$, so

$$N(K^\times) = \varpi^{f\mathbf{Z}} U_F^0$$

by (33). The last assertion follows. $\square$

Corollary 3.11 (Norm characters in cyclic prime degree). *Let K/F be cyclic of prime degree ℓ . Then $F^\times/N_{K/F}(K^\times)$ is cyclic of order ℓ , and $S(K/F)$ has ℓ elements. In particular, $S(K/F)$ is finite.*

Proof. If K/F is ramified, this is Corollary 3.8. Suppose that it is unramified. Lemma 3.10(e) gives a cyclic norm quotient of order ℓ , and (33) shows that the norm group contains the open subgroup U_F^0 . Thus the quotient is discrete, and the remainder of the proof is the same as in Corollary 3.8. $\square$

By (15), we henceforth retain this minimal-conductor choice of χ_F in its $S(K/F)$ -orbit. If K/F is ramified, Proposition 3.9(ii)–(iii) then gives

$$(34) \quad m \leq t + 1 \implies m' = m.$$

4. LAMPRECHT'S FORMULA

Let E be a nonarchimedean local field, let $\psi : E \rightarrow \mathbf{C}^\times$ be a nontrivial additive character, and let θ be a quasi-character of conductor $m = 2d + \varepsilon > 1$, with $\varepsilon \in \{0, 1\}$. Choose $\gamma \in E^\times$ with $\gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n_E(\psi)}$.

Lemma 4.1 (Finite stationary-phase duality). *Let $n = n_E(\psi)$, let $M, q, r \in \mathbf{Z}$ with $r \leq q$, and let $\Gamma \in E^\times$ satisfy*

$$\Gamma \mathcal{O}_E = \mathfrak{p}_E^{M+n}.$$

Then

$$\mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r} \times \mathfrak{p}_E^r/\mathfrak{p}_E^q \longrightarrow \mathbf{C}^\times, \quad (\bar{c}, \bar{x}) \longmapsto \psi(cx/\Gamma),$$

is a perfect pairing.

Proof. The pairing is well defined: changing c by an element of $\mathfrak{p}_E^{M-r}$ or x by an element of $\mathfrak{p}_E^q$ changes cx/Γ by an element of $\mathfrak{p}_E^{-n}$, on which ψ is trivial. The annihilator of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$ in the first factor is zero, because

$$c\mathfrak{p}_E^r/\Gamma \subseteq \mathfrak{p}_E^{-n} \iff c \in \mathfrak{p}_E^{M-r}.$$

Similarly, the annihilator of the first factor in the second is zero, because

$$\mathfrak{p}_E^{M-q}x/\Gamma \subseteq \mathfrak{p}_E^{-n} \iff x \in \mathfrak{p}_E^q.$$

Thus the pairing is nondegenerate in both variables. Both finite additive groups have cardinality q_E^{q-r} , hence the induced maps to the respective character groups are isomorphisms. $\square$

Lemma 4.2 (Stationary numerator class at a chosen depth). *Let $q = m_E(\theta) > 1$, let $n = n_E(\psi)$, choose an integer M and an element $\Gamma \in E^\times$ such that*

$$\Gamma \mathcal{O}_E = \mathfrak{p}_E^{M+n},$$

and let r satisfy

$$\lceil q/2 \rceil \leq r \leq q-1.$$

There is a unique class

$$\text{St}_{\Gamma,r}(\theta) \in \mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r}$$

such that every representative c of this class satisfies

$$(35) \quad \theta(1+x) = \psi(cx/\Gamma) \quad (x \in \mathfrak{p}_E^r).$$

Every representative c has exact valuation

$$(36) \quad v_E(c) = M - q.$$

Proof. For $x, y \in \mathfrak{p}_E^r$, the quotient

$$\frac{(1+x)(1+y)}{1+x+y} = 1 + \frac{xy}{1+x+y}$$

belongs to U_E^q , because $2r \geq q$. Hence $x \mapsto \theta(1+x)$ is an additive character of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$. Lemma 4.1, applied with the present M, q, r, Γ , identifies the character group of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$ with $\mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r}$. This gives the unique class in the statement satisfying (35).

Every representative belongs to $\mathfrak{p}_E^{M-q}$. If it belonged to $\mathfrak{p}_E^{M-q+1}$, then for every $x \in \mathfrak{p}_E^{q-1}$ one would have $cx/\Gamma \in \mathfrak{p}_E^{-n}$. Since $r \leq q-1$, such x lies in the linearization domain, and (35) would make θ trivial on U_E^{q-1} , contrary to the exact conductor q . Thus $v_E(c) = M - q$. $\square$

Lemma 4.3 (Stationary-class calculus). *Use the notation of Lemma 4.2.*

- (a) *If $r \leq s \leq q-1$, restriction from $\mathfrak{p}_E^r$ to $\mathfrak{p}_E^s$ sends $\text{St}_{\Gamma,r}(\theta)$ to the class of the same representative modulo $\mathfrak{p}_E^{M-s}$.*
- (b) *If $\Gamma' = a\Gamma$, then ac represents the stationary class relative to Γ' whenever c represents the class relative to Γ .*
- (c) *Let θ_1, θ_2 be characters and suppose that, on a common depth r , coefficients c_1, c_2 satisfy*

$$\theta_i(1+x) = \psi(c_i x/\Gamma) \quad (x \in \mathfrak{p}_E^r).$$

Then $c_1 + c_2$ represents the restriction of $\theta_1\theta_2$ on that depth, and jc_1 represents the restriction of the integer power θ_1^j . These assertions are equalities of classes modulo $\mathfrak{p}_E^{M-r}$.

(d) Suppose $m_E(\theta_1) = m_E(\theta_2) = q$, take $\lceil q/2 \rceil \leq r \leq q-1$, and choose representatives c_i of $\text{St}_{\Gamma,r}(\theta_i)$ for $i = 1, 2$. Then

$$(37) \quad m_E(\theta_1\theta_2) < q \iff c_1 + c_2 \in \mathfrak{p}_E^{M-q+1}.$$

If $1 < q' = m_E(\theta_1\theta_2) < q$, put $r' = \lceil q'/2 \rceil$ and choose a representative c' of $\text{St}_{\Gamma,r'}(\theta_1\theta_2)$. Then

$$c' \equiv c_1 + c_2 \pmod{\mathfrak{p}_E^{M-r}}.$$

Proof. Parts (a) and (b) follow immediately from (35) and uniqueness. Part (c) follows by multiplying the two character identities; uniqueness at the common depth identifies the resulting class modulo $\mathfrak{p}_E^{M-r}$.

For (d), the product has conductor less than q precisely when it is trivial on U_E^{q-1} . For $x \in \mathfrak{p}_E^{q-1}$, part (c) gives

$$(\theta_1\theta_2)(1+x) = \psi((c_1 + c_2)x/\Gamma).$$

The annihilator of $\mathfrak{p}_E^{q-1}$ is $\mathfrak{p}_E^{M-q+1}$, which proves (37).

Now suppose $1 < q' = m_E(\theta_1\theta_2) < q$. Since $r' \leq r$, the linearization defining c' is valid for every $x \in \mathfrak{p}_E^{r'}$, and part (c) gives the same character there using $c_1 + c_2$. Thus

$$\psi((c' - c_1 - c_2)x/\Gamma) = 1 \quad (x \in \mathfrak{p}_E^{r'}).$$

The annihilator of $\mathfrak{p}_E^{r'}$ is $\mathfrak{p}_E^{M-r}$, proving the asserted congruence. $\square$

Lemma 4.4 (Magnitude of a nondegenerate finite quadratic sum). *Let V be a finite abelian group and let $q : V \rightarrow \mathbf{C}^\times$ satisfy $|q(x)| = 1$, $q(0) = 1$, and*

$$B(x, y) = \frac{q(x+y)}{q(x)q(y)}.$$

Assume that B is a nondegenerate bicharacter. Then

$$(38) \quad \left| \sum_{x \in V} q(x) \right|^2 = |V|.$$

In particular, the sum is nonzero.

Proof. Put $S = \sum_x q(x)$. Since $\overline{q(y)} = q(y)^{-1}$, the change of variables $x = y + h$ gives

$$\begin{aligned} |S|^2 &= \sum_{x, y \in V} q(x) \overline{q(y)} = \sum_{h, y \in V} q(y+h) q(y)^{-1} \\ &= \sum_{h \in V} q(h) \sum_{y \in V} B(h, y). \end{aligned}$$

For $h = 0$ the inner sum is $|V|$. For $h \neq 0$, nondegeneracy says that $B(h, \cdot)$ is a nontrivial character, whose sum is zero. Hence only $h = 0$ survives and $|S|^2 = |V|$. $\square$

Theorem 4.5 (Lamprecht). *Put*

$$\beta = \beta(\theta) := \text{St}_{\gamma, d+\varepsilon}(\theta) \in \mathcal{O}_E/\mathfrak{p}_E^d.$$

Equivalently, β is the unique class satisfying

$$(39) \quad \theta(1+x) = \psi(\beta x/\gamma) \quad (x \in \mathfrak{p}_E^{d+\varepsilon}).$$

Every representative of this class is a unit. Fix any representative, again denoted $\beta \in \mathcal{O}_E^\times$. If $\varepsilon = 0$, then

$$(40) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \psi(\beta/\gamma) \theta(\beta)^{-1}.$$

If $\varepsilon = 1$, choose $\delta \mathcal{O}_E = \mathfrak{p}_E^d$; we call δ a critical element. Via $\bar{x} \mapsto \delta x \pmod{\mathfrak{p}_E^{d+1}}$, identify k_E with $\mathfrak{p}_E^d/\mathfrak{p}_E^{d+1}$, the critical line. For $\bar{x} \in k_E$, choose a lift $x \in \mathcal{O}_E$ and put

$$(41) \quad H_E(\bar{x}) = \psi(\delta \beta x/\gamma) \theta(1 + \delta x)^{-1}.$$

We call H_E the critical function. It is well defined and unimodular, and

$$(42) \quad \Delta_E(\theta, \psi) = \theta(\gamma)\psi(\beta/\gamma)\theta(\beta)^{-1} \text{ph} \left(\sum_{x \in k_E} H_E(x) \right).$$

The right side of (42) is independent of the choice of δ with $\delta\mathcal{O}_E = \mathfrak{p}_E^d$. Moreover, for $\bar{x}, \bar{y} \in k_E$,

$$(43) \quad H_E(\bar{x} + \bar{y}) = H_E(\bar{x})H_E(\bar{y})\psi_0(\bar{x}\bar{y}),$$

where

$$(44) \quad \psi_0(\bar{z}) = \psi(\beta\delta^2 z/\gamma), \quad z \in \mathcal{O}_E \text{ a lift of } \bar{z} \in k_E,$$

is a well-defined nontrivial additive character of k_E . Consequently

$$\left| \sum_{x \in k_E} H_E(x) \right| = q_E^{1/2}.$$

Proof. Apply Lemma 4.2 with

$$q = M = m = 2d + \varepsilon, \quad \Gamma = \gamma, \quad r = d + \varepsilon.$$

Since $M - q = 0$ and $M - r = d$, it gives the class $\beta \in \mathcal{O}_E/\mathfrak{p}_E^d$ characterized by (39); (36) gives $v_E(\beta) = 0$ for every representative. This proves the first assertions of the theorem.

Put

$$I = \int_{U_E^0} \psi(u/\gamma)\theta(u)^{-1} du, \quad A = \mathfrak{p}_E^{d+\varepsilon}/\mathfrak{p}_E^m.$$

For each $x \in A$, substitute $u \mapsto u(1+x)$ in the integral. Haar measure on U_E^0 is translation invariant, and (39) gives

$$I = \int_{U_E^0} \psi(u/\gamma)\theta(u)^{-1} \psi((u-\beta)x/\gamma) du.$$

Average this identity over $x \in A$. Character orthogonality and Lemma 4.1, specialized to $M = q = m$, $r = d + \varepsilon$, and $\Gamma = \gamma$, give

$$\frac{1}{|A|} \sum_{x \in A} \psi((u-\beta)x/\gamma) = \begin{cases} 1, & u - \beta \in \mathfrak{p}_E^d, \\ 0, & u - \beta \notin \mathfrak{p}_E^d. \end{cases}$$

Hence only the compact open subset $\beta(1 + \mathfrak{p}_E^d)$ survives.

Assume first that $m = 2d$. Write $u = \beta(1+z)$ with $z \in \mathfrak{p}_E^d$. The linearization identity gives $\theta(1+z) = \psi(\beta z/\gamma)$. Therefore on the surviving set

$$\psi(u/\gamma)\theta(u)^{-1} = \psi(\beta/\gamma)\theta(\beta)^{-1}.$$

The integral is this nonzero constant times the positive measure of the surviving set. Taking its phase proves (40).

Now assume that $m = 2d + 1$. Since $m > 1$, one has $d \geq 1$. Partition $\beta(1 + \mathfrak{p}_E^d)$ into its q_E cosets of $1 + \mathfrak{p}_E^{d+1}$. They are represented by $u_x = \beta(1 + \delta x)$, where $x \in \mathcal{O}_E$ is a lift of $\bar{x} \in k_E$. If $w \in \mathfrak{p}_E^{d+1}$, then the quotient of the integrand at $u_x(1+w)$ by its value at u_x is

$$\psi(u_x w/\gamma)\theta(1+w)^{-1}.$$

Linearization gives $\theta(1+w) = \psi(\beta w/\gamma)$, while $(u_x - \beta)w \in \mathfrak{p}_E^{2d+1} = \mathfrak{p}_E^m$; hence the displayed quotient is one. Thus the integrand is constant on each of the q_E cosets. In particular, changing the lift x of $\bar{x}$ does not change the value of $H_E(\bar{x})$, so H_E is well defined. It is unimodular because $|\psi(\delta\beta x/\gamma)| = 1$ and the restriction of the quasi-character θ to the compact group U_E^0 has values of absolute value one. The value of the integrand at u_x is

$$\psi(\beta/\gamma)\theta(\beta)^{-1}H_E(\bar{x}).$$

As in the even case, integrating a constant over each coset contributes its value times the common positive Haar measure $c = \text{vol}(1 + \mathfrak{p}_E^{d+1})$. Hence

$$I = c \psi(\beta/\gamma) \theta(\beta)^{-1} \sum_{\bar{x} \in k_E} H_E(\bar{x}).$$

Since $I \neq 0$, one has $\sum_{\bar{x} \in k_E} H_E(\bar{x}) \neq 0$. Taking phases therefore gives (42). If $\delta' = v\delta$ with $v \in \mathcal{O}_E^\times$, and H'_E is formed using δ' , then

$$H'_E(\bar{x}) = H_E(\bar{v}\bar{x}),$$

because vx is a lift of $\bar{v}\bar{x}$. Multiplication by $\bar{v} \in k_E^\times$ permutes k_E , so $\sum_{\bar{x}} H'_E(\bar{x}) = \sum_{\bar{x}} H_E(\bar{x})$. This proves the asserted independence of δ .

For $\bar{x}, \bar{y} \in k_E$, choose lifts $x, y \in \mathcal{O}_E$ and put

$$w = \frac{\delta^2 xy}{1 + \delta(x + y)}.$$

Then

$$(1 + \delta x)(1 + \delta y) = (1 + \delta(x + y))(1 + w).$$

The additive linear factors in (41) cancel, so

$$\frac{H_E(\bar{x} + \bar{y})}{H_E(\bar{x})H_E(\bar{y})} = \theta(1 + w) = \psi(\beta w/\gamma).$$

Since $w - \delta^2 xy \in \mathfrak{p}_E^{3d}$ and $3d \geq 2d + 1 = m$, the last expression equals $\psi(\beta \delta^2 xy/\gamma)$. This proves (43) and (44). Put $a = \beta \delta^2/\gamma$. Since $v_E(a) = -n_E(\psi) - 1$, changing a lift z of $\bar{z}$ by an element of $\mathfrak{p}_E$ changes az by an element of $\mathfrak{p}_E^{-n_E(\psi)}$, on which ψ is trivial. Hence ψ_0 is well defined. If ψ_0 were trivial, then ψ would be trivial on $a\mathcal{O}_E = \mathfrak{p}_E^{-n_E(\psi)-1}$, contradicting the definition of $n_E(\psi)$. Thus ψ_0 is nontrivial. The bicharacter $(x, y) \mapsto \psi_0(xy)$ is nondegenerate: for $x \neq 0$, multiplication by x is a bijection of k_E , so $y \mapsto \psi_0(xy)$ is nontrivial. Equation (38) now gives $|\sum_{x \in k_E} H_E(x)| = q_E^{1/2}$. $\square$

This is [3, Lemma 8.1, pp. 60–62, and Lemma 9.3, pp. 86–87]. For the p -adic stationary evaluation in Dwork's normalization, see [2, Chapter II, pp. 25–27].

Lemma 4.6 (Independence of the stationary representative). *The stationary expression in Lamprecht's formula is independent of the representative of the class $\beta \in \mathcal{O}_E/\mathfrak{p}_E^d$. More precisely, if $\beta' = \beta + h$ with $h \in \mathfrak{p}_E^d$, then the right sides of (40) and (42), respectively, are unchanged when β is replaced by β' and the Hasse function (41) is formed with β' .*

Proof. Suppose first that $m = 2d$. Since β is a unit, $z = h/\beta$ lies in $\mathfrak{p}_E^d$, and the linearization formula gives $\theta(1 + z) = \psi(\beta z/\gamma) = \psi(h/\gamma)$. Hence

$$\frac{\psi(\beta'/\gamma) \theta(\beta')^{-1}}{\psi(\beta/\gamma) \theta(\beta)^{-1}} = \psi(h/\gamma) \theta(1 + h/\beta)^{-1} = 1.$$

Now let $m = 2d + 1$. Choose $\delta \mathcal{O}_E = \mathfrak{p}_E^d$ and write

$$a = \frac{h}{\beta \delta} \in \mathcal{O}_E, \quad \bar{a} \in k_E$$

for its residue class. Let H_β and $H_{\beta'}$ denote the Hasse functions formed with β and β' . For $\bar{x} \in k_E$, choose a lift $x \in \mathcal{O}_E$. Directly from (41),

$$H_{\beta'}(\bar{x}) = H_\beta(\bar{x}) \psi(\delta h x/\gamma).$$

The functional equation (43) gives

$$H_\beta(\bar{x} + \bar{a}) = H_\beta(\bar{x}) H_\beta(\bar{a}) \psi(\beta \delta^2 a x/\gamma).$$

Since $\beta \delta a = h$, one has $\psi(\beta \delta^2 a x/\gamma) = \psi(\delta h x/\gamma)$. Therefore

$$H_{\beta'}(\bar{x}) = H_\beta(\bar{a})^{-1} H_\beta(\bar{x} + \bar{a}), \quad \sum_{\bar{x} \in k_E} H_{\beta'}(\bar{x}) = H_\beta(\bar{a})^{-1} \sum_{\bar{x} \in k_E} H_\beta(\bar{x}).$$

On the other hand,

$$\frac{\psi(\beta'/\gamma)\theta(\beta')^{-1}}{\psi(\beta/\gamma)\theta(\beta)^{-1}} = \psi(h/\gamma)\theta(1+\delta a)^{-1} = H_\beta(\bar{a}).$$

Because the Hasse sums are nonzero, taking phases shows that these two multipliers cancel. Thus the odd-conductor stationary factor is also unchanged. $\square$

For a chosen representative β of the stationary class, set

$$(45) \quad \Delta_{2,E}(\theta) = \psi(\beta/\gamma)\theta(\beta)^{-1}, \quad \Delta_{3,E}(\theta) = \begin{cases} 1, & m_E(\theta) \text{ even,} \\ \text{ph}(\sum H_E), & m_E(\theta) \text{ odd.} \end{cases}$$

Then

$$(46) \quad \Delta_E(\theta, \psi) = \theta(\gamma)\Delta_{2,E}(\theta)\Delta_{3,E}(\theta).$$

When the conductor is odd, $\Delta_{2,E}(\theta)$ and $\Delta_{3,E}(\theta)$ separately depend on the chosen representative; their product is independent by Lemma 4.6. Every later comparison therefore either keeps these two factors together or applies the induced translation to the critical function.

Lemma 4.7 (Stable twist). *Let ν be a quasi-character of $E^\times$. If $m_E(\theta) = 2d + \varepsilon > 1$ and $m_E(\nu) \leq d$, then, for any representative β of the stationary class of θ ,*

$$(47) \quad \Delta_E(\nu\theta, \psi) = \nu(\gamma/\beta)\Delta_E(\theta, \psi).$$

Proof. Since $m_E(\nu) \leq d < m$, the character ν is trivial on U_E^{m-1} . Thus $\nu\theta$ is nontrivial on U_E^{m-1} and trivial on U_E^m , so

$$m_E(\nu\theta) = m_E(\theta) = m.$$

Moreover, ν is trivial on U_E^d , hence on $U_E^{d+\varepsilon}$. Therefore θ and $\nu\theta$ have the same restriction to the stationary layer $U_E^{d+\varepsilon}$, so their stationary classes coincide. Choose the same representative β for both. If m is even, both residual factors are 1; if m is odd, their Hasse functions are identical. Taking the ratio in (46) gives $\nu(\gamma)\nu(\beta)^{-1}$, proving (47). $\square$

Compare [3, Lemma 9.4, pp. 87–88].

Lemma 4.8 (Conductors and leading classes in a minimal norm-character orbit). *Let K/F be ramified cyclic of prime degree ℓ , let t be its break, put $T = t + 1$, and let τ be a nontrivial element of $S(K/F)$. Retain the minimal-conductor choice of χ_F . Put $m = m_F(\chi_F)$. Then, for every nontrivial power τ^j ,*

$$(48) \quad m_F(\tau^j\chi_F) = \max\{m, T\}.$$

If $T > 1$ and $m = T$, choose Γ with

$$\Gamma\mathcal{O}_F = \mathfrak{p}_F^{m+n_F(\psi_F)},$$

choose

$$\lceil m/2 \rceil \leq r \leq m - 1,$$

and choose $a, b \in \mathcal{O}_F^\times$ such that

$$\tau(1+x) = \psi_F(ax/\Gamma), \quad \chi_F(1+x) = \psi_F(bx/\Gamma) \quad (x \in \mathfrak{p}_F^r).$$

Then, for $j = 0, \dots, \ell - 1$, viewed as integers in F ,

$$(49) \quad b + ja \notin \mathfrak{p}_F.$$

Proof. A nontrivial norm character has conductor T by Corollary 3.8. If $m < T$, then χ_F is trivial on U_F^{T-1} , whereas τ^j is not, so the product has conductor T . If $m > T$, then τ^j is trivial on U_F^{m-1} , whereas χ_F is not, so the product has conductor m . If $m = T$, every member of the orbit has conductor at least T by minimality of χ_F in its orbit, and at most T because both factors are trivial on U_F^T . This proves (48).

Assume $T > 1$ and $m = T$. For $x \in \mathfrak{p}_F^r$, the two displayed linearizations give

$$(\tau^j\chi_F)(1+x) = \psi_F((b+ja)x/\Gamma).$$

If $b + ja \in \mathfrak{p}_F$, equation (37), with $M = q = m$, would give conductor less than m , contradicting (48). This proves (49). $\square$

5. FINITE-FIELD IDENTITIES

Let k be a finite field and fix a nontrivial additive character $\psi : (k, +) \rightarrow \mathbf{C}^\times$. Multiplicative characters in this section are characters of $k^\times$, unless another field is specified. We use the normalized Gauss sum $\tau_k(\chi, \psi)$ defined in (8); in particular $\tau_k(1, \psi) = 1$.

Lemma 5.1 (Finite-field Gauss-sum magnitude). *For*

$$G_k(\chi, \psi) = \sum_{x \in k^\times} \chi(x) \psi(x)$$

one has $G_k(1, \psi) = -1$. If χ is nontrivial, then

$$(50) \quad |G_k(\chi, \psi)|^2 = |k|.$$

Consequently every $G_k(\chi, \psi)$ and every $\tau_k(\chi, \psi)$ is nonzero.

Proof. The trivial-character formula is $\sum_{x \in k^\times} \psi(x) = -1$, because the sum over all of k is zero. Assume χ nontrivial and write $q = |k|$. Since the values of χ and ψ have absolute value one,

$$\begin{aligned} |G_k(\chi, \psi)|^2 &= \sum_{x, y \in k^\times} \chi(x/y) \psi(x - y) \\ &= \sum_{t \in k^\times} \chi(t) \sum_{y \in k^\times} \psi((t - 1)y). \end{aligned}$$

The inner sum is $q - 1$ when $t = 1$ and is -1 when $t \neq 1$. Therefore

$$|G_k(\chi, \psi)|^2 = (q - 1) - \sum_{t \neq 1} \chi(t).$$

Character orthogonality gives $\sum_{t \in k^\times} \chi(t) = 0$, so $\sum_{t \neq 1} \chi(t) = -1$ and the last expression is q . $\square$

Theorem 5.2 (Hasse–Davenport lift). *For a finite extension k'/k of degree f ,*

$$(51) \quad \tau_{k'}(\chi \circ N_{k'/k}, \psi \circ \text{Tr}_{k'/k}) = \tau_k(\chi, \psi)^f.$$

Proof. This is the classical lifting relation of Davenport and Hasse [1]. We include the Euler-product proof in the normalization used here.

For a monic polynomial

$$P(X) = X^r + a_1 X^{r-1} + \cdots + a_r \in k[X], \quad a_r \neq 0,$$

put

$$w(P) = \chi((-1)^r a_r) \psi(-a_1).$$

If P, Q are monic with nonzero constant terms, then the constant terms multiply and the negative of the coefficient of degree $\deg(PQ) - 1$ is the sum of the roots; hence $w(PQ) = w(P)w(Q)$. Therefore

$$\begin{aligned} L(T) &= \prod_{\substack{P \text{ monic irreducible} \\ P \neq X}} (1 - w(P)T^{\deg P})^{-1} \\ &= \sum_{\substack{P \text{ monic} \\ P(0) \neq 0}} w(P)T^{\deg P}. \end{aligned}$$

For degree $r \geq 2$, the coefficient is zero, because a_1 ranges freely through k and $\sum_{a_1 \in k} \psi(-a_1) = 0$. The coefficient in degree 1 is

$$\sum_{a_1 \neq 0} \chi(-a_1) \psi(-a_1) = G_k(\chi, \psi).$$

Thus

$$L(T) = 1 + G_k(\chi, \psi)T.$$

On the other hand, taking logarithms of the Euler product and grouping closed points after extension of scalars gives

$$\log L(T) = \sum_{r \geq 1} \frac{T^r}{r} \sum_{x \in k_r^\times} \chi(N_{k_r/k} x) \psi(\text{Tr}_{k_r/k} x),$$

where k_r/k is the degree- r extension. Indeed, an element of k_r of degree $d \mid r$ contributes through its monic minimal polynomial with multiplicity d , as in the usual Euler-product proof for the zeta function of the affine line. Comparing the displayed formulas for $L(T)$ and $\log L(T)$ yields

$$G_{k_r}(\chi \circ N_{k_r/k}, \psi \circ \text{Tr}_{k_r/k}) = (-1)^{r-1} G_k(\chi, \psi)^r.$$

Now apply this identity to χ^{-1} . Since $\tau_k(\chi, \psi) = -G_k(\chi^{-1}, \psi)$, the signs cancel:

$$\tau_{k_r}(\chi \circ N_{k_r/k}, \psi \circ \text{Tr}_{k_r/k}) = -(-1)^{r-1} G_k(\chi^{-1}, \psi)^r = \tau_k(\chi, \psi)^r.$$

Taking $r = f$ proves the theorem. $\square$

Theorem 5.3 (Hasse–Davenport multiplication formula). *Let $\ell \mid |k^\times|$, let $\eta : k^\times \rightarrow \mathbf{C}^\times$ be a multiplicative character of order ℓ , and let $\chi : k^\times \rightarrow \mathbf{C}^\times$ be any multiplicative character. Then*

$$(52) \quad \chi(\ell^\ell) \tau_k(\chi^\ell, \psi) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \psi) = \tau_k(\chi, \psi) \prod_{j=1}^{\ell-1} \tau_k(\chi \eta^j, \psi).$$

Proof. For

$$G(\alpha) = \sum_{x \in k^\times} \alpha(x) \psi(x), \quad G(1) = -1,$$

the classical Davenport–Hasse multiplication formula [1] is

$$\prod_{j=0}^{\ell-1} G(\alpha \eta^j) = \alpha(\ell^{-\ell}) G(\alpha^\ell) \prod_{j=1}^{\ell-1} G(\eta^j).$$

Apply it with $\alpha = \chi^{-1}$. Inversion permutes the nontrivial powers of η , and $G(\alpha) = -\tau_k(\alpha^{-1}, \psi)$. Both sides contain exactly ℓ ordinary Gauss sums when the factor $G(\chi^{-\ell})$ is counted, so the signs cancel. Finally $\chi^{-1}(\ell^{-\ell}) = \chi(\ell^\ell)$, which gives (52) in Langlands’s normalization. $\square$

Theorem 5.4 (Hasse–function lift). *Let k'/k be finite and let a nowhere-zero function $H : k \rightarrow \mathbf{C}^\times$ satisfy*

$$H(x+y) = H(x)H(y)\psi(xy).$$

Put

$$(53) \quad \omega_{k'/k}(x) = \sum_{\{\sigma, \tau\}} x^\sigma x^\tau, \quad H_{k'/k}(x) = H(\text{Tr } x) \psi(-\omega_{k'/k}(x)).$$

Here the sum defining $\omega_{k'/k}$ is over the unordered pairs of distinct k -embeddings $k' \hookrightarrow \bar{k}$. Then

$$(54) \quad - \sum_{x \in k'} H_{k'/k}(x) = \left(- \sum_{x \in k} H(x) \right)^{[k':k]}.$$

Proof. For the sum identity, let

$$P(X) = X^r + a_1 X^{r-1} + a_2 X^{r-2} + \dots$$

be monic and put

$$w(P) = H(-a_1) \psi(-a_2),$$

where $a_2 = 0$ in degree 1. If Q has first coefficients b_1, b_2 , then the first two coefficients of PQ are $a_1 + b_1$ and $a_2 + b_2 + a_1 b_1$. Hence

$$\begin{aligned} w(PQ) &= H(-a_1 - b_1) \psi(-a_2 - b_2 - a_1 b_1) \\ &= H(-a_1) H(-b_1) \psi(a_1 b_1) \psi(-a_2 - b_2 - a_1 b_1) = w(P) w(Q). \end{aligned}$$

Thus

$$L(T) = \prod_{P \text{ monic irreducible}} (1 - w(P)T^{\deg P})^{-1} = \sum_{P \text{ monic}} w(P)T^{\deg P}.$$

For degree $r \geq 2$, the coefficient is zero because a_2 ranges freely and ψ is nontrivial. In degree 1 it is $\sum_{x \in k} H(x)$. Therefore

$$L(T) = 1 + \left(\sum_{x \in k} H(x) \right) T.$$

We next justify the closed-point expansion, including compatibility with repeated closed points. Let P be monic irreducible of degree d , choose a root $\xi \in k_d$, and put

$$s = \text{Tr}_{k_d/k}(\xi), \quad e = \omega_{k_d/k}(\xi).$$

The first two coefficients of P are $-s$ and e , so

$$w(P) = H(s)\psi(-e).$$

If $r = hd$ and ξ is viewed in k_r , each of its d conjugates occurs h times. Consequently

$$\text{Tr}_{k_r/k}(\xi) = hs, \quad \omega_{k_r/k}(\xi) = he + \binom{h}{2}s^2.$$

Iteration of the functional equation for H gives

$$H(hs) = H(s)^h \psi \left(\binom{h}{2} s^2 \right).$$

Combining these identities yields

$$H(\text{Tr}_{k_r/k} \xi) \psi(-\omega_{k_r/k}(\xi)) = (H(s)\psi(-e))^h = w(P)^h.$$

By this identity, the d roots belonging to the closed point P contribute $d w(P)^h$ to the sum over k_r . Expanding the logarithm of the Euler product and regrouping by $r = h \deg P$ therefore gives

$$\log L(T) = \sum_{r \geq 1} \frac{T^r}{r} \sum_{x \in k_r} H(\text{Tr}_{k_r/k} x) \psi(-\omega_{k_r/k}(x)).$$

Comparing the logarithm of the linear formula for $L(T)$ with this expansion gives

$$\sum_{x \in k_r} H_{k_r/k}(x) = (-1)^{r-1} \left(\sum_{x \in k} H(x) \right)^r.$$

Multiplying by -1 gives (54). □

This is [3, Lemma 9.1, pp. 81–84]; see also the finite-field lifting theorem in [2, Chapter II, pp. 31–33].

Lemma 5.5 (Quadratic phase). *Assume $\text{char } k \neq 2$. Define the quadratic character $\nu : k^\times \rightarrow \{\pm 1\}$ by*

$$\nu(a) = \begin{cases} 1, & a \in (k^\times)^2, \\ -1, & a \notin (k^\times)^2, \end{cases}$$

and extend it by $\nu(0) = 0$. Put

$$Q_\psi(a, b) = \text{ph} \left(\sum_{x \in k} \psi \left(\frac{a}{2} x^2 + bx \right) \right), \quad a \neq 0.$$

Then

$$(55) \quad Q_\psi(a, b) = \nu(a) \psi(-b^2/2a) Q_\psi(1, 0), \quad Q_\psi(1, 0)^2 = \nu(-1).$$

The defining sum has absolute value $|k|^{1/2}$.

Proof. Let

$$S(a, b) = \sum_{x \in k} \psi \left(\frac{a}{2} x^2 + bx \right).$$

Completing the square gives

$$S(a, b) = \psi(-b^2/2a) S(a, 0).$$

For $c \neq 0$, every $y \in k^\times$ has $1 + \nu(y)$ square roots, while zero has one square root. Hence

$$(56) \quad \sum_{x \in k} \psi(cx^2) = \sum_{y \in k^\times} \nu(y) \psi(cy) = \nu(c) G_k(\nu, \psi).$$

The additive-character part disappeared because $\sum_{y \in k} \psi(cy) = 0$. Taking $c = a/2$ in (56) and comparing with $a = 1$ proves the first phase identity. Equation (50) shows at once that $|S(a, b)| = |k|^{1/2}$.

It remains to compute the square of the phase. Put $G = G_k(\nu, \psi)$. Changing variables $x = ty$ in the product of the two Gauss sums gives

$$G^2 = \sum_{t \in k^\times} \nu(t) \sum_{y \in k^\times} \psi((t+1)y).$$

Here the inner sum is $|k| - 1$ for $t = -1$ and is -1 for $t \neq -1$. Therefore

$$\begin{aligned} G^2 &= \nu(-1)(|k| - 1) - \sum_{t \neq -1} \nu(t) \\ &= \nu(-1)(|k| - 1) + \nu(-1) = \nu(-1)|k|, \end{aligned}$$

because $\sum_{t \in k^\times} \nu(t) = 0$. Since $S(1, 0) = \nu(1/2)G$ and $\nu(1/2)^2 = 1$, dividing by absolute values gives $Q_\psi(1, 0)^2 = \nu(-1)$. $\square$

Proposition 5.6 (Ramified Hasse-phase comparison). *Let K/F be totally ramified cyclic of odd prime degree ℓ , with single break t . Let $\chi_F : F^\times \rightarrow \mathbf{C}^\times$ be a quasi-character and let $\psi_F : F \rightarrow \mathbf{C}^\times$ be a nontrivial additive character. Put*

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \text{Tr}_{K/F}.$$

Suppose

$$m_F(\chi_F) = 2d + 1 > 1,$$

and assume either

$$t = 0 \quad \text{or} \quad d \geq t + 1.$$

Let $\Gamma \in F^\times$ be admissible for (χ_F, ψ_F) , and let $\beta \in \mathcal{O}_F^\times$ represent the stationary class of χ_F relative to Γ . Then Γ is also admissible for (χ_K, ψ_K) , and β represents its stationary class. With this common denominator and representative, the residual factors satisfy

$$(57) \quad \Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)^\ell.$$

Proof. Write $N = N_{K/F}$. By Proposition 3.9(i),

$$m_K(\chi_K) = \ell(2d + 1) - (\ell - 1)(t + 1) = 2d' + 1, \quad d' = \ell d - \frac{\ell - 1}{2}t.$$

Put $n = n_F(\psi_F)$ and choose uniformizers with $\varpi_F = N(\varpi_K)$. If $t > 0$, choose a generator σ of $\text{Gal}(K/F)$ and define λ by (21). The different exponent is $(\ell - 1)(t + 1)$, and therefore

$$n_K(\psi_K) = \ell n + (\ell - 1)(t + 1), \quad (2d' + 1) + n_K(\psi_K) = \ell(2d + 1 + n).$$

Since $v_K(\Gamma) = \ell(2d + 1 + n)$, the denominator Γ is also admissible for χ_K . For $y \in \mathfrak{p}_K^{d'+1}$, (17) gives $\text{Tr}_{K/F}(y) \in \mathfrak{p}_F^{d+1}$, and the norm expansion gives

$$N(1 + y) \equiv 1 + \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

Indeed, if $t = 0$, then $d' = \ell d$ and every degree- j term with $j \geq 2$ has valuation at least $\lceil j(d' + 1)/\ell \rceil \geq 2d + 1$. If $t > 0$, then $\ell = p$, and (25) gives this bound for $2 \leq j < p$, since

$2(d' + 1) + (p - 1)(t + 1) = p(2d + 1) + 1$; the norm term has valuation at least $d' + 1 \geq 2d + 1$ by (26). The chosen representative β therefore satisfies

$$\chi_K(1 + y) = \chi_F(1 + \text{Tr}_{K/F}(y)) = \psi_F(\beta \text{Tr}_{K/F}(y)/\Gamma) = \psi_K(\beta y/\Gamma) \quad (y \in \mathfrak{p}_K^{d'+1}).$$

Let k be the common residue field and, for $x \in k$, choose a lift $\tilde{x} \in \mathcal{O}_F \subset \mathcal{O}_K$; use the same notation for lifts of other elements of k . Write $E_2(y)$ for the second elementary symmetric function of the conjugates of y . The critical function for χ_F is

$$(58) \quad H_F(x) = \psi_F\left(\frac{\beta \varpi_F^d \tilde{x}}{\Gamma}\right) \chi_F(1 + \varpi_F^d \tilde{x})^{-1},$$

and its polar character is

$$\psi_0(z) = \psi_F\left(\frac{\beta \varpi_F^{2d} \tilde{z}}{\Gamma}\right).$$

Put $Y = \varpi_K^{d'} \tilde{x}$. We claim that the critical function for χ_K is

$$(59) \quad H_K(x) = \psi_F\left(\frac{\beta \text{Tr}_{K/F}(Y)}{\Gamma}\right) \chi_F\left(1 + \text{Tr}_{K/F}(Y) + E_2(Y)\right)^{-1}.$$

Indeed, it is enough to prove

$$N(1 + Y) \equiv 1 + \text{Tr}_{K/F}(Y) + E_2(Y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

If $t = 0$, then $d' = \ell d$. Every term of degree at least three in the norm has valuation at least $3d \geq 2d + 1$, and the norm term has valuation at least $\ell d \geq 2d + 1$. If $t > 0$, then $\ell = p$ is the residue characteristic and $d \geq t + 1$. For $3 \leq j \leq p - 1$, Equation (25) gives

$$v_F(s_j(Y)) \geq \left\lfloor \frac{jd' + (p - 1)(t + 1)}{p} \right\rfloor \geq 2d + 1,$$

while

$$v_F(NY) \geq d' = pd - \frac{p-1}{2}t \geq 2d + 1.$$

This proves the claimed norm congruence. Theorem 4.5 gives the independence of the lift in (59). The same estimates also give

$$\text{Tr}(Y) \in \mathfrak{p}_F^d, \quad E_2(Y) \in \mathfrak{p}_F^{2d},$$

and one extra power of $\mathfrak{p}_F$ when $\tilde{x} \in \mathfrak{p}_K$.

Set

$$\epsilon = \frac{\text{Tr}_{K/F}(\varpi_K^{2d'})}{\varpi_F^{2d}} \in \mathcal{O}_F.$$

The preceding valuation estimates show that this quotient is integral. We now show that it is a unit and determine its residue. If $t = 0$, then

$$c = \overline{\varpi_K^{d'}/\varpi_F^d} \in k^\times$$

is fixed by $\text{Gal}(K/F)$, because $d' = \ell d$. Hence

$$\bar{\epsilon} = \ell c^2.$$

If $t > 0$, equation (28) gives

$$(60) \quad \bar{\epsilon} = -\lambda^{p-1}.$$

In particular, ϵ is a unit in both cases.

The functional equation (43) over K gives directly

$$\begin{aligned} \frac{H_K(x + y)}{H_K(x)H_K(y)} &= \psi_K\left(\frac{\beta \varpi_K^{2d'} \tilde{x}\tilde{y}}{\Gamma}\right) \\ &= \psi_0(\bar{\epsilon}xy), \quad x, y \in k, \end{aligned}$$

with $\epsilon = \text{Tr}_{K/F}(\varpi_K^{2d'})/\varpi_F^{2d}$. Since $\bar{\epsilon} \neq 0$, this polar bicharacter is nondegenerate.

We compare its normalized sum with that of H_F . First assume that k has odd characteristic. The polar character of H_F is $(x, y) \mapsto \psi_0(xy)$, so for a unique $\alpha_0 \in k$ one has

$$H_F(x) = \psi_0(x^2/2 + \alpha_0 x).$$

Put

$$T_x = \overline{\text{Tr}(Y)/\varpi_F^d}, \quad Q_x = \overline{\text{Tr}(Y^2)/\varpi_F^{2d}}.$$

Using $\text{Tr}(Y)/\varpi_F^d$ itself as a lift of T_x in (58), one has

$$H_F(T_x) = \psi_F\left(\frac{\beta \text{Tr}(Y)}{\Gamma}\right) \chi_F(1 + \text{Tr}(Y))^{-1}.$$

Because $E_2(Y) \in \mathfrak{p}_F^{2d}$,

$$\frac{1 + \text{Tr}(Y)}{1 + \text{Tr}(Y) + E_2(Y)} \equiv 1 - E_2(Y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

Equation (39), applied to $-E_2(Y) \in \mathfrak{p}_F^{2d} \subseteq \mathfrak{p}_F^{d+1}$, gives

$$H_K(x) = H_F(T_x) \psi_0\left(-\frac{\overline{E_2(Y)}}{\varpi_F^{2d}}\right).$$

Now $2E_2(Y) = (\text{Tr } Y)^2 - \text{Tr}(Y^2)$, so

$$-\frac{\overline{E_2(Y)}}{\varpi_F^{2d}} = \frac{Q_x - T_x^2}{2}.$$

Substituting $H_F(T_x) = \psi_0(T_x^2/2 + \alpha_0 T_x)$ in $H_K(x) = H_F(T_x) \psi_0(-\overline{E_2(Y)}/\varpi_F^{2d})$ yields

$$H_K(x) = \psi_0(Q_x/2 + \alpha_0 T_x).$$

If $t = 0$, substitute $T_x = \ell c x$ and $Q_x = \ell c^2 x^2$ in this expression for H_K and replace x by $c^{-1}x$:

$$H_K(c^{-1}x) = \psi_0(\ell x^2/2 + \ell \alpha_0 x).$$

Multiplication by c^{-1} permutes k , so $\sum_{x \in k} H_K(c^{-1}x) = \sum_{x \in k} H_K(x)$. If $t > 0$, the trace-ideal formula (17) gives $T_x = 0$, while $Q_x = \bar{\epsilon} x^2$, and hence

$$H_K(x) = \psi_0(\bar{\epsilon} x^2/2).$$

Let ν be the quadratic character of $k^\times$ and put $Q = Q_{\psi_0}(1, 0)$. Suppose first that $t = 0$. Applying (55) to the formula for $H_K(c^{-1}x)$, the phase defining $\Delta_{3,F}(\chi_F)$ is $Q\psi_0(-\alpha_0^2/2)$ and the phase defining $\Delta_{3,K}(\chi_K)$ is

$$\nu(\ell)Q\psi_0(-\ell\alpha_0^2/2).$$

Since $Q^2 = \nu(-1)$, the required power identity is equivalent to

$$\nu(\ell) = \nu(-1)^{(\ell-1)/2}.$$

Let $|k| = p^f$. If f is even, both sides are 1. If f is odd, the order of p modulo ℓ is odd because $\ell \mid p^f - 1$; therefore $(p/\ell) = 1$. Quadratic reciprocity now gives the required sign identity.

Now suppose $t > 0$. Then $\ell = p$ and $\bar{\epsilon} = -\lambda^{p-1}$ by (60), so $\nu(\bar{\epsilon}) = \nu(-1)$. The formula for $H_K(x)$ in the case $t > 0$ and $Q^2 = \nu(-1)$ reduce the desired identity to

$$\nu(-1)^{(p-3)/2} = 1.$$

If $p \equiv 1 \pmod{4}$, then $\nu(-1) = 1$; if $p \equiv 3 \pmod{4}$, the exponent $(p-3)/2$ is even. This proves the comparison in odd residue characteristic.

It remains to treat residue characteristic two. Then $t = 0$, because ℓ is odd. With $c = \overline{\varpi_K^{d'}/\varpi_F^d}$, the norm calculation gives

$$H_K(c^{-1}x) = H_F(\ell x) \psi_0\left(-\binom{\ell}{2} x^2\right) = H_F(x)^\ell.$$

The second equality follows by iterating $H_F(x+y) = H_F(x)H_F(y)\psi_0(xy)$. Put $S = \sum_x H_F(x)$ and $H = \text{ph}(S)$. Choose $c_0 \in k^\times$ so that $\psi_0(x^2) = \psi_0(c_0x)$. Then

$$\overline{H_F(x)} = H_F(x)\psi_0(c_0x), \quad \overline{S} = H_F(c_0)^{-1}S, \quad H^2 = H_F(c_0).$$

As $H_F(x)^4 = 1$, the identity for $H_K(c^{-1}x)$ shows that the desired power identity is equivalent to

$$H_F(c_0)^{(\ell-1)/2} = 1 \quad (\ell \equiv 1 \pmod{4}), \quad H_F(c_0)^{(\ell+1)/2} = 1 \quad (\ell \equiv 3 \pmod{4}).$$

Write $\psi_0(x) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(ax)}$. One may take $c_0 = a^{2^f-1}$, where $f = [k : \mathbb{F}_2]$; hence

$$H_F(c_0)^2 = \psi_0(c_0^2) = (-1)^f.$$

The two displayed powers are automatic for $\ell \equiv 1, 7 \pmod{8}$. If $\ell \equiv 3, 5 \pmod{8}$, then $(2/\ell) = -1$; since $\ell \mid 2^f - 1$, the order of 2 modulo ℓ and therefore f are even. Thus $H_F(c_0)^2 = 1$, and the remaining two residue classes also follow.

In every case the phase defining $\Delta_{3,K}(\chi_K)$ is the ℓ th power of the phase defining $\Delta_{3,F}(\chi_F)$, which is (57). $\square$

Lemma 5.7 (Affine pencil of quadratic phases). *Let k have odd characteristic p , normalize ψ so that $\psi(x^p) = \psi(x)$, take $\rho \in k^\times$ with $1 + j\rho \neq 0$ for every $j \in \mathbb{F}_p$, and let $\sigma, v \in k$. Choose $a_0, b_0 \in k$ with*

$$a_0^p = 1 - \rho^{1-p}, \quad b_0^p = \sigma - \rho^{-1}v.$$

Then

$$Q_\psi(a_0, b_0) \prod_{j \in \mathbb{F}_p^\times} Q_\psi(j\rho, jv) = \prod_{j \in \mathbb{F}_p} Q_\psi(1 + j\rho, \sigma + jv).$$

Proof. Apply (55). The quadratic-character parts agree because

$$\prod_{j \in \mathbb{F}_p} (1 + j\rho) = 1 - \rho^{p-1}, \quad \prod_{j \neq 0} j = -1,$$

and $\nu(a_0) = \nu(1 - \rho^{1-p})$. The additive parts agree by

$$(61) \quad \sum_{j \in \mathbb{F}_p} \frac{(\sigma + jv)^2}{1 + j\rho} = \frac{\rho^{p-3}(\sigma\rho - v)^2}{\rho^{p-1} - 1}.$$

To prove (61), use

$$\sum_j \frac{1}{1 + j\rho} = \frac{\rho^{p-1}}{\rho^{p-1} - 1}, \quad \sum_j \frac{j}{1 + j\rho} = -\frac{\rho^{p-2}}{\rho^{p-1} - 1}, \quad \sum_j \frac{j^2}{1 + j\rho} = \frac{\rho^{p-3}}{\rho^{p-1} - 1},$$

which follow from the logarithmic derivative of $\prod_j (X + j) = X^p - X$. Finally the right side of (61) is $(b_0^2/a_0)^p$, and the normalization $\psi(z^p) = \psi(z)$ finishes the proof. $\square$

Proposition 5.8 (Homogeneous affine-pencil quotient). *Under the hypotheses and notation of Lemma 5.7, let $c \in k^\times$. Then*

$$\frac{Q_\psi(ca_0^p, cb_0^p) \prod_{j \in \mathbb{F}_p^\times} Q_\psi(cj\rho, cjv)}{\prod_{j \in \mathbb{F}_p} Q_\psi(c(1 + j\rho), c(\sigma + jv))} = \frac{Q_\psi(a_0, b_0) \prod_{j \in \mathbb{F}_p^\times} Q_\psi(j\rho, jv)}{\prod_{j \in \mathbb{F}_p} Q_\psi(1 + j\rho, \sigma + jv)}.$$

Thus a common nonzero scale does not change the normalized affine-pencil quotient.

Proof. For $a, c \neq 0$, completion of the square gives

$$Q_\psi(ca, cb) = \nu(c)\psi\left(-\frac{(c-1)b^2}{2a}\right)Q_\psi(a, b),$$

where ν is the quadratic character. Apply this identity to every factor of the quotient on the left. The factors $\nu(c)$ cancel because the numerator and denominator each contain p factors. The additive corrections cancel because

$$\frac{b_0^{2p}}{a_0^p} + \sum_{j \in \mathbf{F}_p^\times} \frac{(j\nu)^2}{j\rho} = \frac{b_0^{2p}}{a_0^p} = \sum_{j \in \mathbf{F}_p} \frac{(\sigma + j\nu)^2}{1 + j\rho},$$

using $\sum_{j \in \mathbf{F}_p^\times} j = 0$ and (61). Finally, $Q_\psi(a_0^p, b_0^p) = Q_\psi(a_0, b_0)$ by Frobenius invariance, which gives the quotient on the right. $\square$

Lemma 5.9 (Artin–Schreier phase). *Let k have characteristic 2 and let $\psi_0(x) = (-1)^{\text{Tr}_{k/\mathbf{F}_2}(x)}$. Then*

$$(62) \quad \psi_0(c + c^2) = 1 \quad (c \in k).$$

Proof. The absolute trace is Frobenius-invariant, so $\text{Tr}(c + c^2) = 0$. $\square$

6. STATIONARY PARAMETERS UNDER NORM

Let K/F be a cyclic extension of nonarchimedean local fields of prime degree ℓ . Let $\chi_F : F^\times \rightarrow \mathbf{C}^\times$ be a quasi-character and let $\psi_F : F \rightarrow \mathbf{C}^\times$ be a nontrivial additive character. Put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \text{Tr}_{K/F}.$$

If K/F is ramified, let t be its unique lower ramification break. Write

$$m = m_F(\chi_F) = 2d + \varepsilon > 1, \quad m' = m_K(\chi_K) = 2d' + \varepsilon', \quad \varepsilon, \varepsilon' \in \{0, 1\}.$$

The preceding conductor results give the following three cases for m' .

- (i) If K/F is unramified, then $m' = m$ by Lemma 3.10(d).
- (ii) If K/F is ramified and $m > t + 1$, then

$$m' = \psi_{K/F}^H(m - 1) + 1$$

by Proposition 3.9(i).

- (iii) If K/F is ramified and $m \leq t + 1$, retain the minimal-conductor choice of χ_F . Then $m' = m$ by (34).

Lemma 6.1 (Norm filtration at stationary depth). *In each of the three cases above,*

$$(63) \quad N_{K/F}(U_K^{d' + \varepsilon'}) \subseteq U_F^{d + \varepsilon},$$

$$(64) \quad N_{K/F}(U_K^{m'}) \subseteq U_F^m,$$

$$(65) \quad N_{K/F}(U_K^{d'}) \subseteq U_F^d.$$

The third inclusion implies that if $b, b' \in \mathcal{O}_K^\times$ and $b' \equiv b \pmod{\mathfrak{p}_K^{d'}}$, then

$$(66) \quad N_{K/F}(b') \equiv N_{K/F}(b) \pmod{\mathfrak{p}_F^d}.$$

Proof. Write $N = N_{K/F}$. Put

$$q = \left\lceil \frac{m}{2} \right\rceil = d + \varepsilon, \quad q' = \left\lceil \frac{m'}{2} \right\rceil = d' + \varepsilon'.$$

In case (i), Lemma 3.10(c) gives all three inclusions.

Now consider case (ii), so K/F is ramified and $m > t + 1$. We first prove (63). Set $r = m - 1 > t$ and $s = \lfloor r/2 \rfloor = q - 1$. Then

$$m' = \psi_{K/F}^H(r) + 1, \quad q' - 1 = \left\lfloor \frac{\psi_{K/F}^H(r)}{2} \right\rfloor.$$

We claim that

$$\left\lfloor \frac{\psi_{K/F}^H(r)}{2} \right\rfloor \geq \psi_{K/F}^H(s).$$

If $s \leq t$, then the right side is s , whereas $\psi_{K/F}^H(r) \geq r \geq 2s$. If $s > t$, write $r = 2s + e$ with $e \in \{0, 1\}$. Equation (16) gives

$$\psi_{K/F}^H(r) - 2\psi_{K/F}^H(s) = (\ell - 1)t + \ell e \geq 0,$$

which again proves the claimed inequality. Hence

$$q' \geq \psi_{K/F}^H(q - 1) + 1.$$

If $q - 1 < t$, norm preserves the filtration below the break, so $N(U_K^q) \subseteq U_F^q$. If $q - 1 = t$, the critical norm map gives $N(U_K^{t+1}) \subseteq U_F^{t+1}$. If $q - 1 > t$, (19), with $r = q - 1$, gives $N(U_K^{\psi_{K/F}^H(q-1)+1}) = U_F^q$. Together with the bound for q' above and the decreasing unit filtration, these three cases prove (63).

For (64), (19), applied with $r = m - 1$, gives $N(U_K^{m'}) = U_F^m$.

We next prove (65). Since $\psi_{K/F}^H(u) \geq u$, one has $m' \geq m$ and hence $d' \geq d$. If $d \leq t$, norm preserves the filtration at depth d , so

$$N(U_K^{d'}) \subseteq N(U_K^d) \subseteq U_F^d.$$

If $d = t + 1$, the same conclusion follows from the critical inclusion $N(U_K^{t+1}) \subseteq U_F^{t+1}$. Finally suppose $d \geq t + 2$ and put

$$R = \psi_{K/F}^H(d - 1) + 1.$$

Using (16) and $m = 2d + \varepsilon$, one obtains

$$\begin{aligned} m' - 2R &= \psi_{K/F}^H(2d + \varepsilon - 1) + 1 - 2(\psi_{K/F}^H(d - 1) + 1) \\ &= (\ell - 1)(t + 1) + \ell\varepsilon \geq 0. \end{aligned}$$

Thus $d' = \lfloor m'/2 \rfloor \geq R$. Equation (19), applied with $r = d - 1$, now gives

$$N(U_K^{d'}) \subseteq N(U_K^R) = U_F^d.$$

This proves all three inclusions in case (ii).

Finally consider case (iii). Here $m' = m \leq t + 1$, so $q' = q$ and $d' = d \leq t$. Below the break, norm maps U_K^i into U_F^i for every $i \leq t$; at the last possible step $m = t + 1$ one uses the critical inclusion $N(U_K^{t+1}) \subseteq U_F^{t+1}$. Applying these statements at $i = q$, $i = m$, and $i = d$ proves all three inclusions.

For the final assertion, write $b' = b(1 + z)$ with $z \in \mathfrak{p}_K^{d'}$. Then $1 + z \in U_K^{d'}$, so (65) gives $N(b')/N(b) \in U_F^d$, which is equivalent to (66). $\square$

For $y \in \mathfrak{p}_K^{d'+\varepsilon'}$, let $\bar{y}$ denote its class modulo $\mathfrak{p}_K^{m'}$.

Lemma 6.2 (Truncated norm homomorphism and adjoint). *In each of cases (i)–(iii), the assignment*

$$P_{K/F}(\bar{y}) = \overline{N_{K/F}(1 + y) - 1}$$

is a well-defined additive homomorphism

$$P_{K/F} : \mathfrak{p}_K^{d'+\varepsilon'} / \mathfrak{p}_K^{m'} \longrightarrow \mathfrak{p}_F^{d+\varepsilon} / \mathfrak{p}_F^m.$$

Let γ_F and γ_K be admissible denominators for (χ_F, ψ_F) and (χ_K, ψ_K) , respectively. The pairings

$$(\bar{x}, \bar{y}) \longmapsto \psi_F(xy/\gamma_F), \quad (\bar{x}', \bar{y}') \longmapsto \psi_K(x'y'/\gamma_K)$$

between $\mathcal{O}_F/\mathfrak{p}_F^d$ and $\mathfrak{p}_F^{d+\varepsilon}/\mathfrak{p}_F^m$, and between $\mathcal{O}_K/\mathfrak{p}_K^{d'}$ and $\mathfrak{p}_K^{d'+\varepsilon'}/\mathfrak{p}_K^{m'}$, are perfect. There is a unique additive adjoint

$$P_{K/F; \gamma_F, \gamma_K}^* : \mathcal{O}_F/\mathfrak{p}_F^d \longrightarrow \mathcal{O}_K/\mathfrak{p}_K^{d'}$$

with the following property: for $x \in \mathcal{O}_F$, an element $z \in \mathcal{O}_K$ represents $P_{K/F; \gamma_F, \gamma_K}^(\bar{x})$ if and only if*

$$(67) \quad \psi_F\left(\frac{x(N_{K/F}(1 + y) - 1)}{\gamma_F}\right) = \psi_K(zy/\gamma_K) \quad (y \in \mathfrak{p}_K^{d'+\varepsilon'}).$$

If

$$N_{K/F}(1+y) - 1 \equiv \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^m} \quad (y \in \mathfrak{p}_K^{d'+\varepsilon'})$$

and $\gamma_K/\gamma_F \in \mathcal{O}_K^\times$, then

$$P_{K/F;\gamma_F,\gamma_K}^*(\bar{x}) = \overline{(\gamma_K/\gamma_F)x} \quad (x \in \mathcal{O}_F).$$

Under the same trace-congruence hypothesis, if $\gamma_K = \gamma_F$, this is the map $\mathcal{O}_F/\mathfrak{p}_F^d \rightarrow \mathcal{O}_K/\mathfrak{p}_K^{d'}$ induced by $F \hookrightarrow K$.

Proof. Write $N = N_{K/F}$ and put

$$q = d + \varepsilon, \quad q' = d' + \varepsilon', \quad f(y) = N(1+y) - 1 \quad (y \in \mathfrak{p}_K^{q'}).$$

Equation (63) gives $f(y) \in \mathfrak{p}_F^q$. If $y' - y \in \mathfrak{p}_K^{m'}$, then $(1+y')/(1+y) \in U_K^{m'}$, so (64) gives $f(y') \equiv f(y) \pmod{\mathfrak{p}_F^m}$. Thus f induces the stated map.

For $y, z \in \mathfrak{p}_K^{q'}$, one has $yz \in \mathfrak{p}_K^{m'}$ because $2q' \geq m'$. Multiplicativity of the norm gives the identity

$$1 + f(y + z + yz) = (1 + f(y))(1 + f(z)).$$

The preceding congruence permits replacing $f(y + z + yz)$ by $f(y + z)$ modulo $\mathfrak{p}_F^m$. Also $f(y)f(z) \in \mathfrak{p}_F^{2q} \subseteq \mathfrak{p}_F^m$. Hence $f(y + z) \equiv f(y) + f(z) \pmod{\mathfrak{p}_F^m}$, proving additivity of $P_{K/F}$.

Lemma 4.1 gives perfectness of both pairings. For $x \in \mathcal{O}_F$, the left side of (67) defines a character of $\mathfrak{p}_K^{q'}/\mathfrak{p}_K^{m'}$, depending only on $\bar{x} \in \mathcal{O}_F/\mathfrak{p}_F^d$. Perfectness over K gives a unique class $\bar{z} \in \mathcal{O}_K/\mathfrak{p}_K^{d'}$ representing this character. Character multiplication and uniqueness make $\bar{x} \mapsto \bar{z}$ additive.

Under the stated trace congruence, the left side of (67) equals

$$\psi_F\left(\frac{x \text{Tr}_{K/F}(y)}{\gamma_F}\right) = \psi_K(xy/\gamma_F) = \psi_K\left(\frac{((\gamma_K/\gamma_F)x)y}{\gamma_K}\right).$$

Since $(\gamma_K/\gamma_F)x \in \mathcal{O}_K$, uniqueness gives $P_{K/F;\gamma_F,\gamma_K}^*(\bar{x}) = \overline{(\gamma_K/\gamma_F)x}$ and its common-denominator specialization. $\square$

The truncated norm and its adjoint are introduced in [3, pp. 62–64].

Corollary 6.3 (Stationary class under norm pullback). *Under the hypotheses and with the denominators of Lemma 6.2, let*

$$\beta_F = \text{St}_{\gamma_F, d+\varepsilon}(\chi_F) \in \mathcal{O}_F/\mathfrak{p}_F^d, \quad \beta_K = \text{St}_{\gamma_K, d'+\varepsilon'}(\chi_K) \in \mathcal{O}_K/\mathfrak{p}_K^{d'}.$$

Then

$$(68) \quad \beta_K = P_{K/F;\gamma_F,\gamma_K}^*(\beta_F).$$

Proof. Let $b_F \in \mathcal{O}_F^\times$ represent β_F , and let $z \in \mathcal{O}_K$ represent $P_{K/F;\gamma_F,\gamma_K}^*(\beta_F)$. For $y \in \mathfrak{p}_K^{d'+\varepsilon'}$, Lemma 6.1 and (67) give

$$\begin{aligned} \chi_K(1+y) &= \chi_F(N_{K/F}(1+y)) \\ &= \psi_F\left(\frac{b_F(N_{K/F}(1+y) - 1)}{\gamma_F}\right) \\ &= \psi_K(zy/\gamma_K). \end{aligned}$$

Uniqueness in Lemma 4.2, applied over K , gives $\bar{z} = \beta_K$. $\square$

In the unramified case, $d' = d$ and $\varepsilon' = \varepsilon$. If $y \in \mathfrak{p}_K^{d+\varepsilon}$, the j th elementary symmetric term in $N_{K/F}(1+y)$ has valuation at least $j(d+\varepsilon)$; hence every term with $j \geq 2$ lies in $\mathfrak{p}_F^m$, and

$$(69) \quad N_{K/F}(1+y) - 1 \equiv \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^m}.$$

Lemma 6.4 (Integer powers and Teichmuller representatives). *Let K/F be wildly ramified cyclic of prime degree $p = \text{char } k_F$, with break t , and put $T = t + 1$ and $s_0 = \lceil T/2 \rceil$. Let τ be a character of $F^\times$ of conductor T . For $j \in \mathbf{F}_p$, let $j\mathbf{z}$ be its representative in $\{0, \dots, p-1\}$, viewed in F , and let $[j] \in \mathcal{O}_F$ be its Teichmuller lift. If $A \in F$ satisfies*

$$\tau(1+x) = \psi_F(Ax) \quad (x \in \mathfrak{p}_F^{s_0}),$$

then

$$\psi_F(j\mathbf{z}Ax) = \psi_F([j]Ax) \quad (x \in \mathfrak{p}_F^{s_0}).$$

Thus $[j]A$ and $j\mathbf{z}A$ give the same linearization of τ^j on $\mathfrak{p}_F^{s_0}$. If p is odd, then

$$(70) \quad [j] - j\mathbf{z} \in \mathfrak{p}_F^{s_0}.$$

The Teichmuller lifts satisfy

$$(71) \quad \prod_{j \in \mathbf{F}_p} (X - [j]) = X^p - X, \quad \prod_{j \in \mathbf{F}_p} (X + [j]) = X^p + (-1)^p X,$$

and

$$(72) \quad \sum_{j \in \mathbf{F}_p} [j] = \begin{cases} 0, & p \text{ odd}, \\ 1, & p = 2. \end{cases}$$

Proof. In equal characteristic, the prime-field element $j\mathbf{z}$ is already its Teichmuller lift. Suppose that F has characteristic zero and put $e = v_F(p)$. Since $p = \text{Tr}_{K/F}(1)$, the trace-ideal formula (17) with $a = 0$ gives

$$e \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor \geq \left\lfloor \frac{T}{2} \right\rfloor.$$

Moreover $[j] - j\mathbf{z} \in p\mathcal{O}_F$. Hence for $x \in \mathfrak{p}_F^{s_0}$,

$$([j] - j\mathbf{z})x \in \mathfrak{p}_F^{e+s_0} \subseteq \mathfrak{p}_F^T.$$

For $x \in \mathfrak{p}_F^T$, the assumed linearization gives $\psi_F(Ax) = \tau(1+x) = 1$. This proves the asserted equality.

For even T , the bound on e proves (70). If p is odd and $T = 2r + 1 > 1$, then $r \geq 1$ and

$$p(r+1) \leq (p-1)(2r+1).$$

Hence

$$e \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor \geq r+1 = \frac{T+1}{2}.$$

Since $[j] - j\mathbf{z} \in p\mathcal{O}_F$, this proves (70).

The elements $[j]$ are the p roots of $X^p - X$ in F . This proves the first polynomial identity; replacing X by $-X$ gives the second. The coefficient of X^{p-1} gives the sum formula. $\square$

Remark 6.5 (Notation for powers of the norm character). In the wild cases below, τ denotes a generator of $S(K/F)$. An exponent j in τ^j denotes the integer representative of a class in $\mathbf{F}_p$. In formulas in F or K , the same symbol denotes its Teichmuller lift; in residue-field formulas it denotes its class in $\mathbf{F}_p$.

Lemma 6.6 (Norm representatives at subcritical precision). *Let K/F be ramified cyclic of prime degree with break t . Let $h \in \mathbf{Z}$, let $0 \leq s \leq t$, and let $c \in F^\times$ have $v_F(c) = h$. There is $c_1 \in K^\times$ with $v_K(c_1) = h$ such that*

$$(73) \quad c - N_{K/F}(c_1) \in \mathfrak{p}_F^{h+s}.$$

Proof. The first isomorphism in (18) supplies c_1 with $c/N_{K/F}(c_1) \in U_F^t$. Total ramification gives $v_K(c_1) = v_F(N_{K/F}(c_1)) = h$. Since c and $N_{K/F}(c_1)$ have the same valuation and their quotient lies in U_F^t , their difference belongs to $\mathfrak{p}_F^{h+t} \subseteq \mathfrak{p}_F^{h+s}$. $\square$

Lemma 6.7 (Norm-character conversion). *Suppose K/F is wildly ramified cyclic of prime degree p , put $T = t + 1$, and let τ generate $S(K/F)$. Put $s_0 = \lceil T/2 \rceil$, and let $\alpha \in F$ and $\gamma_F \in F^\times$ satisfy*

$$(74) \quad \tau(1 + x) = \psi_F(\alpha x / \gamma_F) \quad (x \in \mathfrak{p}_F^{s_0}).$$

Then, for $y \in \mathfrak{p}_K^{s_0}$,

$$\psi_F\left(\frac{\alpha N_{K/F}(y)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \operatorname{Tr}_{K/F}(y)}{\gamma_F}\right).$$

Proof. Put $D = D_{K/F} = (p - 1)T$. Equation (27), with $q = s_0$ and $r = T$, gives

$$N_{K/F}(1 + y) \equiv 1 + \operatorname{Tr}_{K/F}(y) + N_{K/F}(y) \pmod{\mathfrak{p}_F^T},$$

because $2s_0 \geq T$. The trace-ideal formula (17) and total ramification give, respectively,

$$v_F(\operatorname{Tr}_{K/F}(y)) \geq \left\lfloor \frac{s_0 + D}{p} \right\rfloor \geq s_0, \quad v_F(N_{K/F}(y)) = v_K(y) \geq s_0.$$

Since τ is trivial on norms and on U_F^T , applying (74) to the displayed norm congruence gives the asserted identity. $\square$

6.1. High-conductor parameters.

Proposition 6.8 (High-conductor stationary-class table). *Let γ_F be admissible for (χ_F, ψ_F) . Whenever β occurs below, it denotes a representative in $\mathcal{O}_F^\times$ of $\operatorname{St}_{\gamma_F, d+\varepsilon}(\chi_F)$. In each case, $\gamma_K = \gamma_F$, viewed in K , is admissible for (χ_K, ψ_K) . Set $P_{K/F}^* = P_{K/F; \gamma_F, \gamma_K}^*$.*

- (a) *If K/F is unramified, then $P_{K/F}^*(\bar{\beta}) = \bar{\beta}$ in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$.*
- (b) *If K/F is tamely ramified quadratic, then $P_{K/F}^*(\bar{\beta}) = \bar{\beta}$ in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$.*
- (c) *If K/F is ramified of odd prime degree, put $T = t + 1$. If $d \geq T$, equivalently $m = 2d + \varepsilon \geq 2T$, then $P_{K/F}^*(\bar{\beta}) = \bar{\beta}$ in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$.*
- (d) *Suppose K/F is wildly ramified of odd prime degree p . Retain the minimal-conductor choice of χ_F . In the complementary high-conductor range*

$$T \leq m \leq 2T - 1,$$

let τ generate $S(K/F)$. A representative $\beta \in \mathcal{O}_F^\times$ of $\operatorname{St}_{\gamma_F, d+\varepsilon}(\chi_F)$ and elements $\alpha_1 \in K^\times$, $\beta_1 \in \mathcal{O}_K^\times$ can be chosen so that

$$(75) \quad \alpha = N_{K/F}(\alpha_1), \quad v_K(\alpha_1) = m - T,$$

satisfies (74) on $\mathfrak{p}_F^{\lceil T/2 \rceil}$, and

$$(76) \quad \beta = N_{K/F}(\beta_1).$$

For these choices, let $\beta_K \in \mathcal{O}_K^\times$ and, for $j \in \mathbf{F}_p$, $\beta_j \in \mathcal{O}_F^\times$ represent the stationary classes of χ_K and $\tau^j \chi_F$ relative to γ_K and γ_F , respectively. Then

$$(77) \quad \beta_K \equiv \beta - \frac{\beta_1 \alpha}{\alpha_1} \pmod{\mathfrak{p}_K^{d'}}.$$

For $j \in \mathbf{F}_p$, with its Teichmüller representative used in F and K ,

$$(78) \quad \beta_j \equiv N_{K/F}(\beta_1 + j\alpha_1) \equiv \beta + j\alpha \pmod{\mathfrak{p}_F^d}.$$

Moreover,

$$N_{K/F}(\beta_K) \equiv \prod_{j \in \mathbf{F}_p} (\beta + j\alpha) = \beta^p - \beta\alpha^{p-1} \pmod{\mathfrak{p}_F^d}.$$

Proof. Write $N = N_{K/F}$ and put $D = D_{K/F}$. We first verify admissibility of $\gamma_K = \gamma_F$. In the unramified case, $m' = m$ and $n_K(\psi_K) = n_F(\psi_F)$, so an admissible γ_F remains admissible in K . In each ramified case, Proposition 3.9(i), with (34) at $m = t + 1$, and (29) give

$$m' = \ell m - D, \quad n_K(\psi_K) = \ell n_F(\psi_F) + D,$$

and therefore

$$m' + n_K(\psi_K) = \ell(m + n_F(\psi_F)) = v_K(\gamma_F).$$

Thus $\gamma_K = \gamma_F$ is admissible for χ_K in all four parts.

For (a), (69) and Lemma 6.2 identify the adjoint with the map on quotient rings induced by $F \hookrightarrow K$. Applying it to $\bar{\beta}$ proves (a).

For (b), $m' = 2m - 1$, so $d' + \varepsilon' = m$. For $x \in \mathfrak{p}_K^m$, the quadratic norm formula and $v_F(N(x)) = v_K(x) \geq m$ give $N(1+x) - 1 \equiv \text{Tr}_{K/F}(x) \pmod{\mathfrak{p}_F^m}$. Lemma 6.2 again gives $P_{K/F}^*(\bar{\beta}) = \bar{\beta}$.

For (c), put $T = t + 1$ and $q = d' + \varepsilon' = \lceil m'/2 \rceil$, and separate the tame and wild cases. If K/F is tamely ramified of odd degree ℓ , then

$$m' - 1 = \ell(m - 1).$$

For $x \in \mathfrak{p}_K^q$, let $s_j(x)$ be the j th elementary symmetric function of its ℓ conjugates. For $2 \leq j \leq \ell - 1$, every summand defining $s_j(x)$ has K -valuation at least jq . Since $s_j(x) \in F$ and $v_K|_F = \ell v_F$, it follows that

$$v_F(s_j(x)) \geq \left\lceil \frac{jq}{\ell} \right\rceil \geq \left\lceil \frac{2q}{\ell} \right\rceil \geq m.$$

Also $v_F(N(x)) = v_K(x) \geq q \geq m$. Thus all nonlinear norm terms vanish modulo $\mathfrak{p}_F^m$.

If K/F is wildly ramified, its degree is the residue characteristic p , $D = (p - 1)T$, and

$$m' = p(m - T) + T.$$

The assumption $d \geq T$ gives $q \geq m$. Moreover $2q + D \geq m' + D = pm$, so (25) gives $v_F(s_j(x)) \geq m$ for $2 \leq j \leq p - 1$. Again the norm term has valuation at least m . Thus in either case

$$N(1+x) - 1 \equiv \text{Tr}_{K/F}(x) \pmod{\mathfrak{p}_F^m} \quad (x \in \mathfrak{p}_K^q).$$

Lemma 6.2 gives $P_{K/F}^*(\bar{\beta}) = \bar{\beta}$, proving (c).

For (d), put $T = t + 1$, $v = m - T$, $r_0 = \lfloor T/2 \rfloor$, and $s_0 = \lceil T/2 \rceil$. Proposition 3.9(i) and (34) at $m = T$ give

$$m' = pm - D = T + pv = 2d' + \varepsilon,$$

so $\varepsilon' = \varepsilon$.

Relative to γ_F , the stationary class of τ on $\mathfrak{p}_F^{s_0}$ lies in $\mathfrak{p}_F^v / \mathfrak{p}_F^{v+r_0}$. Since $r_0 \leq t$, Lemma 6.6 with $(h, s) = (v, r_0)$ gives $\alpha = N(\alpha_1)$ satisfying (75) and (74). Likewise $d \leq t$, so the same lemma with $(h, s) = (0, d)$ gives $\beta = N(\beta_1)$ representing $\text{St}_{\gamma_F, d+\varepsilon}(\chi_F)$, proving (76). By definition of this stationary class,

$$\chi_F(1+x) = \psi_F(\beta x / \gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

For $j \in \mathbf{F}_p^\times$, equation (48) gives $m_F(\tau^j \chi_F) = m$ because $m \geq T$; for $j = 0$ the character is χ_F .

Put $q = \lceil m'/2 \rceil$. Since $2q + D \geq pm$, equation (27) gives

$$(79) \quad N(1+x) - 1 \equiv \text{Tr}(x) + N(x) \pmod{\mathfrak{p}_F^m} \quad (x \in \mathfrak{p}_K^q).$$

Since

$$q - v = \left\lceil \frac{T + (p-2)v}{2} \right\rceil \geq \left\lceil \frac{T+v}{2} \right\rceil = d + \varepsilon \geq s_0,$$

the element $y = \beta_1 x / \alpha_1$ belongs to $\mathfrak{p}_K^{s_0}$. Since $N\left(\frac{\beta_1 x}{\alpha_1}\right) = \frac{\beta}{\alpha} N(x)$ by (75) and (76), Lemma 6.7, applied to $y = \beta_1 x / \alpha_1$, gives

$$\psi_F\left(\frac{\beta N(x)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \text{Tr}(\beta_1 x / \alpha_1)}{\gamma_F}\right).$$

Equation (79) therefore gives

$$\begin{aligned}\psi_F\left(\frac{\beta(N(1+x)-1)}{\gamma_F}\right) &= \psi_F\left(\frac{\text{Tr}((\beta - \beta_1\alpha/\alpha_1)x)}{\gamma_F}\right) \\ &= \psi_K\left(\frac{(\beta - \beta_1\alpha/\alpha_1)x}{\gamma_K}\right),\end{aligned}$$

where $\gamma_K = \gamma_F$. By the defining adjoint identity (67), $\beta - \beta_1\alpha/\alpha_1$ represents $P_{K/F}^*(\bar{\beta})$. Equation (68) then gives (77).

Since $d + \varepsilon \geq s_0$, equations (39), (74), and Lemma 6.4 give

$$(\tau^j \chi_F)(1+x) = \psi_F((\beta + j\alpha)x/\gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

Equation (48) gives conductor m for $\tau^j \chi_F$. Using (75) and factoring $N(\beta_1 + j\alpha_1) = \beta N(1 + j\alpha_1/\beta_1)$, equation (25) gives every intermediate term valuation at least d , since

$$iv + (p-1)T \geq pd \quad (1 \leq i \leq p-1).$$

The two endpoint terms are β and $j\alpha$, because $j^p = j$ for the chosen Teichmüller lift. This proves (78) as a class congruence.

For the product congruence, choose the representative

$$\beta_K^0 = \beta - \frac{\beta_1\alpha}{\alpha_1}$$

of the class in (77). By (66), it suffices to compute $N(\beta_K^0)$ modulo $\mathfrak{p}_F^d$. Put $y = \beta_1\alpha/\alpha_1$. Then $v_K(y) = (p-1)v$ and $N(y) = \beta\alpha^{p-1}$. In the expansion of $N(\beta - y)$ every intermediate term has valuation at least d , because

$$i(p-1)v + (p-1)T \geq pd \quad (1 \leq i \leq p-1).$$

The endpoints give $\beta^p - \beta\alpha^{p-1}$, which equals $\prod_{j \in \mathbb{F}_p} (\beta + j\alpha)$. This proves the asserted congruence for $N_{K/F}(\beta_K)$. $\square$

Proposition 6.9 (Compatible wild-quadratic stationary classes and representatives). *Suppose K/F is wildly ramified quadratic and put $T = t + 1$. Retain the minimal-conductor choice of χ_F . Assume $m = m_F(\chi_F) \geq T$. Let τ be the nontrivial character in $S(K/F)$, and let γ_F be admissible for (χ_F, ψ_F) . Then $\gamma_K = \gamma_F$, viewed in K , is admissible for (χ_K, ψ_K) . There exist $\alpha_1 \in K^\times$ with $v_K(\alpha_1) = m - T$, $\beta_1 \in \mathcal{O}_K^\times$, and $\delta_h \in U_F^t$ such that the elements*

$$\alpha = N_{K/F}(\alpha_1), \quad \beta = \delta_h N_{K/F}(\beta_1)$$

satisfy

$$\begin{aligned}\tau(1+x) &= \psi_F(\alpha\delta_h x/\gamma_F) \quad (x \in \mathfrak{p}_F^{\lceil T/2 \rceil}), \\ \chi_F(1+x) &= \psi_F(\beta x/\gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).\end{aligned}$$

Let $\beta_K \in \mathcal{O}_K^\times$ and $\beta_{\tau\chi_F} \in \mathcal{O}_F^\times$ represent the stationary classes of χ_K and $\tau\chi_F$ for γ_K and γ_F , respectively. Then

$$(80) \quad \beta_K \equiv \beta - \frac{\beta_1\alpha\delta_h}{\alpha_1} \pmod{\mathfrak{p}_K^{d'}}, \quad \beta_{\tau\chi_F} \equiv \beta + \alpha\delta_h \pmod{\mathfrak{p}_F^d},$$

and

$$N_{K/F}(\beta_K) \equiv \beta(\beta + \alpha\delta_h) \pmod{\mathfrak{p}_F^d}.$$

Proof. Write $N = N_{K/F}$. Put $v = m - T \geq 0$. Proposition 3.9(i), with (34) when $m = T$, gives

$$m' = 2m - T = 2d' + \varepsilon'.$$

Since $D_{K/F} = T$, equation (29) gives $m' + n_K(\psi_K) = 2(m + n_F(\psi_F)) = v_K(\gamma_F)$. Thus $\gamma_K = \gamma_F$ is admissible. Set

$$r = \left\lfloor \frac{T}{2} \right\rfloor, \quad s = \left\lceil \frac{T}{2} \right\rceil.$$

Then

$$(81) \quad d' + \varepsilon' = m - r, \quad d' + \varepsilon' - v = s, \quad r + s = T.$$

Choose any representative β of the stationary class of χ_F . By the second isomorphism in (18), with $r = 0$, choose $\beta_1 \in \mathcal{O}_K^\times$ such that $N(\beta_1)/\beta \in U_F^t$, and put

$$\delta_h = \frac{\beta}{N(\beta_1)} \in U_F^t.$$

Let c_τ be any representative of the stationary coefficient class of τ relative to γ_F on $\mathfrak{p}_F^s$. That class lies in

$$\mathfrak{p}_F^v / \mathfrak{p}_F^{v+r}.$$

Apply Lemma 6.6 to the element $c_\tau \delta_h^{-1}$ with $(h, s) = (v, r)$. It supplies $\alpha_1 \in K^\times$ with $v_K(\alpha_1) = v$ such that, for $\alpha = N(\alpha_1)$,

$$\alpha \delta_h \equiv c_\tau \pmod{\mathfrak{p}_F^{v+r}}.$$

Thus $\alpha \delta_h$ is a permitted representative and

$$(82) \quad \tau(1+x) = \psi_F\left(\frac{\alpha \delta_h x}{\gamma_F}\right) \quad (x \in \mathfrak{p}_F^s).$$

This constructs the required simultaneous choices.

Lemma 6.7, applied to (82) with coefficient $\alpha \delta_h$ and denominator γ_F , gives, for $y \in \mathfrak{p}_K^s$,

$$(83) \quad \psi_F\left(\frac{\alpha \delta_h N(y)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \delta_h \text{Tr}(y)}{\gamma_F}\right).$$

Let $x \in \mathfrak{p}_K^{d'+\varepsilon'}$ and put $y = \beta_1 x / \alpha_1$. By (81), $y \in \mathfrak{p}_K^s$, and the identities $\beta = \delta_h N(\beta_1)$ and $\alpha = N(\alpha_1)$ give

$$\beta N(x) = \alpha \delta_h N(y).$$

Applying (83) to $y = \beta_1 x / \alpha_1$, and using the quadratic identity $N(1+x) - 1 = \text{Tr}_{K/F}(x) + N(x)$, yields

$$\begin{aligned} \psi_F\left(\frac{\beta(N(1+x) - 1)}{\gamma_F}\right) &= \psi_F\left(\frac{\beta \text{Tr}(x) - \alpha \delta_h \text{Tr}(\beta_1 x / \alpha_1)}{\gamma_F}\right) \\ &= \psi_K\left(\frac{(\beta - \beta_1 \alpha \delta_h / \alpha_1)x}{\gamma_F}\right). \end{aligned}$$

By (68), this proves the first congruence in (80).

Since $d + \varepsilon \geq s$, multiplying (82) by the linearization of χ_F gives

$$(\tau \chi_F)(1+x) = \psi_F((\beta + \alpha \delta_h)x / \gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

Since $m \geq T$, equation (48) gives $m_F(\tau \chi_F) = m$. This proves the second congruence in (80).

Finally choose the representative

$$\beta_K^0 = \beta - \frac{\beta_1 \alpha \delta_h}{\alpha_1}$$

of the stationary class of χ_K , and put $c = \beta_1 \alpha \delta_h / \alpha_1$. By (66), it is enough to compute $N(\beta_K^0)$ modulo $\mathfrak{p}_F^d$. The quadratic norm formula gives

$$N(\beta - c) = \beta^2 - \beta \text{Tr}(c) + N(c), \quad N(c) = \beta \alpha \delta_h.$$

Now $v_K(c) = v$, and the trace-ideal formula (17), with different exponent T , gives

$$v_F(\text{Tr}(c)) \geq \left\lfloor \frac{v+T}{2} \right\rfloor = d.$$

Consequently

$$N(\beta_K) \equiv N(\beta_K^0) \equiv \beta^2 + \beta \alpha \delta_h = \beta(\beta + \alpha \delta_h) \pmod{\mathfrak{p}_F^d},$$

which proves the asserted congruence for $N_{K/F}(\beta_K)$. $\square$

6.2. Low-conductor parameters. Let K/F be wildly ramified cyclic of prime degree $p = \text{char } k_F$, with break t , and put $T = t + 1$. Let τ be a nontrivial element of $S(K/F)$. Retain the minimal-conductor choice of χ_F . Assume

$$2 \leq m = 2d + \varepsilon \leq T, \quad v = T - m.$$

By (34), $m_K(\chi_K) = m$. Put

$$r_0 = \left\lfloor \frac{T}{2} \right\rfloor, \quad s_0 = \left\lceil \frac{T}{2} \right\rceil.$$

Choose

$$\Gamma_\tau \mathcal{O}_F = \mathfrak{p}_F^{T+n_F(\psi_F)}, \quad \xi_1 \mathcal{O}_K = \mathfrak{p}_K^v, \quad \xi = N_{K/F}(\xi_1),$$

and

$$(84) \quad \gamma_F = \Gamma_\tau / \xi, \quad \gamma_K = \Gamma_\tau / \xi_1.$$

Relative to Γ_τ , the stationary class of τ on $\mathfrak{p}_F^{s_0}$ lies in $\mathcal{O}_F / \mathfrak{p}_F^{r_0}$; relative to γ_F , the stationary class of χ_F lies in $\mathcal{O}_F / \mathfrak{p}_F^d$. Since $r_0, d \leq t$, Lemma 6.6 permits independent, hence simultaneous, choices $\alpha_1, \beta_1 \in U_K^0$ and exact norm representatives

$$(85) \quad \alpha = N_{K/F}(\alpha_1), \quad \beta = N_{K/F}(\beta_1),$$

such that

$$(86) \quad \tau(1+x) = \psi_F(\alpha x / \Gamma_\tau) \quad (x \in \mathfrak{p}_F^{s_0}), \quad \chi_F(1+x) = \psi_F(\beta x / \gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

When $m = T$, equation (49) of Lemma 4.8, applied with $\Gamma = \Gamma_\tau$, gives

$$j\alpha + \xi\beta \notin \mathfrak{p}_F \quad (j \neq 0).$$

Proposition 6.10 (Low-conductor stationary-class table). *Retain the minimal-conductor choice of χ_F . Let $\beta_K \in \mathcal{O}_K^\times$ represent the stationary class of χ_K relative to γ_K . Then*

$$\beta_K \equiv \frac{\beta\xi}{\xi_1} - \frac{\beta_1\alpha}{\alpha_1} \pmod{\mathfrak{p}_K^d}.$$

Relative to Γ_τ , the stationary class of χ_F is

$$\text{St}_{\Gamma_\tau, d+\varepsilon}(\chi_F) = \xi\beta + \mathfrak{p}_F^{v+d} \in \mathfrak{p}_F^v / \mathfrak{p}_F^{v+d}, \quad \xi\beta = N_{K/F}(\xi_1\beta_1).$$

For $j \in \mathbf{F}_p^\times$, let $\tilde{\beta}_j \in \mathcal{O}_F^\times$ represent $\text{St}_{\Gamma_\tau, s_0}(\tau^j \chi_F)$. With the Teichmüller representative used for j in F and K ,

$$(87) \quad \tilde{\beta}_j \equiv N_{K/F}(j\alpha_1 + \xi_1\beta_1) \equiv j\alpha + \xi\beta \pmod{\mathfrak{p}_F^{r_0}} \quad (j \in \mathbf{F}_p^\times).$$

Moreover,

$$N_{K/F}(\beta_K) \equiv \beta \prod_{j \in \mathbf{F}_p^\times} (\xi\beta + j\alpha) \pmod{\mathfrak{p}_F^d}.$$

Proof. Write $N = N_{K/F}$. The denominators in (84) are admissible, since

$$\begin{aligned} v_F(\gamma_F) &= T + n_F(\psi_F) - v = m + n_F(\psi_F), \\ v_K(\gamma_K) &= p(T + n_F(\psi_F)) - v = m + n_K(\psi_K), \end{aligned}$$

where the second equality uses (29). Put $q = d + \varepsilon = \lceil m/2 \rceil$. Since $2q + (p-1)T \geq pm$, equation (27) gives

$$(88) \quad N(1+x) - 1 \equiv \text{Tr}(x) + N(x) \pmod{\mathfrak{p}_F^m} \quad (x \in \mathfrak{p}_K^q).$$

For $y = \xi_1\beta_1 x / \alpha_1$ one has

$$v_K(y) \geq q + v = T - \lfloor m/2 \rfloor \geq s_0.$$

Lemma 6.7, applied to the linearization of τ in (86) with denominator Γ_τ , gives

$$\psi_F\left(-\frac{\alpha \text{Tr}(y)}{\Gamma_\tau}\right) = \psi_F\left(\frac{\alpha N(y)}{\Gamma_\tau}\right).$$

By (85), $\alpha N(y) = \xi \beta N(x)$, and hence

$$(89) \quad \psi_F\left(-\frac{\alpha \operatorname{Tr}(y)}{\Gamma_\tau}\right) = \psi_F\left(\frac{\xi \beta N(x)}{\Gamma_\tau}\right).$$

Put $\beta_K^0 = \beta \xi / \xi_1 - \beta_1 \alpha / \alpha_1$. Then

$$\begin{aligned} \psi_K\left(\frac{\beta_K^0 x}{\gamma_K}\right) &= \psi_F\left(\frac{\operatorname{Tr}_{K/F}(\xi_1 \beta_K^0 x)}{\Gamma_\tau}\right) \\ &= \psi_F\left(\frac{\xi \beta \operatorname{Tr}(x) - \alpha \operatorname{Tr}(\xi_1 \beta_1 x / \alpha_1)}{\Gamma_\tau}\right). \end{aligned}$$

Equations (89) and (88) therefore give

$$\psi_K(\beta_K^0 x / \gamma_K) = \psi_F\left(\frac{\beta(N(1+x) - 1)}{\gamma_F}\right) = \chi_K(1+x).$$

Corollary 6.3 now gives the stated congruence for β_K .

The formula for the stationary class of χ_F follows from $\gamma_F = \Gamma_\tau / \xi$ and $v_F(\xi) = v$. For $j \neq 0$, put $w = \xi_1 \beta_1 / (j \alpha_1)$, so $v_K(w) = v$. Equation (25) gives

$$v_F(s_i(w)) \geq \left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor \geq r_0 \quad (1 \leq i \leq p-1),$$

and therefore $N(j \alpha_1 + \xi_1 \beta_1) \equiv j \alpha + \xi \beta \pmod{\mathfrak{p}_F^{r_0}}$. Since $s_0 \geq d + \varepsilon$, the two linearizations in (86) and Lemma 6.4 give

$$(\tau^j \chi_F)(1+x) = \psi_F((j \alpha + \xi \beta)x / \Gamma_\tau) \quad (x \in \mathfrak{p}_F^{s_0}).$$

For $j \neq 0$, the character has conductor T by Lemma 4.8, so this proves (87).

Put

$$u = \frac{\xi_1 \beta_1}{\alpha_1}, \quad n = N(u) = \frac{\xi \beta}{\alpha},$$

so the representative already constructed is

$$\beta_K^0 = \frac{\alpha}{\xi_1}(n - u).$$

By (66), $N(\beta_K)$ modulo $\mathfrak{p}_F^d$ may be computed from this representative. Thus

$$N(\beta_K) \equiv N(\beta_K^0) = \frac{\alpha^p}{\xi} N(n - u) \pmod{\mathfrak{p}_F^d}.$$

In the expansion of $N(n - u)$ the two endpoints are $n^p + (-1)^p n$. For $1 \leq i \leq p-1$, the intermediate term $n^{p-i} s_i(u)$, after multiplication by α^p / ξ , has valuation at least

$$L_i = (p-i-1)v + \left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor.$$

Because T is an integer,

$$\left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor = T - \left\lceil \frac{T - iv}{p} \right\rceil.$$

Moreover $i \geq 1$ and $v = T - m \geq 0$, so $T - iv \leq T - v = m = 2d + \varepsilon$. Since $p \geq 2$,

$$\left\lceil \frac{T - iv}{p} \right\rceil \leq \left\lceil \frac{m}{2} \right\rceil = d + \varepsilon.$$

Consequently

$$\begin{aligned} L_i &\geq (p-i-1)v + T - d - \varepsilon \\ &= (p-i)v + d \\ &\geq d. \end{aligned}$$

Thus every intermediate term vanishes modulo $\mathfrak{p}_F^d$, and only the endpoints survive. The Teichmüller product identity (71) gives

$$N(\beta_K) \equiv \frac{\alpha^p}{\xi} (n^p + (-1)^p n) = \beta \prod_{j \neq 0} (\xi \beta + j \alpha) \pmod{\mathfrak{p}_F^d}.$$

This proves the asserted congruence for $N_{K/F}(\beta_K)$. $\square$

7. UNRAMIFIED AND TAME CASES

Let K/F be cyclic of prime degree ℓ , let χ_F be a quasi-character of $F^\times$, and let ψ_F be a nontrivial additive character of F . Put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \text{Tr}_{K/F}.$$

Retain the minimal-conductor choice of χ_F .

If both multiplicative conductors exceed 1, a common admissible denominator $\gamma \in F^\times$ and a common stationary representative $\beta \in \mathcal{O}_F^\times$ give

$$(90) \quad \chi_K(\gamma) \Delta_{2,K}(\chi_K) = (\chi_F(\gamma) \Delta_{2,F}(\chi_F))^\ell.$$

Here both Δ_2 factors use these choices. Indeed, $N_{K/F}(a) = a^\ell$ and $\text{Tr}_{K/F}(a) = \ell a$ for $a \in F$; substitution in (45) proves (90).

Proposition 7.1 (Unramified case). *The First Main Lemma holds when K/F is unramified.*

Proof. Write $N = N_{K/F}$. Put

$$k = k_F, \quad k' = k_K, \quad n = n_F(\psi_F), \quad m = m_F(\chi_F).$$

Choose a uniformizer ϖ of F . By Lemma 3.10, it is also a uniformizer of K , the residue extension k'/k has degree ℓ , and

$$n_K(\psi_K) = n, \quad m_K(\chi_K) = m.$$

Step 1: reduction to a power identity. Lemma 3.10(e) identifies $F^\times/N(K^\times)$ with the cyclic group generated by the class of ϖ , of order ℓ . Hence every $\mu \in S(K/F)$ is unramified, and the values $\mu(\varpi)$ run through all ℓ th roots of unity. Twisting by such a μ does not change the conductor of χ_F . Use the same admissible element $\gamma_F = \varpi^{m+n}$ for χ_F and $\mu\chi_F$. Since μ is trivial on U_F^0 , their unit integrals are equal, and therefore

$$(91) \quad \Delta_F(\mu\chi_F, \psi_F) = \mu(\varpi)^{m+n} \Delta_F(\chi_F, \psi_F).$$

For μ itself, (9) gives

$$\Delta_F(\mu, \psi_F) = \mu(\varpi)^n.$$

Both formulas remain valid when $m = 0$. The values $\mu(\varpi)$ are the roots of $X^\ell - 1$, whose product is $(-1)^{\ell-1}$; hence

$$\prod_{\mu \in S(K/F)} \mu(\varpi) = (-1)^{\ell-1}.$$

Equations (91) and (9) give the quotient of the product on the right side of (1) by the norm-character product on its left as

$$(-1)^{m(\ell-1)} \Delta_F(\chi_F, \psi_F)^\ell.$$

Thus the First Main Lemma is equivalent to

$$(92) \quad \Delta_K(\chi_K, \psi_K) = (-1)^{m(\ell-1)} \Delta_F(\chi_F, \psi_F)^\ell.$$

Step 2: conductor $m = 0$. Formula (9) gives

$$\Delta_F(\chi_F, \psi_F) = \chi_F(\varpi)^n, \quad \Delta_K(\chi_K, \psi_K) = \chi_K(\varpi)^n.$$

By Lemma 3.10(c), $N(\varpi) = \varpi^\ell$, so $\chi_K(\varpi) = \chi_F(\varpi)^\ell$. This proves (92) for $m = 0$.

Step 3: conductor $m = 1$. For $\bar{u} \in k^\times$ and $\bar{x} \in k$, choose lifts $u \in \mathcal{O}_F^\times$ and $x \in \mathcal{O}_F$. Define

$$\bar{\chi}(\bar{u}) = \chi_F(u), \quad \bar{\psi}(\bar{x}) = \psi_F(x/\varpi^{n+1}).$$

These are the residual multiplicative and additive characters on k . By (31), the residual characters on k' are

$$\bar{\chi}_K = \bar{\chi} \circ N_{k'/k}, \quad \bar{\psi}_K = \bar{\psi} \circ \text{Tr}_{k'/k}.$$

The conductor-one formula (10) gives

$$\Delta_F(\chi_F, \psi_F) = -\chi_F(\varpi)^{n+1} \text{ph}(\tau_k(\bar{\chi}, \bar{\psi}))$$

and

$$\Delta_K(\chi_K, \psi_K) = -\chi_K(\varpi)^{n+1} \text{ph}(\tau_{k'}(\bar{\chi}_K, \bar{\psi}_K)).$$

Equations (51) and (31) give

$$\tau_{k'}(\bar{\chi}_K, \bar{\psi}_K) = \tau_k(\bar{\chi}, \bar{\psi})^\ell.$$

Since $\text{ph}(z^\ell) = \text{ph}(z)^\ell$ for $z \neq 0$, and since $\chi_K(\varpi) = \chi_F(\varpi)^\ell$, this yields

$$\Delta_K(\chi_K, \psi_K) = (-1)^{\ell-1} \Delta_F(\chi_F, \psi_F)^\ell,$$

which is (92) because $m = 1$.

Step 4: conductor $m > 1$. Write $m = 2d + \varepsilon$, with $\varepsilon \in \{0, 1\}$, and put $\gamma = \varpi^{m+n}$. Proposition 6.8(a) provides a common stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F and χ_K with this common admissible denominator. If $\varepsilon = 0$, both residual factors are 1 by (45). Equations (90) and (46) then prove (92), whose sign is 1 for even m .

It remains to treat $\varepsilon = 1$. Then $m = 2d + 1$ and $d \geq 1$. Take $\delta = \varpi^d$ and form $H_F : k \rightarrow \mathbf{C}^\times$ and $H_K : k' \rightarrow \mathbf{C}^\times$ by (41), using γ , β , and δ for both characters. The residual additive character (44) is

$$(93) \quad \psi_0(\bar{z}) = \psi_F(\beta \delta^2 z / \gamma), \quad z \in \mathcal{O}_F \text{ a lift of } \bar{z} \in k.$$

For $x \in \mathcal{O}_K$, let $E_j(x)$ be the j th elementary symmetric function of its F -conjugates. Thus $E_1(x) = T(x) := \text{Tr}_{K/F}(x)$ and

$$E_2(x) = \sum_{\{\sigma, \tau\}} x^\sigma x^\tau,$$

the sum being over unordered pairs of distinct F -embeddings of K . The norm expansion is

$$N(1 + \delta x) = 1 + \delta T(x) + \delta^2 E_2(x) + \sum_{j=3}^{\ell} \delta^j E_j(x),$$

where the last sum is empty if $\ell = 2$. Each term in that sum lies in $\mathfrak{p}_F^{3d} \subseteq \mathfrak{p}_F^{2d+1}$, and the same is true of the cross term $\delta^3 T(x) E_2(x)$. Hence

$$(94) \quad N(1 + \delta x) \equiv (1 + \delta T(x))(1 + \delta^2 E_2(x)) \pmod{\mathfrak{p}_F^m}.$$

Because $\delta^2 E_2(x) \in \mathfrak{p}_F^{2d} \subseteq \mathfrak{p}_F^{d+1}$, equations (39) and (93) give

$$(95) \quad \chi_F(1 + \delta^2 E_2(x))^{-1} = \psi_F(-\beta \delta^2 E_2(x) / \gamma) = \psi_0(-\overline{E_2(x)}).$$

Substituting (94) and (95) in (41) over K gives

$$(96) \quad H_K(\bar{x}) = H_F(\text{Tr}_{k'/k}(\bar{x})) \psi_0(-\omega_{k'/k}(\bar{x})).$$

Here (31) identifies $\overline{T(x)}$ with the residue-field trace. The same reduction of the characteristic polynomial, now using its second coefficient, identifies $\overline{E_2(x)}$ with the function $\omega_{k'/k}$ in (53). Taking phases in (54), using (96) and (45), gives

$$(97) \quad \Delta_{3,K}(\chi_K) = (-1)^{\ell-1} \Delta_{3,F}(\chi_F)^\ell.$$

Equations (90), (46), and (97) prove (92), since m is odd. □

Proposition 7.2 (Tame conductor-one case). *Let K/F be tamely ramified cyclic of prime degree ℓ . Retain the minimal-conductor choice of χ_F . Suppose $m_F(\chi_F) = 1$. Then the First Main Lemma holds.*

Proof. Since $t = 0$ and $m_F(\chi_F) = 1$, equation (34) gives $m_K(\chi_K) = 1$.

Choose ϖ_K and put $\varpi_F = N_{K/F}(\varpi_K)$. Let $n = n_F(\psi_F)$. Since the tame different has exponent $\ell - 1$,

$$n_K(\psi_K) = \ell n + \ell - 1.$$

The element

$$\Gamma = \varpi_F^{n+1}$$

has K -valuation $\ell(n + 1) = n_K(\psi_K) + 1$, so it is an admissible element for every conductor-one factor over both fields. Let k be the common residue field. For $x \in k$, choose a lift $\tilde{x} \in \mathcal{O}_F$ and define

$$\bar{\psi}(x) = \psi_F(\tilde{x}/\Gamma) \quad (x \in k), \quad \bar{\chi}(x) = \chi_F(\tilde{x}) \quad (x \in k^\times).$$

On residue classes the norm and trace are

$$\overline{N_{K/F}(u)} = \bar{u}^\ell, \quad \overline{\text{Tr}_{K/F}(u)} = \ell \bar{u}.$$

Hence the residual characters attached to (χ_K, ψ_K) are

$$(98) \quad \bar{\chi}_K = \bar{\chi}^\ell, \quad \bar{\psi}_K(x) = \bar{\psi}(\ell x).$$

Choose a generator μ_1 of $S(K/F)$, put $\mu_j = \mu_1^j$ for $0 \leq j < \ell$, and let η be the residue character of μ_1 . Then η has order ℓ and the residue character of μ_j is η^j . Moreover $\mu_j(\varpi_F) = 1$, because $\varpi_F = N_{K/F}(\varpi_K)$. For $j \neq 0$, equation (48) of Lemma 4.8 gives $m_F(\mu_j \chi_F) = 1$; the case $j = 0$ is the given conductor of χ_F .

The finite-realization formula (5), with admissible element Γ , definition (8), and (98) give

$$(99) \quad \begin{aligned} \Delta_K(\chi_K, \psi_K) &= -\chi_F(\Gamma)^\ell \text{ph}(\tau_k(\bar{\chi}^\ell, \bar{\psi}(\ell \cdot))), \\ \Delta_F(\mu_j \chi_F, \psi_F) &= -\chi_F(\Gamma) \text{ph}(\tau_k(\bar{\chi} \eta^j, \bar{\psi})), \\ \Delta_F(\mu_j, \psi_F) &= -\text{ph}(\tau_k(\eta^j, \bar{\psi})) \quad (1 \leq j < \ell). \end{aligned}$$

By (11), $\Delta_F(1, \psi_F) = 1$. The powers of $\chi_F(\Gamma)$ and the signs are the same on the two sides of the First Main Lemma and cancel.

The substitution $x \mapsto \ell^{-1}x$ in (8) gives

$$(100) \quad \tau_k(\bar{\chi}^\ell, \bar{\psi}(\ell \cdot)) = \bar{\chi}(\ell^\ell) \tau_k(\bar{\chi}^\ell, \bar{\psi}).$$

Taking phases in the Hasse–Davenport multiplication formula (52), with $\chi = \bar{\chi}$ and $\psi = \bar{\psi}$, and substituting (99) and (100) proves the proposition. $\square$

Proposition 7.3 (Tamely ramified odd degree). *The First Main Lemma holds for a tamely ramified cyclic extension of odd prime degree.*

Proof. Here $t = 0$ and every nontrivial norm character has conductor 1. Put $m = m_F(\chi_F)$ and $n = n_F(\psi_F)$. For $m = 0$, choose a uniformizer ϖ_K of K and set $\varpi_F = N_{K/F}(\varpi_K)$; this is a uniformizer of F . Since $n_K(\psi_K) = \ell n + (\ell - 1)$, formula (9) gives

$$\Delta_K(\chi_K) = \chi_F(\varpi_F)^{\ell n + \ell - 1}.$$

Every nontrivial $\mu \in S(K/F)$ is trivial on ϖ_F and has conductor one, so (10) gives

$$\Delta_F(\mu \chi_F) = \chi_F(\varpi_F)^{n+1} \Delta_F(\mu).$$

Multiplying this for the $\ell - 1$ nontrivial μ and including $\Delta_F(\chi_F) = \chi_F(\varpi_F)^n$ gives exponent $n + (\ell - 1)(n + 1) = \ell n + \ell - 1$; the factors $\prod_{\mu \neq 1} \Delta_F(\mu)$ agree on both sides. This proves the conductor-zero case; the same calculation is valid for $\ell = 2$.

For $m = 1$, Proposition 7.2 applies. Now let $m = 2d + \varepsilon > 1$, with $\varepsilon \in \{0, 1\}$. Since $t = 0$ and $d \geq 1$, Proposition 6.8(c) provides a common admissible denominator $\gamma \in F^\times$ and a common stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F and χ_K . Equation (47) gives

$$\prod_{\mu \in S(K/F)} \Delta_F(\mu \chi_F, \psi_F) = \Delta_F(\chi_F, \psi_F)^\ell,$$

because the product of all characters in the odd-order group $S(K/F)$ is trivial. By (13), it remains to prove $\Delta_K(\chi_K, \psi_K) = \Delta_F(\chi_F, \psi_F)^\ell$. Apply (90) to these choices of γ and β . By (30), $m_K(\chi_K) = \ell(m-1) + 1$, which has the same parity as m . If m is even, both residual factors are 1 by (45). If m is odd, (57) gives $\Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)^\ell$. In either case, (46) proves the required identity. $\square$

Proposition 7.4 (Tamely ramified quadratic case). *The First Main Lemma holds for a tamely ramified quadratic extension.*

Proof. Put $m = m_F(\chi_F)$, $n = n_F(\psi_F)$, and $k = k_F = k_K$, and write $S(K/F) = \{1, \mu\}$. For $m = 0$, the conductor-zero calculation in the proof of Proposition 7.3 applies with $\ell = 2$. For $m = 1$, Proposition 7.2 applies.

Assume $m = 2d + \varepsilon \geq 2$, with $\varepsilon \in \{0, 1\}$. The residue characteristic is odd. Let σ be the nontrivial element of $\text{Gal}(K/F)$. Choose a uniformizer ϖ_K with $\sigma(\varpi_K) = -\varpi_K$ and put $\varpi_F = N_{K/F}(\varpi_K) = -\varpi_K^2$. Then $\mu(\varpi_F) = 1$, and $\mu|_{\mathcal{O}_F^\times}$ is induced by the quadratic character ν of $k^\times$. Put

$$\gamma = \varpi_F^{m+n}.$$

This element is admissible for χ_F . Proposition 6.8(b) gives its admissibility for χ_K and a common stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F and χ_K . By (30), $m_K(\chi_K) = 2m - 1$. Since $m_F(\mu) = 1 \leq d$, Lemma 4.7 applies to $\theta = \chi_F$ and twisting character μ . The equalities $\mu(\gamma) = 1$ and $\mu(\beta)^{-1} = \nu(\bar{\beta})$ reduce (47) to

$$(101) \quad \Delta_F(\mu\chi_F) = \nu(\bar{\beta})\Delta_F(\chi_F).$$

Equations (90), with $\ell = 2$, (101), (46), and (11) reduce (1) to

$$(102) \quad \Delta_{3,K}(\chi_K)\Delta_F(\mu) = \nu(\bar{\beta})\Delta_{3,F}(\chi_F)^2.$$

For $x \in k$, choose a lift $\tilde{x} \in \mathcal{O}_F$ and define

$$\psi_0(x) = \psi_F(\tilde{x}/\varpi_F^{n+1}).$$

Put $Q = Q_{\psi_0}(1, 0)$. Equations (10) and (56), with $\mu(\varpi_F) = 1$, give

$$(103) \quad \Delta_F(\mu) = \nu(2)Q.$$

By (55), $Q^2 = \nu(-1)$.

If $m = 2d$, then $m_K(\chi_K) = 4d - 1$. Taking $\delta_K = \varpi_K^{2d-1}$, let $x \in k$ and choose a lift $\tilde{x} \in \mathcal{O}_F$. Since $\sigma(\varpi_K) = -\varpi_K$ and $2d - 1$ is odd,

$$\text{Tr}_{K/F}(\delta_K \tilde{x}) = 0,$$

so the additive factor in $H_K(x)$ is 1. Moreover

$$N_{K/F}(1 + \delta_K \tilde{x}) = 1 + \varpi_F^{2d-1} \tilde{x}^2.$$

Hence, using $\chi_K = \chi_F \circ N_{K/F}$ and the stationary linearization of χ_F ,

$$\begin{aligned} H_K(x) &= \chi_F \left(1 + \varpi_F^{2d-1} \tilde{x}^2 \right)^{-1} \\ &= \psi_F \left(-\frac{\beta \varpi_F^{2d-1} \tilde{x}^2}{\gamma} \right) \\ &= \psi_F \left(-\frac{\beta \tilde{x}^2}{\varpi_F^{n+1}} \right) = \psi_0(-\bar{\beta} x^2), \end{aligned}$$

because $\gamma = \varpi_F^{2d+n}$. Equation (55) gives

$$\Delta_{3,K} = Q_{\psi_0}(-2\bar{\beta}, 0) = \nu(-2\bar{\beta})Q.$$

Using (103) and $Q^2 = \nu(-1)$,

$$\Delta_{3,K}\Delta_F(\mu) = \nu(-2\bar{\beta})\nu(2)\nu(-1) = \nu(\bar{\beta}).$$

Since $\Delta_{3,F}(\chi_F) = 1$ by (45), this proves (102).

If $m = 2d + 1$, take $\delta_F = \varpi_F^d$. Then (43) gives

$$\frac{H_F(x+y)}{H_F(x)H_F(y)} = \psi_0(\bar{\beta}xy) \quad (x, y \in k),$$

because $\beta\delta_F^2/\gamma = \beta/\varpi_F^{n+1}$. The function $x \mapsto H_F(x)\psi_0(-\bar{\beta}x^2/2)$ is therefore an additive character of k . Since ψ_0 is nontrivial, there is a unique $c \in k$ such that

$$H_F(x) = \psi_0(\bar{\beta}x^2/2 + cx).$$

Here $m_K(\chi_K) = 4d + 1$. With $\delta_K = \varpi_K^{2d} = (-1)^d \varpi_F^d$, substitution gives

$$H_K(x) = H_F((-1)^d x)^2 = \psi_0(\bar{\beta}x^2 + 2(-1)^d cx).$$

Equation (55) gives

$$(104) \quad \begin{aligned} \Delta_{3,K} &= \nu(2\bar{\beta})\psi_0(-c^2/\bar{\beta})Q, \\ \Delta_{3,F}^2 &= \psi_0(-c^2/\bar{\beta})\nu(-1). \end{aligned}$$

Equations (103) and (104), with $Q^2 = \nu(-1)$, prove (102). $\square$

8. WILD ODD DEGREE: STABLE RANGE

Proposition 8.1. *Let K/F be wildly ramified of odd prime degree p , with break t . If $m_F(\chi_F) = 2d + \varepsilon > 1$ and $d \geq t + 1$, then the First Main Lemma holds.*

Proof. Every nontrivial norm character has conductor $t + 1 \leq d$. The stable twist formula (47) and (13) reduce the theorem to

$$\Delta_K(\chi_K, \psi_K) = \Delta_F(\chi_F, \psi_F)^p.$$

Proposition 6.8(c) supplies a common admissible denominator and stationary representative, so (90) gives the p th-power identity for the elementary factors. Here $m' + n_K(\psi_K) = p(m + n_F(\psi_F))$ by (30) and (29), and $m' \equiv m \pmod{2}$ by (30). For even m , both residual factors are 1 by (45); for odd m , apply (57). $\square$

9. THE NON-STABLE WILD RANGE

9.1. The endpoint conductors in wild prime degree. Let K/F be wildly ramified cyclic of prime degree p , let t be its break, and put $n = n_F(\psi_F)$, and choose uniformizers with $\varpi_F = N_{K/F}(\varpi_K)$. Then by (29),

$$(105) \quad n_K(\psi_K) = pn + (p-1)(t+1).$$

Conductor 0. If χ_F is unramified, then

$$\Delta_K(\chi_K, \psi_K) = \chi_F(\varpi_F)^{n_K(\psi_K)}.$$

For $j \in \mathbf{F}_p$, let τ^j run through $S(K/F)$, with $\tau^0 = 1$. The character τ^j has conductor $t + 1$ for $j \neq 0$, whereas τ^0 has conductor 0. Since an unramified character is constant on the unit group, substitution in (3) gives

$$\Delta_F(\chi_F \tau^j, \psi_F) = \chi_F(\varpi_F)^{m_F(\tau^j) + n} \Delta_F(\tau^j, \psi_F).$$

Multiplication over j gives the exponent

$$n + (p-1)(t+1+n) = pn + (p-1)(t+1) = n_K(\psi_K),$$

so the First Main Lemma follows.

Conductor 1. Set $k = k_F = k_K$ and $\gamma_F = \varpi_F^{n+1}$. By Corollary 2.9 and Lemma 2.7, we may scale ψ_F by a unit so that the residual character (6) is

$$\psi_0(\bar{x}) = \psi_F(x/\gamma_F) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{k/\mathbf{F}_p}(\bar{x})\right),$$

where $x \in \mathcal{O}_F$ is any lift of $\bar{x} \in k$. Frobenius invariance of $\operatorname{Tr}_{k/\mathbf{F}_p}$ gives $\psi_0(\bar{x}^p) = \psi_0(\bar{x})$.

Let $f(T)$ be the minimal polynomial of ϖ_K and put

$$(106) \quad \gamma_K = \gamma_F \frac{f'(\varpi_K)}{\varpi_K^{p-1}}.$$

The different formula $v_K(f'(\varpi_K)) = (p-1)(t+1)$ and (105) give $v_K(\gamma_K) = 1 + n_K(\psi_K)$; hence γ_K is admissible for (χ_K, ψ_K) . For $0 \leq i \leq p-1$, partial fractions for $T^i/f(T)$ at infinity give

$$\mathrm{Tr}_{K/F} \left(\frac{\varpi_K^i}{f'(\varpi_K)} \right) = \begin{cases} 0, & i < p-1, \\ 1, & i = p-1. \end{cases}$$

For $x \in \mathcal{O}_K$, choose $a \in \mathcal{O}_F$ with $\bar{a} = \bar{x} \in k$. Then $x - a \in \mathfrak{p}_K$, and the trace-ideal formula (17) shows that

$$\mathrm{Tr}_{K/F} \left(\frac{(x-a)\varpi_K^{p-1}}{f'(\varpi_K)} \right) \in \mathfrak{p}_F.$$

The identity for $i = p-1$ therefore gives

$$\mathrm{Tr}_{K/F} \left(\frac{x\varpi_K^{p-1}}{f'(\varpi_K)} \right) \equiv a \pmod{\mathfrak{p}_F}.$$

Since $v_F(\gamma_F) = n+1$ and ψ_F is trivial on $\mathfrak{p}_F^{-n}$,

$$\psi_K(x/\gamma_K) = \psi_F(a/\gamma_F) = \psi_0(\bar{x}).$$

Thus the residual additive character determined by γ_K is ψ_0 . If $\bar{\chi}$ denotes the residue-field character of χ_F , then that of χ_K is $x \mapsto \bar{\chi}(x^p)$. Since $\psi_0(x^p) = \psi_0(x)$ and $x \mapsto x^p$ permutes $k^\times$,

$$\sum_{x \in k^\times} \psi_0(x) \bar{\chi}(x^p)^{-1} = \sum_{x \in k^\times} \psi_0(x) \bar{\chi}(x)^{-1}.$$

Then by (5),

$$(107) \quad \frac{\Delta_K(\chi_K, \psi_K)}{\Delta_F(\chi_F, \psi_F)} = \chi_F \left(\frac{N_{K/F}(\gamma_K)}{\gamma_F} \right).$$

It remains to identify the residue class of the element on the right. Let σ generate $\mathrm{Gal}(K/F)$ and let $\lambda \in k^\times$ be defined by (21). The ramification map sends σ^j to $j\lambda$. Equations (22) and (23) give, in these source and target coordinates,

$$(108) \quad P(Z) = Z^p - \lambda^{p-1}Z, \quad \mathrm{im} P = \lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p}).$$

Choose an admissible γ_τ for τ and a representative b_τ of its stationary numerator class such that

$$\tau(1+x) = \psi_F(b_\tau x / \gamma_\tau)$$

on the Lamprecht layer, and put $c_\tau = \gamma_\tau / b_\tau$.

Lemma 9.1 (Endpoint annihilator and discriminant congruence). *With the notation of the conductor-1 endpoint calculation above,*

$$\left(\frac{c_\tau}{\varpi_F^t \gamma_F} \right)^{p-1} = \lambda^{p(p-1)}.$$

Moreover, for $D = f'(\varpi_K) / \varpi_K^{(p-1)(t+1)}$, one has

$$\bar{D} = (-1)^p \lambda^{p-1}, \quad \overline{N_{K/F}(D)} = (-1)^p \lambda^{p(p-1)}.$$

Proof. By (108), the critical norm polynomial is $P(Z) = Z^p - \lambda^{p-1}Z$. Its image is $\lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p})$, and the annihilator of this image under the trace pairing is $\lambda^{-p} \mathbf{F}_p$. On the critical unit quotient, let $z \in k$, choose a lift $\tilde{z} \in \mathcal{O}_F$, and put $x = \varpi_F^t \tilde{z}$. Since $\tau(1+x) = \psi_F(x/c_\tau)$ and, for a lift $y \in \mathcal{O}_F$ of $\bar{y} \in k$, $\psi_0(\bar{y}) = \psi_F(y/\gamma_F)$, the residual coefficient of τ is

$$a_\tau = \frac{\overline{\varpi_F^t \gamma_F}}{c_\tau}.$$

The character τ is trivial on the norm image. Hence $\psi_0(a_\tau P(z)) = 1$ for every $z \in k$, so $a_\tau \in \lambda^{-p}\mathbf{F}_p$. It is nonzero because τ has conductor $t+1$; consequently $a_\tau = \lambda^{-p}u$ for some $u \in \mathbf{F}_p^\times$. Taking inverses and then $(p-1)$ st powers gives

$$\left(\frac{c_\tau}{\varpi_F^t \gamma_F}\right)^{p-1} = (\lambda^p u^{-1})^{p-1} = \lambda^{p(p-1)},$$

which proves the first assertion.

For the second assertion, write

$$f'(\varpi_K) = \prod_{j=1}^{p-1} (\varpi_K - \sigma^j \varpi_K).$$

After division by $\varpi_K^{(p-1)(t+1)}$, the residues of the factors are the nonzero elements $-j\lambda$ of the ramification line. Their product is $(-1)^p \lambda^{p-1}$. Taking the norm to F raises this residue to the p th power and gives the last identity. $\square$

Put

$$D = \frac{f'(\varpi_K)}{\varpi_K^{(p-1)(t+1)}} \in \mathcal{O}_K^\times.$$

Using (106) and $N_{K/F}(\varpi_K) = \varpi_F$, we obtain

$$\frac{N_{K/F}(\gamma_K)}{\gamma_F} = (\gamma_F \varpi_F^t)^{p-1} N_{K/F}(D).$$

Lemma 9.1 gives the congruence

$$(109) \quad \frac{N_{K/F}(\gamma_K)}{\gamma_F} \equiv (-1)^p c_\tau^{p-1} \pmod{U_F^1}.$$

Finally, Lemma 4.3(c) shows that $j b_\tau$ represents the stationary numerator class of τ^j on the same layer. Its conductor remains $t+1$ because τ^j is a nontrivial norm character. Choose this representative. The stable-twist formula (47), applied with $\theta = \tau^j$ and $\nu = \chi_F$, gives

$$\Delta_F(\tau^j \chi_F, \psi_F) = \chi_F\left(\frac{c_\tau}{j}\right) \Delta_F(\tau^j, \psi_F).$$

The product of the Teichmüller representatives of $\mathbf{F}_p^\times$ is -1 for odd p and is 1 for $p=2$; hence

$$\prod_{j \in \mathbf{F}_p^\times} \frac{c_\tau}{j} = (-1)^p c_\tau^{p-1}.$$

Combining (109) and (107), we obtain (1), the First Main Lemma in the conductor-one case.

9.2. Odd residue characteristic. Retain the minimal-conductor choice of χ_F .

Let k have odd characteristic. Recall from (7) that

$$\psi_{\text{can}}(z) = \exp\left(\frac{2\pi i}{p} \text{Tr}_{k/\mathbf{F}_p}(z)\right).$$

By Corollary 2.9 and Lemma 2.7, normalize the residual additive character once and for all by $\psi_0 = \psi_{\text{can}}$. In particular $\psi_0(z^p) = \psi_0(z)$. For the quadratic phase of Lemma 5.5, its magnitude assertion gives, when $A \neq 0$ and $B \in k$,

$$(110) \quad Q_{\psi_0}(A, B) = |k|^{-1/2} \sum_{x \in k} \psi_0(Ax^2/2 + Bx).$$

Equation (55) gives

$$(111) \quad Q_{\psi_0}(A, B) = q_0 \nu(A) \psi_0\left(-\frac{B^2}{2A}\right),$$

where ν is the quadratic character and $q_0 = |k|^{-1/2} \sum_x \psi_0(x^2/2)$.

9.2.1. Critical-coordinate transport and residual phase cancellation.

Lemma 9.2 (Powers of the basic quadratic phase). *One has*

$$(112) \quad q_0^2 = \nu(-1), \quad q_0^{p-1} = \nu(-1), \quad q_0^p = q_0 \nu(-1).$$

Proof. The first identity is (55). In particular $q_0^4 = 1$. If $p \equiv 1 \pmod{4}$, then -1 is a square in $\mathbf{F}_p \subset k$, so $\nu(-1) = 1$, and $p-1$ is divisible by 4; hence $q_0^{p-1} = 1 = \nu(-1)$. If $p \equiv 3 \pmod{4}$, then $p-1 \equiv 2 \pmod{4}$, and therefore $q_0^{p-1} = q_0^2 = \nu(-1)$. Multiplication by q_0 gives the last identity. $\square$

Let E be a nonarchimedean local field with residue field k , let ψ_E be a nontrivial additive character, and let θ have conductor $m_E(\theta) > 1$. Choose an admissible denominator γ and a stationary representative β . Put $d = \lfloor m_E(\theta)/2 \rfloor$, choose δ with $v_E(\delta) = d$, and write H_θ for (41) with $\psi = \psi_E$. If $m_E(\theta) = 2d+1$, define $A_\theta \in k^\times$ by

$$\psi_E(\beta \delta^2 \tilde{z}/\gamma) = \psi_0(A_\theta z) \quad (z \in k),$$

where $\tilde{z} \in \mathcal{O}_E$ lifts z . Since $\psi_0 = \psi_{\text{can}}$, existence and uniqueness follow from (44) and the nondegeneracy of the trace pairing $k \times k \rightarrow \mathbf{F}_p$. Equation (43) makes $x \mapsto H_\theta(x) \psi_0(-A_\theta x^2/2)$ an additive character. Define $B_\theta \in k$ by

$$(113) \quad H_\theta(x) = \psi_0(A_\theta x^2/2 + B_\theta x) \quad (x \in k).$$

If $m_E(\theta) = 2d$, equation (39) gives

$$(114) \quad H_\theta(x) = 1 \quad (x \in k).$$

Thus (45) and (110) give

$$(115) \quad \Delta_{3,E}(\theta) = \begin{cases} Q_{\psi_0}(A_\theta, B_\theta), & m_E(\theta) \text{ odd}, \\ 1, & m_E(\theta) \text{ even}. \end{cases}$$

Let K/F be wildly ramified cyclic of odd prime degree p , with lower break t , and put $T = t+1$ and $k = k_F = k_K$. Choose a uniformizer ϖ_K , put $\varpi_F = N_{K/F}(\varpi_K)$, choose a generator σ of $\text{Gal}(K/F)$, and recall from (21) that

$$\lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

A tilde denotes a lift from k to $\mathcal{O}_F$. Proposition 3.3 supplies the norm polynomial (22) and trace coefficient (24) in these coordinates. For $w \in \mathcal{O}_K$ this trace coefficient gives

$$(116) \quad \text{Tr}_{K/F}(w \varpi_K^t \tilde{z}) \equiv \varpi_F^t (-\widetilde{\lambda^{p-1} \bar{w} z}) \pmod{\mathfrak{p}_F^{t+1}} \quad (z \in k),$$

because $w - \tilde{w} \in \mathfrak{p}_K$ and $\text{Tr}_{K/F}(\mathfrak{p}_K^{t+1}) = \mathfrak{p}_F^{t+1}$ by (17).

Lemma 9.3 (Normalization of the norm-character generator). *A generator τ of $S(K/F)$ can be chosen so that*

$$(117) \quad \tau(1 + \varpi_F^t \tilde{z}) = \psi_0(\lambda^{-p} z) \quad (z \in k).$$

Put

$$(118) \quad \eta_0 = \lambda^{-p}.$$

If T is odd, the Hasse function of τ with critical element $\delta_\tau = \varpi_F^{(T-1)/2}$ has quadratic coefficient η_0 .

Proof. By Corollary 3.8, a generator τ has conductor $T = t+1$. Hence its restriction to $U_F^t/U_F^{t+1} \simeq (k, +)$ is a nontrivial additive character. Since $\psi_0 = \psi_{\text{can}}$, the nondegeneracy of the trace pairing gives a unique $a \in k^\times$ such that

$$\tau(1 + \varpi_F^t \tilde{z}) = \psi_0(az).$$

By Proposition 3.3, the annihilator of the image in (23) is $\lambda^{-p} \mathbf{F}_p$. Since τ is trivial on norms, $a = j \lambda^{-p}$ for some $j \in \mathbf{F}_p^\times$. Replacing τ by $\tau^{j^{-1}}$ gives (117).

Assume T is odd. For the stationary representative β_τ and admissible denominator γ_τ used to form H_τ , the equality $\delta_\tau^2 = \varpi_F^t$ gives

$$\frac{H_\tau(x+y)}{H_\tau(x)H_\tau(y)} = \psi_F\left(\frac{\beta_\tau \delta_\tau^2 \widetilde{xy}}{\gamma_\tau}\right) = \tau(1 + \varpi_F^t \widetilde{xy}) = \psi_0(\eta_0 xy),$$

by (43), (39), and (117). Thus the quadratic coefficient is η_0 . $\square$

Lemma 9.4 (Changes of critical coordinate). *Let H_θ have coefficient pair (A_θ, B_θ) in (113). If a critical element δ is replaced by $\widetilde{a}\delta$, where $a \in k^\times$ and $\widetilde{a}$ lifts a , the pair becomes $(a^2 A_\theta, a B_\theta)$. Precomposing H_θ with $x \mapsto x^p$ replaces the pair by $(A_\theta^{1/p}, B_\theta^{1/p})$; precomposing with $x \mapsto x^{1/p}$ replaces it by (A_θ^p, B_θ^p) . Each substitution preserves $\sum_{x \in k} H_\theta(x)$.*

Proof. Equation (41) replaces $H_\theta(x)$ by $H_\theta(ax)$ when δ is replaced by $\widetilde{a}\delta$. The other two rules follow from

$$\psi_0(Ax^{2p}/2 + Bx^p) = \psi_0(A^{1/p}x^2/2 + B^{1/p}x),$$

since $\psi_0(z^p) = \psi_0(z)$. Multiplication by a and Frobenius are bijections of k . $\square$

Lemma 9.5 (Symmetric terms of degree at least three at the χ_K -critical depth). *Assume that K/F is wildly ramified cyclic of odd prime degree p , with break t , and put $T = t + 1$. Suppose $m = m_F(\chi_F) = 2d + 1 > 1$, set $a = T - m$, and suppose that $m_K(\chi_K)$ is odd, say $m_K(\chi_K) = 2d_K + 1$, with*

$$2d_K = \begin{cases} m - 1, & m < T, \\ t, & m = T, \\ t + p(m - T), & m > T. \end{cases}$$

Let $u \in K^\times$ satisfy $v_K(u) = a$, put $\mathcal{N} = N_{K/F}(u)$, and let $x \in \mathfrak{p}_K^{d_K}$. Then, for every $3 \leq j \leq p-1$,

$$(119) \quad s_j(ux) \in \mathfrak{p}_F^T, \quad \mathcal{N} s_j(x) \in \mathfrak{p}_F^T.$$

Moreover

$$(120) \quad \sum_{j=2}^p (s_j(ux) - \mathcal{N} s_j(x)) \equiv s_2(ux) - \mathcal{N} s_2(x) \pmod{\mathfrak{p}_F^T}.$$

Proof. Write $N = N_{K/F}$. Since $v_F(\mathcal{N}) = v_K(u) = a$ by (26), Equation (25) gives

$$\begin{aligned} v_F(s_j(ux)) &\geq \left\lfloor \frac{j(d_K + a) + (p-1)T}{p} \right\rfloor, \\ v_F(\mathcal{N} s_j(x)) &\geq a + \left\lfloor \frac{j d_K + (p-1)T}{p} \right\rfloor. \end{aligned}$$

It is therefore enough to prove

$$j(d_K + a) \geq T, \quad j d_K + pa \geq T.$$

If $p = 3$, the range $3 \leq j \leq p-1$ is empty and $s_3(ux) = N(u)N(x) = \mathcal{N} s_3(x)$, so both assertions of the lemma follow immediately. We may therefore suppose $p \geq 5$. The formulas below show in particular that $d_K + a > 0$; hence both left sides increase with j , and it suffices to take $j = 3$. For $m < T$, one has $a = T - m > 0$ and $d_K = (t - a)/2$; for $m > T$, put $q = m - T = -a > 0$, so $d_K = (t + pq)/2$. Substitution gives

	$\frac{3(d_K + a) - T}{t + 3a - 2}$	$\frac{3d_K + pa - T}{t + (2p-3)a - 2}$
$m < T$	$\frac{2}{t - 2}$	$\frac{2}{t - 2}$
$m = T$	$\frac{2}{t + 3(p-2)q - 2}$	$\frac{2}{t + pq - 2}$
$m > T$	$\frac{2}{2}$	$\frac{2}{2}$

Every entry is nonnegative. When $m = T$, one has $t = 2d \geq 2$; when $m < T$, use $t \geq 1$ and $a \geq 1$; when $m > T$, use $t \geq 1$, $q \geq 1$, and $p \geq 3$. This proves the two required inequalities and hence (119).

Finally,

$$s_p(ux) = N(ux) = N(u)N(x) = \mathcal{N} s_p(x),$$

so the degree- p terms cancel. Removing them and the terms of degrees $3, \dots, p-1$ proves (120). The remaining term is homogeneous of degree two in x . $\square$

Lemma 9.6 (Odd-prime critical-coordinate transport). *Retain the normalized generator τ and uniformizers of Lemma 9.3. Put $m = m_F(\chi_F)$ and assume $2 \leq m < 2T$. Use the notation of Proposition 6.10 when $m \leq T$ and Proposition 6.8(d) when $m > T$, and put*

$$\mathcal{N} = \begin{cases} \xi\beta/\alpha, & m \leq T, \\ \beta/\alpha, & m > T, \end{cases} \quad d_0 = \overline{\varpi_F^{m-T}\mathcal{N}} \in k^\times.$$

When m is odd, let (A_{χ_F}, B_{χ_F}) be the coefficient pair of H_{χ_F} for the critical element $\delta_\chi = \varpi_F^{(m-1)/2}$. Then $A_{\chi_F} = \eta_0 d_0$.

The conductor $m_K(\chi_K)$ has the same parity as m . When m is odd, write $m_K(\chi_K) = 2d_K + 1$, take $\delta_K = \varpi_K^{d_K}$, and define

$$(121) \quad H_{\chi_K}(X) = \psi_K \left(\frac{\beta_K \delta_K \widetilde{X^{1/p}}}{\gamma_K} \right) \chi_K (1 + \delta_K \widetilde{X^{1/p}})^{-1}, \quad X \in k.$$

Let (A_K, B_K) be the coefficient pair of this function in (113). Then

$$(122) \quad (A_K, B_K) = \begin{cases} (A_{\chi_F}, B_{\chi_F}), & m < T, \\ (-\eta_0(d_0^p - d_0), B_{\chi_F} - d_0 B_\tau), & m = T, \\ (-d_0^{p-1} A_{\chi_F}, 0), & m > T. \end{cases}$$

Proof. Write $N = N_{K/F}$. For $z \in k$, write $\tilde{z} \in \mathcal{O}_F$ for a lift, also viewed in $\mathcal{O}_K$. Put

$$u = \begin{cases} \xi_1 \beta_1 / \alpha_1, & m \leq T, \\ \beta_1 / \alpha_1, & m > T, \end{cases} \quad a = T - m,$$

and then

$$u_0 = \varpi_K^{-a} u, \quad \mathcal{N}_0 = \varpi_F^{-a} \mathcal{N}, \quad \zeta = \bar{u}_0.$$

Propositions 6.10 and 6.8(d), in the respective ranges, give $N(u) = \mathcal{N}$ and $v_K(u) = v_F(\mathcal{N}) = a$. Hence $u_0, \mathcal{N}_0$ are units and $d_0 = \overline{\mathcal{N}_0}$. Since the extension is totally ramified, the residue norm is Frobenius; hence

$$(123) \quad d_0 = \overline{\mathcal{N}_0} = \bar{u}_0^p = \zeta^p.$$

There is no additional residue scalar when $\mathcal{N}$ is viewed as an element of K . Indeed, if σ generates $\text{Gal}(K/F)$, then

$$\frac{\varpi_F}{\varpi_K^p} = \prod_{i=0}^{p-1} \frac{\sigma^i(\varpi_K)}{\varpi_K} \equiv 1 \pmod{\mathfrak{p}_K},$$

because the extension is wildly ramified. Hence, after division by ϖ_K^{pa} , the initial residue of $\mathcal{N} = \varpi_F^a \mathcal{N}_0$ is d_0 for every integer a , including $a < 0$.

Put

$$(124) \quad \mathcal{A} = \begin{cases} \alpha/\Gamma_\tau, & m \leq T, \\ \alpha/\gamma_F, & m > T. \end{cases}$$

Here Γ_τ is the denominator for τ in (84). For the representatives in Propositions 6.8 and 6.10, the stationary ratios are

$$(125) \quad \frac{\beta(\tau)}{\gamma_\tau} = \mathcal{A}, \quad \frac{\beta(\chi_F)}{\gamma_\chi} = \mathcal{AN}, \quad \frac{\beta_K}{\gamma_K} = \mathcal{A}(\mathcal{N} - u).$$

The normalization of τ in Lemma 9.3 gives

$$\psi_F(\mathcal{A}\varpi_F^t \tilde{z}) = \psi_0(\eta_0 z) \quad (z \in k).$$

Suppose first that $m_F(\chi_F) = 2d + 1$. Take $\delta_\chi = \varpi_F^d$. Since $a + 2d = t$, the polar quotient of the χ_F critical function is, by (43) and (125),

$$\begin{aligned} (X, Y) &\mapsto \psi_F(\mathcal{AN}\delta_\chi^2 \tilde{X}\tilde{Y}) \\ &= \psi_F(\mathcal{A}\varpi_F^t \mathcal{N}_0 \tilde{X}\tilde{Y}) = \psi_0(\eta_0 \overline{\mathcal{N}_0} XY). \end{aligned}$$

Together with (123), this gives $A_{\chi_F} = \eta_0 d_0$.

To compute A_K , use the trace coefficient (116). The parity of $m_K(\chi_K)$ is the parity of $m_F(\chi_F)$: in the low range the conductors are equal, while in the high range

$$m_K(\chi_K) - 1 = t + p(m - T) \equiv m - 1 \pmod{2}.$$

Assume this parity is odd and write $m_K(\chi_K) = 2d_K + 1$. Put $r = v_K(\mathcal{N} - u)$. The exact conductor of χ_K implies $r = 0$ at the boundary; away from the boundary the two terms have distinct valuations. Hence

	r	$2d_K$
$m < T$	$T - m$	$m - 1$
$m = T$	0	t
$m > T$	$p(T - m)$	$t + p(m - T)$.

In every row,

$$r + 2d_K = t.$$

Take $\delta_K = \varpi_K^{d_K}$ and put $w_0 = \varpi_K^{-r}(\mathcal{N} - u) \in \mathcal{O}_K^\times$. In the natural residue coordinate Z for χ_K , define

$$H_K^{\text{nat}}(Z) = \psi_K\left(\frac{\beta_K \delta_K \tilde{Z}}{\gamma_K}\right) \chi_K(1 + \delta_K \tilde{Z})^{-1}.$$

For $Z, W \in k$, Lemma 9.4 and (125) give

$$\frac{H_K^{\text{nat}}(Z + W)}{H_K^{\text{nat}}(Z)H_K^{\text{nat}}(W)} = \psi_F\left(\mathcal{A} \text{Tr}_{K/F}((\mathcal{N} - u)\delta_K^2 \tilde{Z}\tilde{W})\right).$$

Since $r + 2d_K = t$, the argument of the trace is $w_0 \varpi_K^t \tilde{Z}\tilde{W}$. Formula (116) and the fact that $x \mapsto \psi_F(\mathcal{A}x)$ is trivial on $\mathfrak{p}_F^{t+1}$ therefore give

$$\begin{aligned} \frac{H_K^{\text{nat}}(Z + W)}{H_K^{\text{nat}}(Z)H_K^{\text{nat}}(W)} &= \psi_0\left(-\eta_0 \lambda^{p-1} \bar{w}_0 ZW\right) \\ &= \psi_0\left(\lambda^{-1} \text{in}(u - \mathcal{N})ZW\right), \end{aligned}$$

where $\text{in}(y) = \overline{\varpi_K^{-v_K(y)}} y$. The last equality uses $\eta_0 = \lambda^{-p}$ and $\text{in}(u - \mathcal{N}) = -\bar{w}_0$.

By (121), $H_{\chi_K}(X) = H_K^{\text{nat}}(X^{1/p})$. Since Frobenius permutes k ,

$$\sum_{X \in k} H_{\chi_K}(X) = \sum_{Z \in k} H_K^{\text{nat}}(Z).$$

By Lemma 9.4, applied to the expression (113), the quadratic coefficient in this coordinate is the p th power of the coefficient in the displayed polar-quotient calculation. Thus

$$(126) \quad \frac{A_K}{\eta_0} = \text{in}(u - \mathcal{N})^p.$$

The initial term of $u - \mathcal{N}$ is determined by the two valuations. If $m < T$, then $a > 0$ and $v_K(u) = a < pa = v_K(\mathcal{N})$, so it is ζ . If $m = T$, both terms are units and it is $\zeta - d_0$. If $m > T$, then $a < 0$ and $v_K(\mathcal{N}) = pa < a = v_K(u)$, so it is $-d_0$. Consequently

$$A_K = \eta_0 \begin{cases} \zeta^p, & m < T, \\ (\zeta - d_0)^p, & m = T, \\ (-d_0)^p, & m > T, \end{cases} = \begin{cases} A_{\chi_F}, & m < T, \\ -\eta_0(d_0^p - d_0), & m = T, \\ -d_0^{p-1}A_{\chi_F}, & m > T, \end{cases}$$

using (123), $A_{\chi_F} = \eta_0 d_0$, and the fact that p is odd.

It remains to determine the affine coefficient. We do this from an identity of critical functions, retaining the stationary ratios (125).

Let

$$P(y) = N(1 + y) - 1, \quad H(y) = P(y) - \text{Tr}(y) = \sum_{j=2}^p s_j(y).$$

For $X \in k$, let $Z = X^{1/p}$, choose a lift $\tilde{Z}$, and put $x = \delta_K \tilde{Z}$. Thus X is the displayed coordinate for χ_K in (121). Equation (65) gives $P(x) \in \mathfrak{p}_F^{[m_F(\chi_F)/2]}$. In all three ranges $a + d_K \geq \lfloor T/2 \rfloor$; since $\lfloor T/2 \rfloor \leq t$, Theorem 3.1(b) gives $P(ux) \in \mathfrak{p}_F^{[T/2]}$. For each odd conductor, use the fixed critical elements $\delta_\chi = \varpi_F^{(m_F(\chi_F)-1)/2}$ and $\delta_\tau = \varpi_F^{(T-1)/2}$, respectively, and define

$$r_\chi(X) = \overline{P(x)/\delta_\chi}, \quad r_\tau(X) = \overline{P(ux)/\delta_\tau}.$$

If m is even, set $r_\chi = 0$; if T is even, set $r_\tau = 0$. Then

$$(127) \quad H_{\chi_K}(X) = H_{\chi_F}(r_\chi(X)) H_\tau(r_\tau(X))^{-1} \psi_F(\mathcal{A}\{H(ux) - \mathcal{N}H(x)\}).$$

To verify this identity, first note that $1 + P(ux) = N(1 + ux)$ and hence $\tau(1 + P(ux)) = 1$. Therefore the right-hand side of (127) is

$$\begin{aligned} & \psi_F(\mathcal{A}\mathcal{N}P(x))\chi_F(1 + P(x))^{-1}\psi_F(-\mathcal{A}P(ux))\psi_F(\mathcal{A}\{H(ux) - \mathcal{N}H(x)\}) \\ &= \psi_F(\mathcal{A}\{\mathcal{N}\text{Tr}(x) - \text{Tr}(ux)\})\chi_F(N(1 + x))^{-1} \\ &= \psi_K\left(\frac{\beta_K x}{\gamma_K}\right)\chi_K(1 + x)^{-1} = H_{\chi_K}(X), \end{aligned}$$

where the second equality uses $P = \text{Tr} + H$ and the third uses (125).

We now compute the two classes in (127). Only odd conductors need be considered. If $m < T$, write $m = 2d + 1$, so $2d < t$ and $m_K(\chi_K) = m$. Here $\delta_K = \varpi_K^d$ and $\delta_\chi = \varpi_F^d$. For $x = \delta_K \widetilde{X^{1/p}}$, the norm term is

$$N(x) = \delta_\chi \tilde{X} \pmod{\mathfrak{p}_F^{d+1}},$$

whereas Equation (25) gives $v_F(s_j(x)) \geq d + 1$ for $1 \leq j < p$. Thus $r_\chi(X) = X$ in the coordinate (121). If the conductor $T = t + 1$ of τ is odd, put $d_\tau = t/2$. Since $v_K(u) = T - m$,

$$v_K(ux) = d + T - m = t - d \geq d_\tau + 1,$$

and the same symmetric-function estimate puts every term of $P(ux)$ in $\mathfrak{p}_F^{d_\tau+1}$. If T is even there is no τ critical line. In either case $r_\tau(X) = 0$.

If $m = T$ and T is odd, then $m = T = 2d + 1$ and $v_K(u) = 0$. In this case

$$\delta_K = \varpi_K^d, \quad \delta_\chi = \delta_\tau = \varpi_F^d.$$

Modulo $\mathfrak{p}_F^{d+1}$,

$$P(x) \equiv N(x) = \delta_\chi \tilde{X}, \quad P(ux) \equiv N(u)N(x) = \delta_\tau \widetilde{d_0 X}.$$

Hence $r_\chi(X) = X$ and $r_\tau(X) = d_0 X$ with no further coordinate choice.

Finally suppose $m > T$ and write again $m = 2d + 1$. The critical depth for χ_K is

$$d_K = \frac{t + p(m - T)}{2}.$$

The norm term of $P(x)$ lies beyond the χ_F critical line because

$$d_K - d = \frac{(p-1)(m-T)}{2} \geq 1.$$

For the remaining symmetric terms, the smallest lower bound occurs at $j = 1$, and

$$2\{d_K + (p-1)T - p(d+1)\} = (p-1)t - 2 \geq 0.$$

Hence Equation (25) gives $P(x) \in \mathfrak{p}_F^{d+1}$, so $r_\chi(X) = 0$. If T is odd, write $t = 2d_\tau$. Then

$$v_K(ux) = d_K + T - m = d_\tau + (p-2)(d - d_\tau) \geq d_\tau + 1,$$

and again all terms of $P(ux)$ lie beyond the τ critical line. Thus $r_\tau(X) = 0$; the assertion is automatic when T is even. We have proved

	$r_\chi(X)$	$r_\tau(X)$
$m < T$	X	0
$m = T$	X	$d_0 X$
$m > T$	0	0

The polynomial $H(ux) - \mathcal{N}H(x)$ has no terms of degree zero or one. Equations (119) and (120) show that, modulo $\mathfrak{p}_F^T$, its degree- p terms cancel and every term of degree $3, \dots, p-1$ vanishes. Since $z \mapsto \psi_F(\mathcal{A}z)$ is trivial on $\mathfrak{p}_F^T$, the correction factor in (127) is therefore homogeneous quadratic and contributes no affine character. With A_K given by (126), substitution of the calculated values of $r_\chi(X)$ and $r_\tau(X)$ into (127) gives the linear-character factor

$$L_{\chi_K}(X) = \begin{cases} \psi_0(B_{\chi_F} X), & m < T, \\ \psi_0((B_{\chi_F} - d_0 B_\tau) X), & m = T, \\ 1, & m > T. \end{cases}$$

Together with the quadratic coefficients already computed, this proves (122). $\square$

Lemma 9.7 (Integer stationary representatives). *Retain the minimal-conductor choice of χ_F and the parameter choices of Lemma 9.6. For $j \in \mathbf{F}_p^\times$, the following are admissible denominators and stationary representatives:*

	range	θ	γ_θ	β_θ
(128)	$m > T$	τ^j	γ_F/α	$[j]$
	$m > T$	$\tau^j \chi_F$	γ_F	$\beta + [j]\alpha$
	$m \leq T$	τ^j	Γ_τ	$[j]\alpha$
	$m \leq T$	$\tau^j \chi_F$	Γ_τ	$\xi\beta + [j]\alpha$

Replacing $[j]$ by its integer representative $j\mathbf{z} \in \{1, \dots, p-1\}$ leaves the stationary class unchanged. For odd conductor, it also leaves the critical function (41) unchanged, with any fixed critical element.

Proof. The $[j]$ -representatives of $\tau^j \chi_F$ are given by (78) and (87). For τ^j , apply Lemma 6.4 to (74) or (86), respectively. By Corollary 3.8 and (48), $q := m_F(\theta) \geq T$; the displayed denominators satisfy $v_F(\gamma_\theta) = q + n_F(\psi_F)$. Let h be the old numerator minus the new one. In every row, $h/\gamma_\theta = \mathcal{A}([j] - j\mathbf{z})$, where $\mathcal{A}$ is given by (124) and $v_F(\mathcal{A}) = -T - n_F(\psi_F)$. Thus (70) gives

$$v_F(h) \geq q - \lfloor T/2 \rfloor \geq \lceil q/2 \rceil.$$

The two numerators therefore represent the same stationary class. For odd q , any critical element has valuation $(q-1)/2$, so

$$v_F(h\delta_\theta/\gamma_\theta) \geq \frac{q+1}{2} + \frac{q-1}{2} - q - n_F(\psi_F) = -n_F(\psi_F).$$

By (41), the quotient of the two critical functions is $\psi_F(h\delta_\theta \tilde{z}/\gamma_\theta) = 1$ for every $z \in k$, since ψ_F is trivial on $\mathfrak{p}_F^{-n_F(\psi_F)}$. $\square$

Proposition 9.8 (Odd-prime coefficient table). *Retain the minimal-conductor choice of χ_F and the notation of Lemma 9.6. For $j \in \mathbf{F}_p$, let (A_j, B_j) be the coefficient pair of $\tau^j \chi_F$, using the integer sum representative in (128) for $j \neq 0$. Each pair is taken only when its conductor is odd. Use $\delta_\tau = \varpi_F^{(T-1)/2}$ when $m \leq T$ and $j \neq 0$, and $\delta_\chi = \varpi_F^{(m-1)/2}$ otherwise.*

$$(129) \quad (A_j, B_j) = \begin{cases} (j\eta_0, jB_\tau), & m < T, j \neq 0, \\ (A_{\chi_F}, B_{\chi_F}), & m < T, j = 0, \\ (\eta_0(j + d_0), B_{\chi_F} + jB_\tau), & m = T, \\ (A_{\chi_F}, B_{\chi_F}), & m > T. \end{cases}$$

The coefficient pair for χ_K is given by (122).

Proof. Use the sum representatives in (128). Lemma 9.7 permits replacing $[j]$ by $j\mathbf{z}$ without changing a critical function. Put

$$\Gamma = \begin{cases} \Gamma_\tau, & m \leq T, \\ \gamma_F, & m > T, \end{cases} \quad \beta_\tau = \Gamma\mathcal{A}, \quad \beta_\chi = \Gamma\mathcal{AN},$$

where $\mathcal{A}$ is defined in (124). By (125), these are the coefficients of τ and χ_F relative to Γ . The selected coefficient of $\tau^j \chi_F$ is $j\mathbf{z}\beta_\tau + \beta_\chi$. For $x \in k$, let $\tilde{x} \in \mathcal{O}_F$ be a lift.

Suppose $m < T$ and $j \neq 0$. A coefficient pair is needed only when T is odd. Put $d_\tau = (T-1)/2$ and use $\delta_\tau = \varpi_F^{d_\tau}$, as in Lemma 9.3. Since $m_F(\chi_F) \leq T-1$,

$$v_F(\delta_\tau) = \frac{T-1}{2} \geq \left\lfloor \frac{m_F(\chi_F)}{2} \right\rfloor.$$

Thus the stationary linearization of χ_F is valid at $\delta_\tau \tilde{x}$. With the sum representative just chosen,

$$\begin{aligned} \frac{H_{\tau^j \chi_F}(x)}{H_{\tau^j}(x)} &= \psi_F\left(\frac{\beta_\chi \delta_\tau \tilde{x}}{\Gamma}\right) \chi_F(1 + \delta_\tau \tilde{x})^{-1} \\ &= 1. \end{aligned}$$

Hence the two critical functions agree. Since $H_{\tau^j} = H_\tau^j$, its pair is $(j\eta_0, jB_\tau)$. The case $j = 0$ is χ_F itself and has pair (A_{χ_F}, B_{χ_F}) .

Suppose $m > T$. A coefficient pair is needed only when $m_F(\chi_F)$ is odd. Put $d_\chi = (m_F(\chi_F) - 1)/2$ and use $\delta_\chi = \varpi_F^{d_\chi}$, as in Lemma 9.6. Since $m_F(\chi_F) > T$,

$$v_F(\delta_\chi) = \frac{m_F(\chi_F) - 1}{2} \geq \left\lfloor \frac{T}{2} \right\rfloor.$$

The stationary linearization of every τ^j is therefore valid at $\delta_\chi \tilde{x}$, and for every $j \in \mathbf{F}_p$,

$$\begin{aligned} \frac{H_{\tau^j \chi_F}(x)}{H_{\chi_F}(x)} &= \psi_F\left(\frac{j\mathbf{z}\beta_\tau \delta_\chi \tilde{x}}{\Gamma}\right) \tau^j(1 + \delta_\chi \tilde{x})^{-1} \\ &= 1. \end{aligned}$$

Thus every critical function has the pair (A_{χ_F}, B_{χ_F}) .

At $m = T$, only odd T requires coefficient pairs. Both conductors are T , and $\Gamma = \Gamma_\tau$ is admissible for that conductor. Here $\beta_\tau = \alpha$ and $\beta_\chi = \xi\beta$ are units, so (49) gives $j\mathbf{z}\alpha + \xi\beta \notin \mathfrak{p}_F$. Use this sum representative and the common critical element $\delta_\tau = \delta_\chi = \varpi_F^{(T-1)/2}$. The critical function of the product is the pointwise product of the two critical functions, so the coefficient pairs add:

$$j(\eta_0, B_\tau) + (A_{\chi_F}, B_{\chi_F}).$$

This proves the three cases for (A_j, B_j) . □

Proposition 9.9 (Odd-prime finite phase cancellation). *Retain the minimal-conductor choice of χ_F . The residual factors satisfy*

$$(130) \quad \mathcal{R}_{\text{odd}} := \frac{\Delta_{3,K}(\chi_K) \prod_{j \in \mathbf{F}_p^\times} \Delta_{3,F}(\tau^j)}{\prod_{j \in \mathbf{F}_p} \Delta_{3,F}(\tau^j \chi_F)} = 1.$$

Proof. Write $R_j = \Delta_{3,F}(\tau^j)$, $T_j = \Delta_{3,F}(\tau^j \chi_F)$, and $U = \Delta_{3,K}(\chi_K)$. For odd conductor, evaluate $Q_{\psi_0}(A_\theta, B_\theta)$ by (111); for even conductor the factor is 1.

If $m < T$, (129) gives $T_j = R_j$ for $j \neq 0$, while the $j = 0$ pair and the pair for χ_K agree, so $T_0 = U$.

Suppose $m > T$. If $m_F(\chi_F)$ is even, then every T_j and U is 1. If it is odd, (129) gives the same phase $Q_\chi = Q_{\psi_0}(A_{\chi_F}, B_{\chi_F})$ for all T_j . By (111), the additive factor has p th power 1, because its value lies in the group of p th roots of unity. Hence (112) gives

$$Q_\chi^p = q_0^p \nu(A_{\chi_F})^p = q_0 \nu(-1) \nu(A_{\chi_F}) = q_0 \nu(-A_{\chi_F}) = U,$$

because d_0^{p-1} is a square and the affine coefficient of χ_K is zero by (122). Independently, if the conductor $T = t + 1$ of τ is even then all R_j are 1. If it is odd, then

$$(131) \quad \begin{aligned} \prod_{j \neq 0} R_j &= q_0^{p-1} \nu(\eta_0)^{p-1} \nu\left(\prod_{j \neq 0} j\right) \psi_0\left(-\frac{B_\tau^2}{2\eta_0} \sum_{j \neq 0} j\right) \\ &= q_0^{p-1} \nu(-1) = 1. \end{aligned}$$

Here $\sum_{j \neq 0} j = 0$, $\prod_{j \neq 0} j = -1$, and $q_0^{p-1} = \nu(-1)$ by (112). Hence $U \prod R_j = \prod T_j$ in every parity pattern above the break.

Finally suppose $m = T$. All four kinds of character have conductor T . If T is even, every residual phase is 1. Suppose T is odd. Put

$$c = A_{\chi_F} = \eta_0 d_0, \quad \rho = d_0^{-1}, \quad \sigma = \frac{B_{\chi_F}}{A_{\chi_F}}, \quad v = \frac{B_\tau}{A_{\chi_F}}.$$

Here $c \neq 0$. At $m = T$, the stationary coefficients relative to Γ_τ are α and $\xi\beta = \alpha\mathcal{N}$, with $\alpha \in \mathcal{O}_F^\times$ and $\bar{\mathcal{N}} = d_0$. Equation (49) gives $j + d_0 \neq 0$ for every $j \in \mathbf{F}_p$, hence $1 + j\rho \neq 0$. Choose $a_0, b_0 \in k$ with

$$a_0^p = 1 - \rho^{1-p} = 1 - d_0^{p-1}, \quad b_0^p = \sigma - \rho^{-1}v = \frac{B_{\chi_F}}{A_{\chi_F}} - \frac{B_\tau}{\eta_0}.$$

The boundary rows of (129) and (122), together with the pair $(j\eta_0, jB_\tau)$ for τ^j , are

$$\begin{aligned} (A_j, B_j) &= c(1 + j\rho, \sigma + jv), \quad j \in \mathbf{F}_p, \\ (A(\tau^j), B(\tau^j)) &= c(j\rho, jv), \quad j \in \mathbf{F}_p^\times, \\ (A_K, B_K) &= c(a_0^p, b_0^p). \end{aligned}$$

Therefore Proposition 5.8 reduces the quotient in (130) to

$$\frac{Q_{\psi_0}(a_0, b_0) \prod_{j \in \mathbf{F}_p^\times} Q_{\psi_0}(j\rho, jv)}{\prod_{j \in \mathbf{F}_p} Q_{\psi_0}(1 + j\rho, \sigma + jv)}.$$

This quotient is 1 by Lemma 5.7. Hence (130) also holds at $m = T$. $\square$

9.2.2. Stationary-factor assembly. Put $m = m_F(\chi_F)$ and $a = T - m$. Using the parameter choices of Propositions 6.10 and 6.8(d), put

$$(132) \quad \begin{aligned} (A, u, \mathcal{N}) &= \begin{cases} (\alpha/\Gamma_\tau, \xi_1\beta_1/\alpha_1, \xi\beta/\alpha), & m \leq T, \\ (\alpha/\gamma_F, \beta_1/\alpha_1, \beta/\alpha), & m > T, \end{cases} \\ \mathcal{N} &= N_{K/F}(u), \quad v_K(u) = v_F(\mathcal{N}) = a. \end{aligned}$$

Represent $\mathbf{F}_p^\times$ by its Teichmüller lifts. Define

$$(133) \quad z_0 = \frac{N_{K/F}(1 - \mathcal{N}/u)}{1 - \mathcal{N}^{p-1}}, \quad z_j = \frac{N_{K/F}(u + j)}{\mathcal{N} + j} \quad (j \neq 0).$$

If $a \neq 0$, the denominators are nonzero by valuation. If $a = 0$, then $m_F(\chi_F) = T$ and $\xi\beta = \alpha\mathcal{N}$, with $\alpha \in \mathcal{O}_F^\times$. Applying (49) to the coefficients α and $\xi\beta$ gives

$$(134) \quad \overline{\mathcal{N}} + j \neq 0 \ (j \in \mathbf{F}_p), \quad \mathcal{N} + j \in \mathcal{O}_F^\times, \quad 1 - \mathcal{N}^{p-1} \in \mathcal{O}_F^\times \quad (a = 0).$$

Lemma 9.10 (Parameter-ratio identities). *One has*

$$(135) \quad N_{K/F}(u + j) = (\mathcal{N} + j)z_j, \quad N_{K/F}(\mathcal{N} - u) = \left(\prod_{j \in \mathbf{F}_p} (\mathcal{N} + j) \right) z_0.$$

Proof. The first identity is the definition. Since p is odd,

$$N_{K/F}(1 - \mathcal{N}/u) = -\frac{N_{K/F}(\mathcal{N} - u)}{\mathcal{N}}, \quad 1 - \mathcal{N}^{p-1} = -\frac{\mathcal{N}^p - \mathcal{N}}{\mathcal{N}}.$$

Hence

$$z_0 = \frac{N_{K/F}(\mathcal{N} - u)}{\mathcal{N}^p - \mathcal{N}}.$$

Now (71) gives $\mathcal{N}^p - \mathcal{N} = \prod_{j \in \mathbf{F}_p} (\mathcal{N} + j)$. □

Lemma 9.11 (Depth of the correction units). *One has*

$$(136) \quad \begin{aligned} z_j - 1 &\in \mathfrak{p}_F^{\lfloor (|a| + (p-1)T)/p \rfloor} \subseteq \mathfrak{p}_F^{s_0} \quad (j \neq 0), \\ z_0 - 1 &\in \mathfrak{p}_F^{\lfloor (p-1)(T+|a|)/p \rfloor} \subseteq \mathfrak{p}_F^{\lceil m/2 \rceil}. \end{aligned}$$

Thus $z_j - 1$ lies in the linearization domain of τ and $z_0 - 1$ lies in the linearization domain of χ_F .

Proof. Write $N = N_{K/F}$ and, for $c \geq 0$, put

$$R(c) = \left\lfloor \frac{c + (p-1)T}{p} \right\rfloor.$$

For $w \in K$ with $1 + N(w) \neq 0$, put $Z(w) = N(1 + w)/(1 + N(w))$. If $v_K(w) = c > 0$, then

$$Z(w) - 1 = \frac{s_1(w) + \cdots + s_{p-1}(w)}{1 + N(w)},$$

so (25) gives $v_F(Z(w) - 1) \geq R(c)$. If $c < 0$, the identity $Z(w) = Z(w^{-1})$ reduces to the positive case. If $c = 0$, the unit assertions (134) show that $1 + N(w)$ is a unit in each occurrence of $Z(w)$: for $w = u/j$ this is equivalent to $\overline{\mathcal{N}} + j \neq 0$, and for $w = -\mathcal{N}/u$ to $1 - \overline{\mathcal{N}}^{p-1} \neq 0$. Equation (25) then gives

$$v_F(s_i(w)) \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor = R(0) \quad (1 \leq i \leq p-1),$$

so the same conclusion holds when $c = 0$.

Now $z_j = Z(u/j)$, while $z_0 = Z(-\mathcal{N}/u)$ and $v_K(-\mathcal{N}/u) = (p-1)a$ by (132). This proves the first inclusions.

We use the elementary integer inequality

$$\left\lfloor \frac{(p-1)M}{p} \right\rfloor \geq \left\lceil \frac{M}{2} \right\rceil \quad (p \geq 3, \ M \geq 2).$$

If M is even, the real inequality $(p-1)M/p \geq M/2$ already has an integral right side. If M is odd, then $M \geq 3$, and

$$\frac{(p-1)M}{p} \geq \frac{M+1}{2} \iff (p-2)M \geq p,$$

which follows from $M \geq 3$ and $p \geq 3$.

Since $R(|a|) \geq R(0)$, applying $\lfloor (p-1)M/p \rfloor \geq \lceil M/2 \rceil$ with $M = T$ gives $R(|a|) \geq \lceil T/2 \rceil = s_0$. For z_0 , if $a \geq 0$, then $m = T - a$ and

$$R((p-1)a) = \left\lfloor \frac{(p-1)(T+a)}{p} \right\rfloor \geq \left\lceil \frac{T+a}{2} \right\rceil \geq \left\lceil \frac{T-a}{2} \right\rceil = \left\lceil \frac{m}{2} \right\rceil.$$

If $a < 0$, put $c = -a$, so $m = T + c$; applying $\lfloor (p-1)M/p \rfloor \geq \lceil M/2 \rceil$ with $M = T + c$ gives

$$R((p-1)c) = \left\lfloor \frac{(p-1)(T+c)}{p} \right\rfloor \geq \left\lceil \frac{T+c}{2} \right\rceil = \left\lceil \frac{m}{2} \right\rceil.$$

This proves (136). $\square$

Proposition 9.12 (Correction-depth character linearization). *With A , $\mathcal{N}$, z_0 , and z_j defined by (132) and (133),*

$$\chi_F(z_0) = \psi_F(A\mathcal{N}(z_0 - 1)), \quad \tau^j(z_j) = \psi_F(Aj(z_j - 1)) \quad (j \neq 0).$$

Proof. The stationary linearization of τ is valid on $\mathfrak{p}_F^{s_0}$, and that of χ_F is valid on $\mathfrak{p}_F^{[m/2]}$. Equation (136) puts $z_j - 1$ and $z_0 - 1$ in these two domains. The coefficients of the two linearizations are A and $A\mathcal{N}$, respectively. Applying the linearizations and taking the j th power of the first identity gives the two asserted identities. $\square$

Proposition 9.13 (Assembly of the stationary quotient). *The error quotient (14)*

$$\mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{odd}} \psi_F(-AX),$$

where

$$(137) \quad X = \text{Tr}_{K/F}(u) + \mathcal{N}(z_0 - 1) + \sum_{j \neq 0} j(z_j - 1).$$

Proof. Use the factorization (46), the sum representatives in (128), and the critical functions in Proposition 9.8.

In the high range $m > T$, the admissible-element and stationary-parameter pairs, with their sources, are

character	$(\gamma_\theta, \beta_\theta)$	source
χ_K	$(\gamma_F, \alpha(\mathcal{N} - u))$	Proposition 6.8(d), (77)
τ^j	$(\gamma_F/\alpha, j) \quad (j \neq 0)$	Lemma 9.7
$\tau^j \chi_F$	$(\gamma_F, \alpha(\mathcal{N} + j)) \quad (j \in \mathbf{F}_p)$	(78), Lemma 9.7.

In the low range $m \leq T$, the pairs are

character	$(\gamma_\theta, \beta_\theta)$	source
χ_K	$(\Gamma_\tau/\xi_1, (\alpha/\xi_1)(\mathcal{N} - u))$	Proposition 6.10
τ^j	$(\Gamma_\tau, \alpha j) \quad (j \neq 0)$	Lemma 9.7
χ_F	$(\Gamma_\tau/\xi, (\alpha/\xi)\mathcal{N})$	(84), (86)
$\tau^j \chi_F$	$(\Gamma_\tau, \alpha(\mathcal{N} + j)) \quad (j \neq 0)$	(87), Lemma 9.7.

The factors contributed by the admissible elements cancel. In the high range their quotient is

$$\frac{\chi_K(\gamma_F) \prod_{j \neq 0} \tau^j(\gamma_F/\alpha)}{\prod_j (\tau^j \chi_F)(\gamma_F)} = 1,$$

and in the low range it is

$$\frac{\chi_K(\Gamma_\tau/\xi_1) \prod_{j \neq 0} \tau^j(\Gamma_\tau)}{\chi_F(\Gamma_\tau/\xi) \prod_{j \neq 0} (\tau^j \chi_F)(\Gamma_\tau)} = 1.$$

Here $N_{K/F}(\xi_1) = \xi$ and α is a norm.

It remains to collect the factors $\theta(\beta)^{-1}$ from (45). In the high range they give

$$\chi_K(\alpha(\mathcal{N} - u))^{-1} \prod_{j \neq 0} \tau^j(j)^{-1} \prod_{j \in \mathbf{F}_p} (\tau^j \chi_F)(\alpha(\mathcal{N} + j)).$$

Since $\chi_K = \chi_F \circ N_{K/F}$, $N_{K/F}(\alpha) = \alpha^p$, and $\tau(\alpha) = 1$ because α is a norm, this becomes

$$H = \chi_F \left(\frac{\prod_{j \in \mathbf{F}_p} (\mathcal{N} + j)}{N_{K/F}(\mathcal{N} - u)} \right) \prod_{j \neq 0} \tau^j \left(\frac{\mathcal{N} + j}{j} \right).$$

The low-range contribution is

$$\chi_K\left(\frac{\alpha}{\xi_1}(\mathcal{N} - u)\right)^{-1} \prod_{j \neq 0} \tau^j(\alpha j)^{-1} \chi_F\left(\frac{\alpha}{\xi} \mathcal{N}\right) \prod_{j \neq 0} (\tau^j \chi_F)(\alpha(\mathcal{N} + j)).$$

Using $N_{K/F}(\xi_1) = \xi$, $\chi_K = \chi_F \circ N_{K/F}$, and $\tau(\alpha) = 1$, this reduces to the same expression H . By (135), the first ratio is z_0^{-1} . Moreover

$$\frac{\mathcal{N} + j}{j} = N_{K/F}\left(\frac{u + j}{j}\right) z_j^{-1},$$

so the norm character kills the norm and

$$H = \chi_F(z_0)^{-1} \prod_{j \neq 0} \tau^j(z_j)^{-1}.$$

The additive stationary values have signed sum

$$\mathrm{Tr}_{K/F}(\mathcal{N} - u) + \sum_{j \neq 0} j - \sum_{j \in \mathbb{F}_p} (\mathcal{N} + j) = -\mathrm{Tr}_{K/F}(u),$$

using the Teichmuller sum (72). Their additive factor is therefore $\psi_F(-A \mathrm{Tr}_{K/F}(u))$. Together with the expression for H and the residual quotient $\mathcal{R}_{\mathrm{odd}}$, this gives

$$\mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\mathrm{odd}} \left[\psi_F(A \mathrm{Tr}_{K/F}(u)) \chi_F(z_0) \prod_{j \neq 0} \tau^j(z_j) \right]^{-1}.$$

Substituting Proposition 9.12 gives $\mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\mathrm{odd}} \psi_F(-AX)$, with X as in (137). $\square$

9.2.3. The norm-trace formula and its valuation. Let $s_r = s_r(u)$ be the elementary symmetric functions of the conjugates of u , with $s_1 = \mathrm{Tr}_{K/F}(u)$ and $s_p = \mathcal{N}$. Put

$$B = s_1 + \sum_{r=1}^{p-2} (-\mathcal{N})^r s_{p-r}.$$

Proposition 9.14 (Exact norm-trace formula). *The element X in (137) satisfies*

$$(138) \quad X = \frac{pB}{1 - \mathcal{N}^{p-1}}.$$

Proof. Put $x_0 = z_0 - 1$ and $x_j = z_j - 1$ for $j \neq 0$. For $j \neq 0$, multiplication of the conjugates of $u + j$ gives

$$N_{K/F}(u + j) = \sum_{r=0}^p j^{p-r} s_r, \quad s_0 = 1, \quad s_p = \mathcal{N}.$$

Hence

$$x_j = \frac{N_{K/F}(u + j) - (\mathcal{N} + j)}{\mathcal{N} + j} = \frac{\sum_{r=1}^{p-1} j^{p-r} s_r}{\mathcal{N} + j}.$$

Because $j^{p-1} = 1$,

$$\frac{1}{\mathcal{N} + j} = \frac{\sum_{q=0}^{p-2} (-\mathcal{N})^q j^{p-2-q}}{1 - \mathcal{N}^{p-1}}.$$

Indeed, multiplying the numerator by $\mathcal{N} + j$ telescopes to $1 - \mathcal{N}^{p-1}$. Using this expansion to sum over the $(p-1)$ Teichmuller roots, together with

$$\sum_{j \neq 0} j^r = \begin{cases} p-1, & p-1 \mid r, \\ 0, & p-1 \nmid r, \end{cases}$$

shows that only the indices $r+q=1$ and $r+q=p$ survive. With $B = s_1 + \sum_{q=1}^{p-2} (-\mathcal{N})^q s_{p-q}$, this gives

$$\sum_{j \neq 0} j x_j = \frac{(p-1)B}{1 - \mathcal{N}^{p-1}}.$$

On the other hand,

$$N_{K/F}(\mathcal{N} - u) - (\mathcal{N}^p - \mathcal{N}) = \sum_{r=1}^{p-1} (-1)^r s_r \mathcal{N}^{p-r}.$$

By (71), $z_0 = N_{K/F}(\mathcal{N} - u)/(\mathcal{N}^p - \mathcal{N})$, so

$$\begin{aligned} \mathcal{N}x_0 &= -\frac{\sum_{r=1}^{p-1} (-1)^r s_r \mathcal{N}^{p-r}}{1 - \mathcal{N}^{p-1}} \\ &= \frac{s_1 \mathcal{N}^{p-1} + \sum_{q=1}^{p-2} (-\mathcal{N})^q s_{p-q}}{1 - \mathcal{N}^{p-1}} \\ &= \frac{B}{1 - \mathcal{N}^{p-1}} - s_1. \end{aligned}$$

Adding s_1 , this identity, and the formula for $\sum_{j \neq 0} jx_j$ above proves (138). This is an identity in the field F itself. In equal characteristic the same finite sums have $p = 0$ and $p - 1 = -1$ in F , so no lifting to characteristic zero is being used. $\square$

In mixed characteristic, the closed formula (138) appears in [2, Chapter III, proof of Lemma 8, pp. 89–90].

Proposition 9.15 (Vanishing or depth of the norm–trace error). *Retain the minimal-conductor choice of χ_F . One has*

$$(139) \quad X \in \mathfrak{p}_F^T, \quad \psi_F(A\mathfrak{p}_F^T) = 1.$$

If $\text{char } F = p$, then $X = 0$. In all cases $\psi_F(-AX) = 1$.

Proof. The denominator $1 - \mathcal{N}^{p-1}$ in (138) is nonzero. If $v_F(\mathcal{N}) \neq 0$, this follows from valuations; if $v_F(\mathcal{N}) = 0$, (134) shows that it is a unit.

Suppose first that $\text{char } F = p$. In (138), the integer p is zero in F . Therefore

$$X = \frac{pB}{1 - \mathcal{N}^{p-1}} = 0,$$

which proves the first assertion in equal characteristic.

Now assume $\text{char } F = 0$. Put $a = v_K(u) = v_F(\mathcal{N})$, $e = v_F(p)$, and $e_0 = \lfloor (p-1)T/p \rfloor = T - \lceil T/p \rceil$. Lemma 6.4 gives $e \geq e_0$; for odd p , its proof gives $e_0 \geq \lceil T/2 \rceil$, hence $2e_0 \geq T$.

If $a \geq 0$, the denominator in (138) is a unit, as shown by (134) when $a = 0$ and by $v_F(\mathcal{N}) > 0$ when $a > 0$. By (25), every $s_j(u)$ has valuation at least e_0 , and the powers of $\mathcal{N}$ do not decrease it. Hence $v_F(pB) \geq e + e_0 \geq 2e_0 \geq T$.

If $a < 0$, put $c = -a > 0$, $y = u^{-1}$, and for $1 \leq r \leq p-1$ set

$$L_r(c) = \left\lfloor \frac{rc + (p-1)T}{p} \right\rfloor.$$

Then $v_F(1 - \mathcal{N}^{p-1}) = (p-1)a$, so division by the denominator is multiplication, up to a unit, by $\mathcal{N}^{1-p}$. The reciprocal identity is

$$s_j(u) = \mathcal{N} s_{p-j}(y) \quad (1 \leq j \leq p-1).$$

Applying this reciprocal identity to the s_1 term gives

$$v_F(p\mathcal{N}^{1-p}s_1(u)) = v_F(p\mathcal{N}^{2-p}s_{p-1}(y)) \geq e + (p-2)c + L_{p-1}(c) \geq e + e_0 \geq T.$$

For a term $\mathcal{N}^{p-j}s_j(u)$ with $2 \leq j \leq p-1$,

$$v_F(p\mathcal{N}^{1-j}s_j(u)) = v_F(p\mathcal{N}^{2-j}s_{p-j}(y)) \geq e + (j-2)c + L_{p-j}(c) \geq e + e_0 \geq T.$$

Thus $X \in \mathfrak{p}_F^T$ also in mixed characteristic.

Finally τ has conductor T and $\tau(1+x) = \psi_F(Ax)$ on a layer containing $\mathfrak{p}_F^T$. Since τ is trivial on $1 + \mathfrak{p}_F^T$, one has $\psi_F(A\mathfrak{p}_F^T) = 1$. The conclusion follows. $\square$

Corollary 9.16 (Wild odd-prime non-stable case). *For odd p , the First Main Lemma holds in every non-stable wildly ramified case.*

Proof. Propositions 9.9 and 9.13, together with (139), give the result. $\square$

9.3. Wild quadratic extensions. Assume $p = 2$ and K/F is wildly ramified quadratic, with $S(K/F) = \{1, \tau\}$ and $k = k_F = k_K$. Retain the minimal-conductor choice of χ_F . Put $m = m_F(\chi_F) > 1$, $T = t + 1$, and $a = T - m$. Use the stationary parameters of Proposition 6.10 for $m \leq T$ and Proposition 6.9 for $m > T$, and put

$$(140) \quad \begin{aligned} (A, u, \mathcal{N}) &= \begin{cases} (\alpha/\Gamma_\tau, \xi_1\beta_1/\alpha_1, \xi\beta/\alpha), & m \leq T, \\ (\alpha\delta_h/\gamma_F, \beta_1/\alpha_1, \beta/(\alpha\delta_h)), & m > T, \end{cases} \\ \mathcal{N} &= N_{K/F}(u), \quad v_K(u) = v_F(\mathcal{N}) = a. \end{aligned}$$

Equations (86), (82), and (39) give

$$(141) \quad \begin{aligned} \tau(1+z) &= \psi_F(Az) \quad (z \in \mathfrak{p}_F^{\lceil T/2 \rceil}), \\ \chi_F(1+z) &= \psi_F(A\mathcal{N}z) \quad (z \in \mathfrak{p}_F^{\lceil m/2 \rceil}). \end{aligned}$$

Define the correction units by

$$(142) \quad z_0 = \frac{N_{K/F}(1 - \mathcal{N}/u)}{1 + \mathcal{N}}, \quad z_1 = \frac{N_{K/F}(u + 1)}{\mathcal{N} + 1}.$$

If $a \neq 0$, the denominator is nonzero by valuation. If $a = 0$, then $m = T$, $\xi\beta = \alpha\mathcal{N}$, and α is a unit, so (49) gives $1 + \mathcal{N} \in \mathcal{O}_F^\times$.

Lemma 9.17 (Exact quadratic norm correction). *Put $s = \text{Tr}_{K/F}(u)$ and $X = s + \mathcal{N}(z_0 - 1) + (z_1 - 1)$. Then*

$$(143) \quad z_0 - 1 = -\frac{s}{1 + \mathcal{N}}, \quad z_1 - 1 = \frac{s}{1 + \mathcal{N}}, \quad X = \frac{2s}{1 + \mathcal{N}}.$$

Proof. Since $\mathcal{N}/u$ is the nontrivial conjugate of u , the quadratic norm identity gives

$$N_{K/F}(1 + u) = 1 + s + \mathcal{N}, \quad N_{K/F}(1 - \mathcal{N}/u) = 1 - s + \mathcal{N}.$$

Substitution in (142) proves the first two identities, and their substitution in the definition of X proves the third. $\square$

The formulas (143) occur in Dwork's p -adic calculation [2, Chapter III, proof of Lemma 8', p. 93].

Lemma 9.18 (Quadratic graded trace). *Let K/F be wildly ramified quadratic and let $T = v_K(\mathfrak{D}_{K/F}) = t + 1$. If $r \geq 0$ and $r + T$ is odd, trace induces a k -linear isomorphism*

$$\text{Tr}_{K/F} : \mathfrak{p}_K^r / \mathfrak{p}_K^{r+1} \xrightarrow{\sim} \mathfrak{p}_F^{(r+T-1)/2} / \mathfrak{p}_F^{(r+T+1)/2}.$$

In particular:

(a) *If T is odd and $d_\tau = (T - 1)/2$, then $v_F(2) = d_\tau$. For every $w \in \mathcal{O}_K^\times$ and every lift $\tilde{w} \in \mathcal{O}_F$ of its residue,*

$$(144) \quad \text{Tr}(w) \equiv 2\tilde{w} \pmod{\mathfrak{p}_F^{d_\tau+1}}.$$

(b) *If $m = 2d + 1 > T$ and $r = m - T$, then trace identifies $\mathfrak{p}_K^r / \mathfrak{p}_K^{r+1}$ with $\mathfrak{p}_F^d / \mathfrak{p}_F^{d+1}$.*

Proof. For a quadratic extension, (17) reads

$$\text{Tr}(\mathfrak{p}_K^j) = \mathfrak{p}_F^{\lfloor (j+T)/2 \rfloor}.$$

If $r + T = 2q + 1$, then

$$\text{Tr}(\mathfrak{p}_K^r) = \mathfrak{p}_F^q, \quad \text{Tr}(\mathfrak{p}_K^{r+1}) = \mathfrak{p}_F^{q+1}.$$

Thus the displayed map induced by trace is surjective. Both source and target are one-dimensional k -vector spaces, and the map is k -linear because the residue extension is trivial. It is therefore an isomorphism.

If T is odd, apply the isomorphism with $r = 0$. The class of 1 is nonzero and maps to the class of $\text{Tr}(1) = 2$, proving $v_F(2) = d_\tau$. Since $w - \tilde{w} \in \mathfrak{p}_K$, the trace-ideal formula (17) at $j = 1$ gives

$$\text{Tr}(w) - 2\tilde{w} \in \text{Tr}(\mathfrak{p}_K) = \mathfrak{p}_F^{(T+1)/2} = \mathfrak{p}_F^{d_\tau+1},$$

which proves (a). In particular, a wildly ramified quadratic extension with T odd is necessarily of mixed characteristic; in equal characteristic two the equality $v_F(2) = d_\tau$ is impossible. If $m = 2d + 1$ and $r = m - T$, then $r + T = m$ and $q = d$, proving (b). $\square$

9.4. Quadratic refinement in characteristic two. Recall the canonical character of (7):

$$\psi_0(z) = \psi_{\text{can}}(z) = (-1)^{\text{Tr}_{k/\mathbf{F}_2}(z)}.$$

Lemma 9.19 (Normalized refinement). *Let $h : k \rightarrow \mathbf{C}^\times$ be unimodular and satisfy $h(r + s) = h(r)h(s)\psi_0(rs)$. Multiplying h by an additive character gives a unimodular function $\mathfrak{q} : k \rightarrow \mathbf{C}^\times$ such that*

$$(145) \quad \mathfrak{q}(r + s) = \mathfrak{q}(r)\mathfrak{q}(s)\psi_0(rs), \quad \mathfrak{q}(r^2) = \mathfrak{q}(r).$$

Every other function with the same polar character is uniquely

$$(146) \quad h_\lambda(z) = \mathfrak{q}(z)\psi_0(\lambda z) \quad (\lambda \in k).$$

Write

$$(147) \quad H(\lambda) = \text{ph} \left(\sum_{z \in k} h_\lambda(z) \right), \quad H_0 = H(0).$$

We have

$$(148) \quad H(\lambda) = H_0 \overline{\mathfrak{q}(\lambda)}, \quad H_0^2 = \mathfrak{q}(1), \quad \mathfrak{q}(r)^2 = \psi_0(r).$$

Proof. The quotient

$$\ell(z) = \frac{h(z^2)}{h(z)}$$

is additive. Indeed, using the defining functional equation for h and Frobenius invariance of ψ_0 ,

$$\ell(r + s) = \frac{h((r + s)^2)}{h(r + s)} = \frac{h(r^2)}{h(r)} \frac{h(s^2)}{h(s)} = \ell(r)\ell(s).$$

The trace pairing on k is perfect, so there is a unique $c \in k$ such that $\ell(z) = \psi_0(cz)$. Let $f = [k : \mathbf{F}_2]$. For every $z \in k$, telescoping around the Frobenius orbit gives

$$\begin{aligned} 1 &= \frac{h(z^{2^f})}{h(z)} = \prod_{i=0}^{f-1} \frac{h(z^{2^{i+1}})}{h(z^{2^i})} \\ &= \psi_0 \left(c \sum_{i=0}^{f-1} z^{2^i} \right) = \psi_0 \left(c \text{Tr}_{k/\mathbf{F}_2}(z) \right). \end{aligned}$$

Choose z of absolute trace 1. Then $\psi_0(c) = 1$, or equivalently $\text{Tr}_{k/\mathbf{F}_2}(c) = 0$.

Consider the $\mathbf{F}_2$ -linear map

$$L : k \longrightarrow k, \quad L(d) = d + d^{2^{f-1}}.$$

Writing $\text{Fr}(z) = z^2$, one has $L = (\text{Fr} - 1) \circ \text{Fr}^{-1}$. Lemma 3.2 therefore gives $\ker L = \mathbf{F}_2$ and $\text{im } L = \ker \text{Tr}_{k/\mathbf{F}_2}$. Since $\text{Tr}_{k/\mathbf{F}_2}(c) = 0$, there is $d \in k$ such that

$$c = d + d^{2^{f-1}}.$$

Replace $h(z)$ by

$$\mathfrak{q}(z) = h(z)\psi_0(dz).$$

The polar character is unchanged. Moreover

$$\begin{aligned}\frac{\mathbf{q}(z^2)}{\mathbf{q}(z)} &= \psi_0(cz)\psi_0(dz^2 + dz) \\ &= \psi_0((c + d^{2^{f-1}} + d)z) = 1,\end{aligned}$$

where $\psi_0(dz^2) = \psi_0(d^{2^{f-1}}z)$ and the displayed choice of d were used. This proves $\mathbf{q}(z^2) = \mathbf{q}(z)$. If h' has the same polar character as $\mathbf{q}$, then $h'/\mathbf{q}$ is an additive character of k , hence is uniquely $z \mapsto \psi_0(\lambda z)$. This proves (146).

From (145),

$$\mathbf{q}(z)\psi_0(\lambda z) = \mathbf{q}(z + \lambda)\mathbf{q}(\lambda)^{-1},$$

so substituting $z \mapsto z + \lambda$ in $\sum_{z \in k} h_\lambda(z)$ gives the first formula in (148). Setting $r = s$ in (145) gives

$$\mathbf{q}(r)^2 = \psi_0(r^2) = \psi_0(r),$$

and therefore $\overline{\mathbf{q}(r)} = \mathbf{q}(r)\psi_0(r)$.

Put $S_0 = \sum_{r \in k} \mathbf{q}(r)$. Its polar bicharacter is $(r, s) \mapsto \psi_0(rs)$, which is nondegenerate by the perfect trace pairing. Equation (38) therefore gives

$$|S_0|^2 = |k|,$$

so $S_0 \neq 0$. Finally, translation by 1 gives

$$\overline{S_0} = \sum_r \mathbf{q}(r)\psi_0(r) = \frac{1}{\mathbf{q}(1)} \sum_r \mathbf{q}(r+1) = \frac{S_0}{\mathbf{q}(1)}.$$

Taking phases yields $H_0^{-1} = H_0/\mathbf{q}(1)$, and hence $H_0^2 = \mathbf{q}(1)$. $\square$

9.5. The four critical functions. Recall the setup in Theorem 4.5.

Proposition 9.20 (Wild-quadratic critical functions). *Retain the minimal-conductor choice of χ_F . For*

$$\theta \in \{\tau, \chi_F, \tau\chi_F, \chi_K\}, \quad E_\theta = \begin{cases} F, & \theta \neq \chi_K, \\ K, & \theta = \chi_K, \end{cases} \quad \psi_\theta = \begin{cases} \psi_F, & \theta \neq \chi_K, \\ \psi_K, & \theta = \chi_K, \end{cases}$$

if $m_{E_\theta}(\theta) = 2d_\theta + 1$ is odd, choose an admissible denominator γ_θ and a stationary representative β_θ from the classes given by (86) and Proposition 6.10 when $m \leq T$, and by (82) and Proposition 6.9 when $m > T$. Choose a critical element δ_θ with $\delta_\theta \mathcal{O}_{E_\theta} = \mathfrak{p}_{E_\theta}^{d_\theta}$. The choices can be made so that, for every $z \in k$ and every lift $\tilde{z} \in \mathcal{O}_F \subset \mathcal{O}_K$,

$$\psi_\theta\left(\frac{\beta_\theta \delta_\theta^2 \tilde{z}}{\gamma_\theta}\right) = \psi_0(z).$$

The critical function $H_\theta(z) = \psi_\theta\left(\frac{\delta_\theta \beta_\theta \tilde{z}}{\gamma_\theta}\right) \theta(1 + \delta_\theta \tilde{z})^{-1}$ then satisfies

$$H_\theta(r+s) = H_\theta(r)H_\theta(s)\psi_0(rs) \quad (r, s \in k).$$

For even conductor put $H_\theta = 1$, in accordance with (114).

For each of $\tau, \chi_F, \tau\chi_F, \chi_K$ of odd conductor, the coefficient in (146) is unique. Denote these coefficients by $\gamma_0, \gamma, \gamma_1, \gamma'$, respectively, so that

$$H_\tau = h_{\gamma_0}, \quad H_{\chi_F} = h_\gamma, \quad H_{\tau\chi_F} = h_{\gamma_1}, \quad H_{\chi_K} = h_{\gamma'},$$

with h_λ as in (146). At $m = T$, put $\rho = \bar{u} \in k^\times \setminus \{1\}$, and let $\sqrt{\rho}$ denote its unique square root in k . These coefficients satisfy

$$(149) \quad (\gamma_1, (\gamma')^2) = \begin{cases} (\gamma_0, \gamma), & m < T, \\ \left(\frac{\gamma_0 + \rho\gamma + \sqrt{\rho}}{1 + \rho}, \frac{\gamma + \rho\gamma_0 + \rho + \sqrt{\rho}}{1 + \rho}\right), & m = T, \\ (\gamma, 1 + \gamma_0), & m > T. \end{cases}$$

Proof. Write $N = N_{K/F}$. Equations (48), (34), and (30) give

$$(150) \quad \begin{array}{c|cccc} & m_F(\tau) & m_F(\chi_F) & m_F(\tau\chi_F) & m_K(\chi_K) \\ \hline m \leq T & T & m & T & m \\ m > T & T & m & m & 2m - T. \end{array}$$

All four conductors are even exactly when m and T are even.

Equation (68), Proposition 6.10, and Proposition 6.9 identify the stationary classes in the two ranges. To realize the choices asserted in the statement, take representatives satisfying the following equalities, with the parameters (140):

$$(151) \quad \frac{\beta(\chi_F)}{\gamma_F} = A\mathcal{N}, \quad \frac{\beta_K}{\gamma_K} = A(\mathcal{N} - u), \quad \mathcal{N} = N(u).$$

Use the sum representative, relative to a common denominator, for $\tau\chi_F$.

For an odd-conductor character, form (41) with the representative just fixed. Equations (43) and (44) give a nondegenerate polar character $\psi_0(cxy)$ with $c \in k^\times$. Multiplying the critical element by a unit with residue $c^{-1/2}$ gives the normalization stated in the proposition, since Frobenius is bijective on k . For even conductor use $H_\theta = 1$, as given by (114). Write the normalized functions for τ and χ_F as h_{γ_0} and h_γ . At $m = T$, u is a unit and $\mathcal{N} = \rho^2$. Equation (49), with $j = 1$, gives $1 + \rho^2 \neq 0$, hence $1 + \rho \neq 0$. In the normalized δ_τ coordinate, the stationary ratios A and $A\mathcal{N}$ give polar characters $\psi_0(xy)$ and $\psi_0(\rho^2 xy)$. Thus, for a lift $\tilde{\rho} \in \mathcal{O}_F^\times$, take $\delta_\chi = \delta_\tau/\tilde{\rho}$; the χ_F -function in the δ_τ coordinate is then $h_\gamma(\rho z)$. The product has polar character $\psi_0((1 + \rho)^2 xy)$, so its normalized coordinate is obtained by replacing z by $z/(1 + \rho)$.

We first establish the assertions concerning τ , χ_F , and $\tau\chi_F$, independently of whether $m_K(\chi_K)$ is odd. Suppose $m < T$ and T is odd. Put $d_\tau = (T - 1)/2$ and retain the normalized critical element δ_τ , of valuation d_τ . Since $m < T$,

$$d_\tau \geq \left\lfloor \frac{m}{2} \right\rfloor,$$

so the stationary linearization of χ_F is valid on the τ -critical line. Choose the sum representative for the stationary class of $\tau\chi_F$. Then, for $z \in k$,

$$\frac{H_{\tau\chi_F}(z)}{H_\tau(z)} = \psi_F(A\mathcal{N}\delta_\tau\tilde{z})\chi_F(1 + \delta_\tau\tilde{z})^{-1} = 1.$$

Thus the critical functions of $\tau\chi_F$ and τ agree, and $\gamma_1 = \gamma_0$.

Suppose $m > T$ and m is odd. Put $d_\chi = (m - 1)/2$ and retain the normalized critical element δ_χ , of valuation d_χ . Since $m > T$,

$$d_\chi \geq \left\lfloor \frac{T}{2} \right\rfloor,$$

so the stationary linearization of τ is valid on the χ_F -critical line. With the sum representative for $\tau\chi_F$ one obtains

$$\frac{H_{\tau\chi_F}(z)}{H_{\chi_F}(z)} = \psi_F(A\delta_\chi\tilde{z})\tau(1 + \delta_\chi\tilde{z})^{-1} = 1.$$

Hence the critical functions of $\tau\chi_F$ and χ_F agree, and $\gamma_1 = \gamma$.

At the boundary $m = T$, the conductors of τ , χ_F , and $\tau\chi_F$ are all T . When T is odd, the common-coordinate convention $\delta_\chi = \delta_\tau/\tilde{\rho}$ gives the critical functions of τ and χ_F

$$(152) \quad \mathbf{q}(z)\psi_0(\gamma_0 z), \quad \mathbf{q}(\rho z)\psi_0(\rho\gamma z).$$

We now assume that $m_K(\chi_K)$ is odd and compute the critical function of χ_K . Let δ_K be a critical element for χ_K . For $Z \in k$, choose a lift $\tilde{Z} \in \mathcal{O}_F$, put

$$x_Z = \delta_K \tilde{Z}, \quad P(y) = N(1 + y) - 1 = \text{Tr}(y) + N(y),$$

and put $m_K(\chi_K) = 2d_K + 1$. Equation (65) gives $P(x_Z) \in \mathfrak{p}_F^{[m/2]}$. Also $v_K(ux_Z) \geq a + d_K \geq [T/2]$, so Theorem 3.1(b) gives $P(ux_Z) \in \mathfrak{p}_F^{[T/2]}$. Define

$$p_{\text{nat}}(Z) = \overline{P(x_Z)/\delta_\chi} \quad (m \text{ odd}), \quad q_{\text{nat}}(Z) = \overline{P(ux_Z)/\delta_\tau} \quad (T \text{ odd}).$$

If m is even, set $p_{\text{nat}} = 0$; if T is even, set $q_{\text{nat}} = 0$.

Write $H_K^{\text{nat}}(Z)$ for the critical function of χ_K in this natural coordinate. Its definition and (151) give

$$\begin{aligned} H_K^{\text{nat}}(Z) &= \psi_F(A \operatorname{Tr}((\mathcal{N} - u)x_Z)) \chi_F(1 + P(x_Z))^{-1} \\ &= H_{\chi_F}(p_{\text{nat}}(Z)) \psi_F(-A\{\operatorname{Tr}(ux_Z) + \mathcal{N}N(x_Z)\}) \\ &= H_{\chi_F}(p_{\text{nat}}(Z)) \psi_F(-AP(ux_Z)). \end{aligned}$$

For the second equality we used the stationary-layer identity

$$H_{\chi_F}(p_{\text{nat}}(Z)) = \psi_F(A\mathcal{N}P(x_Z)) \chi_F(1 + P(x_Z))^{-1}$$

and cancelled $\mathcal{N} \operatorname{Tr}(x_Z)$; the last equality uses $\mathcal{N}N(x_Z) = N(ux_Z)$. Since $1 + P(ux_Z) = N(1 + ux_Z)$ and τ is trivial on norms, the stationary-layer identity for τ is

$$H_{\tau}(q_{\text{nat}}(Z)) = \psi_F(AP(ux_Z)).$$

For odd T , (146) and (148) give $H_{\tau}(q_{\text{nat}}(Z))^{-1} = H_{\tau}(q_{\text{nat}}(Z)) \psi_0(q_{\text{nat}}(Z))$. For even T , this also holds because $q_{\text{nat}}(Z) = 0$ and $H_{\tau} \equiv 1$. Therefore

$$(153) \quad H_K^{\text{nat}}(Z) = H_{\chi_F}(p_{\text{nat}}(Z)) H_{\tau}(q_{\text{nat}}(Z)) \psi_0(q_{\text{nat}}(Z)).$$

The trace-ideal formula (17) and total ramification give

$$(154) \quad v_F(\operatorname{Tr}(\delta_K \tilde{Z})) \geq \left\lfloor \frac{d_K + T}{2} \right\rfloor, \quad v_F(N(\delta_K \tilde{Z})) \geq d_K.$$

The norm bound is an equality for $Z \neq 0$. When $m_F(\chi_F) = 2d_{\chi} + 1$, $d_K = d_{\chi}$, and $\lfloor (d_K + T)/2 \rfloor \geq d_{\chi} + 1$, reduction of $P(\delta_K \tilde{Z}) = \operatorname{Tr}(\delta_K \tilde{Z}) + N(\delta_K \tilde{Z})$ modulo $\mathfrak{p}_F^{d_{\chi}+1}$ gives

$$(155) \quad p_{\text{nat}}(Z) = \lambda_{\chi} Z^2, \quad \lambda_{\chi} = \frac{\overline{N(\delta_K)}}{\delta_{\chi}} \in k^{\times}.$$

When $T = 2d_{\tau} + 1$, $a + d_K = d_{\tau}$, and $\lfloor (a + d_K + T)/2 \rfloor \geq d_{\tau} + 1$, the same reduction of $P(u\delta_K \tilde{Z})$, with $v_K(u) = a$ and $N(u) = \mathcal{N}$, gives

$$(156) \quad q_{\text{nat}}(Z) = \lambda_{\tau} Z^2, \quad \lambda_{\tau} = \frac{\overline{\mathcal{N}N(\delta_K)}}{\delta_{\tau}} \in k^{\times}.$$

Both displayed scalars are nonzero when defined. Replacing δ_K by $v_0 \delta_K$, with $v_0 \in \mathcal{O}_K^{\times}$, multiplies both scalars by $\bar{v}_0^2$. Since Frobenius is an automorphism of k , we may therefore normalize either nonzero scalar to 1 without changing any valuation or the ratio of the two scalars.

Put $a = T - m = v_K(u)$. Suppose first that $m < T$. Then $m_K = m$, and when m is odd one has $d_K = d_{\chi} = (m - 1)/2$. Equations (154) and (155) show that $p_{\text{nat}}(Z) = \lambda_{\chi} Z^2$. Scale δ_K so that $\lambda_{\chi} = 1$. If T is odd, put $d_{\tau} = (T - 1)/2$. Since the odd conductors $m < T$ differ by at least two,

$$v_K(u\delta_K) = d_K + a = d_{\tau} + \frac{a}{2} \geq d_{\tau} + 1.$$

The norm and trace terms of $P(u\delta_K \tilde{Z})$ then lie in $\mathfrak{p}_F^{d_{\tau}+1}$, so $q_{\text{nat}}(Z) = 0$; if T is even there is no τ -critical factor.

Suppose next that $m > T$ and that $m_K(\chi_K)$ is odd. Then T is odd. With $d_{\tau} = (T - 1)/2$ and $m_K = 2m - T$, one has

$$d_K = m - d_{\tau} - 1, \quad v_K(u\delta_K) = a + d_K = d_{\tau}.$$

The trace of $u\delta_K \tilde{Z}$ lies in $\mathfrak{p}_F^{d_{\tau}+1}$, while its norm has valuation at least d_{τ} , with equality for $Z \neq 0$; hence (156) applies. Scale δ_K so that $\lambda_{\tau} = 1$. If m is odd and $d_{\chi} = (m - 1)/2$, then

$$d_K - d_{\chi} = \frac{m - T}{2} \geq 1,$$

so $P(\delta_K \tilde{Z})$ is beyond the χ_F critical line; if m is even there is no such critical factor.

Finally suppose $m = T$. Then $a = 0$ and $m_K = m = T$. When T is odd, $d_K = d_\chi = d_\tau$. Both initial terms are norm terms. Equations (155) and (156) give

$$p_{\text{nat}}(Z) = \lambda_\chi Z^2, \quad q_{\text{nat}}(Z) = \lambda_\tau Z^2.$$

The critical coordinates for χ_F and τ were fixed by $\delta_\chi = \delta_\tau / \tilde{\rho}$, and $\bar{\mathcal{N}} = \rho^2$. Therefore

$$\frac{\lambda_\tau}{\lambda_\chi} = \bar{\mathcal{N}} \frac{\delta_\chi}{\delta_\tau} = \rho^2 \rho^{-1} = \rho.$$

Scale δ_K so that $\lambda_\chi = 1$. This single scaling preserves this ratio, and hence simultaneously normalizes the two maps to $p_{\text{nat}}(Z) = Z^2$ and $q_{\text{nat}}(Z) = \rho Z^2$. Thus

$$(p_{\text{nat}}(Z), q_{\text{nat}}(Z)) = \begin{cases} (Z^2, 0), & m < T, \\ (Z^2, \rho Z^2), & m = T, \\ (0, Z^2), & m > T. \end{cases}$$

Define $H_{\chi_K}^{\text{raw}}(z) = H_K^{\text{nat}}(z^{1/2})$; this is well defined because Frobenius is bijective. Substitution of the three cases into (153), followed by (146), gives

$$(157) \quad H_{\chi_K}^{\text{raw}}(z) = \begin{cases} \mathbf{q}(z)\psi_0(\gamma z), & m < T, \\ \mathbf{q}(z)\mathbf{q}(\rho z)\psi_0((\gamma + \rho\gamma_0 + \rho)z), & m = T, \\ \mathbf{q}(z)\psi_0((1 + \gamma_0)z), & m > T. \end{cases}$$

Put

$$H_{\chi_K}(Z) = \begin{cases} H_{\chi_K}^{\text{raw}}(Z^2), & m \neq T, \\ H_{\chi_K}^{\text{raw}}(Z^2/(1 + \rho)), & m = T. \end{cases}$$

Since $H_{\chi_K}^{\text{raw}}(z) = H_K^{\text{nat}}(z^{1/2})$, this is $H_K^{\text{nat}}(Z)$ for $m \neq T$ and $H_K^{\text{nat}}(Z/\sqrt{1 + \rho})$ for $m = T$. By the change-of-critical-element formula following (41), the latter is again the critical function (41), with the critical element rescaled by a unit whose residue is $(1 + \rho)^{-1/2}$. For $m \neq T$, the raw function has polar quotient $\psi_0(xy)$, which is unchanged by precomposition with Frobenius. At $m = T$, (145) gives

$$\frac{H_{\chi_K}^{\text{raw}}(x + y)}{H_{\chi_K}^{\text{raw}}(x)H_{\chi_K}^{\text{raw}}(y)} = \psi_0((1 + \rho^2)xy).$$

Since $1 + \rho^2 = (1 + \rho)^2$ and $1 + \rho \neq 0$, the function $Z \mapsto H_{\chi_K}^{\text{raw}}(Z^2/(1 + \rho))$ has polar quotient $\psi_0((XY)^2) = \psi_0(XY)$. Thus H_{χ_K} has normalized polar character in every case. Both changes of variable are bijections of k , hence $\sum_{Z \in k} H_{\chi_K}(Z) = \sum_{z \in k} H_{\chi_K}^{\text{raw}}(z)$.

At $m = T$, put $r = \sqrt{\rho}$. Multiplying the two functions in (152) and using (145) gives

$$\mathbf{q}((1 + \rho)z)\psi_0((\gamma_0 + \rho\gamma)z + \rho z^2).$$

Since $\psi_0(\rho z^2) = \psi_0(rz)$, normalization by the coordinate $(1 + \rho)z$ gives the displayed formula for γ_1 .

For the critical function of χ_K at $m = T$, the boundary row of (157) becomes

$$\mathbf{q}((1 + \rho)z)\psi_0((\gamma + \rho\gamma_0 + \rho)z + \rho z^2).$$

Using $\psi_0(\rho z^2) = \psi_0(rz)$ and substituting $z \mapsto z/(1 + \rho)$ gives the linear coefficient

$$L = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}.$$

The normalized function for χ_K is obtained by precomposition with Frobenius. If $L_0^2 = L$, then

$$\mathbf{q}(z^2)\psi_0(Lz^2) = \mathbf{q}(z)\psi_0(L_0z),$$

by (145) and Frobenius invariance of the absolute trace. Thus Frobenius replaces L by its unique square root γ' , so $(\gamma')^2 = L$. Applying the same argument to the nonboundary rows of (157) gives $L = \gamma$ and $L = 1 + \gamma_0$. This proves (149). $\square$

Proposition 9.21 (Wild-quadratic residual quotient). *Retain the setup and notation of Proposition 9.20. Then*

$$(158) \quad \begin{aligned} \mathcal{R}_{\text{even}} &:= \frac{\Delta_{3,K}(\chi_K)\Delta_{3,F}(\tau)}{\Delta_{3,F}(\chi_F)\Delta_{3,F}(\tau\chi_F)} \\ &= \begin{cases} 1, & m < T \text{ or } m \text{ even}, \\ h_\gamma(1)^{-1}, & m > T, \text{ } m \text{ odd}, \\ h_{\gamma_1+1}\left(\frac{\rho}{1+\rho}\right), & m = T, \text{ } T \text{ odd}. \end{cases} \end{aligned}$$

Proof. For odd conductor use the phase $H(\lambda)$ in (147); for even conductor the factor is 1 by (45). Equations (145) and (148) give, for $\lambda \in k$,

$$H(\lambda) = H(\lambda^2), \quad H(\lambda)^2 = \mathbf{q}(1)\psi_0(\lambda) = h_\lambda(1), \quad H(1+\lambda)H(\lambda) = 1.$$

For the last identity, (145) and (148) give

$$H(1+\lambda)H(\lambda) = H_0^2 \overline{\mathbf{q}(1+\lambda)\mathbf{q}(\lambda)} = 1.$$

If $m < T$, the conductors in (150) and the coefficients in (149) give $\Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)$ and $\Delta_{3,F}(\tau\chi_F) = \Delta_{3,F}(\tau)$. Thus $\mathcal{R}_{\text{even}} = 1$.

If $m > T$, the product $\Delta_{3,K}(\chi_K)\Delta_{3,F}(\tau)$ is 1: for even T both factors are 1, and for odd T it is $H(1+\gamma_0)H(\gamma_0) = 1$ by (149). The denominator is 1 for even m and is $H(\gamma)^2 = h_\gamma(1)$ for odd m . This proves the asserted formulas when $m > T$.

If $m = T$ and T is even, all four conductors are even, so again $\mathcal{R}_{\text{even}} = 1$.

It remains to assume $m = T$ and T odd. Put $\rho = \bar{u} \in k^\times \setminus \{1\}$ and $r = \sqrt{\rho}$. All four critical factors are present. Put

$$L = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}, \quad A_\rho = \frac{\rho}{1 + \rho}.$$

Since $(\gamma')^2 = L$, equations (145) and (148) give $H(\gamma') = H_0 \overline{\mathbf{q}(\gamma')} = H_0 \overline{\mathbf{q}(L)} = H(L)$. Hence

$$\begin{aligned} \mathcal{R}_{\text{even}} &= \frac{\overline{\mathbf{q}(L)\mathbf{q}(\gamma_0)}}{\overline{\mathbf{q}(\gamma)\mathbf{q}(\gamma_1)}} \\ &= \mathbf{q}(\gamma)\mathbf{q}(\gamma_1)\overline{\mathbf{q}(L)\mathbf{q}(\gamma_0)}. \end{aligned}$$

Set $X_0 = \gamma + \gamma_1$ and $Y_0 = L + \gamma_0$. In characteristic two, substitution of the $m = T$ formulas in (149) gives

$$\begin{aligned} X_0 + Y_0 &= \gamma + \gamma_0 + \frac{\gamma_0 + \rho\gamma + r + \gamma + \rho\gamma_0 + \rho + r}{1 + \rho} \\ &= \frac{\rho}{1 + \rho} = A_\rho. \end{aligned}$$

Thus $X_0 = Y_0 + A_\rho$. The refinement law therefore gives

$$\mathcal{R}_{\text{even}} = \mathbf{q}(A_\rho)\psi_0(E), \quad E = Y_0 A_\rho + \gamma\gamma_1 + L\gamma_0.$$

Let $a_\rho = r/(1+r)$, so $a_\rho^2 = A_\rho$. An expansion in characteristic two gives

$$E + \gamma_1 A_\rho + A_\rho = (a_\rho(\gamma + \gamma_0) + A_\rho)^2 + (a_\rho(\gamma + \gamma_0) + A_\rho).$$

The right side of this identity has absolute trace zero. Substituting this into $\mathcal{R}_{\text{even}} = \mathbf{q}(A_\rho)\psi_0(E)$ gives

$$\mathcal{R}_{\text{even}} = \mathbf{q}(A_\rho)\psi_0(\gamma_1 A_\rho + A_\rho) = h_{\gamma_1+1}(A_\rho).$$

This proves (158). □

9.6. Stationary-factor assembly.

Lemma 9.22 (Depth of the quadratic correction units). *The correction units in (142) satisfy*

$$(159) \quad v_F(z_0 - 1) = v_F(z_1 - 1) \geq \left\lfloor \frac{T + |a|}{2} \right\rfloor.$$

Proof. Put $s = \text{Tr}_{K/F}(u)$. Since $v_F(1 + \mathcal{N}) = \min\{a, 0\}$, equations (143) and (17) give

$$\begin{aligned} v_F(z_0 - 1) = v_F(z_1 - 1) &= v_F(s) - \min\{a, 0\} \\ &\geq \left\lfloor \frac{a + T}{2} \right\rfloor - \min\{a, 0\} = \left\lfloor \frac{T + |a|}{2} \right\rfloor. \end{aligned}$$

□

Proposition 9.23 (Character values on the quadratic correction units). *Retain the setup and notation of Proposition 9.20. One has*

$$(160) \quad \begin{aligned} &(\psi_F(A\mathcal{N}(z_0 - 1))\chi_F(z_0)^{-1}, \psi_F(A(z_1 - 1))\tau(z_1)^{-1}) \\ &= \begin{cases} (1, 1), & m < T \text{ or } m \text{ even,} \\ (h_\gamma(1), 1), & m > T, \text{ } m \text{ odd,} \\ \left(h_\gamma\left(\frac{\rho^2}{1 + \rho^2}\right), h_{\gamma_0}\left(\frac{\rho}{1 + \rho^2}\right)\right), & m = T, \text{ } T \text{ odd.} \end{cases} \end{aligned}$$

Proof. Write $N = N_{K/F}$ and $\text{Tr} = \text{Tr}_{K/F}$. Put $x = z_0 - 1$ and $y = z_1 - 1$. For a character θ of $F^\times$ of odd conductor $2d_\theta + 1$, write $H_\theta = h_\lambda$ using its normalized critical element δ_θ . Substitution of the lift w/δ_θ in (41) gives

$$(161) \quad \theta(1 + w)^{-1} = \psi_F\left(-\frac{\beta_\theta w}{\gamma_\theta}\right) h_\lambda\left(\overline{w/\delta_\theta}\right) \quad (w \in \mathfrak{p}_F^{d_\theta}).$$

Since $m = T - a$, the bound (159) is at least both $\lfloor m/2 \rfloor$ and $\lfloor T/2 \rfloor$. Thus (161) applies to $(\theta, w) = (\chi_F, x)$ and (τ, y) when the respective conductor is odd. For even conductor, the bound reaches the linearization domain in (141). For odd conductor put $x' = \overline{x/\delta_\chi}$ or $y' = \overline{y/\delta_\tau}$, respectively. The two residual factors are $h_\gamma(x')$ and $h_{\gamma_0}(y')$; for even conductor the respective factor is 1.

For a fixed stationary representative, normalized critical elements give the same critical function. Indeed, if $\delta'_\theta = v\delta_\theta$ with $v \in \mathcal{O}_F^\times$, their normalization implies $\psi_0((\bar{v}^2 - 1)z) = 1$ for every $z \in k$. Nondegeneracy of the trace pairing gives $\bar{v}^2 = 1$, hence $\bar{v} = 1$. By (41), $H'_\theta(z) = H_\theta(\bar{v}z) = H_\theta(z)$.

If $m < T$, the bound (159) is at least $(m + 1)/2$ when m is odd and at least $(T + 1)/2$ when T is odd. Thus $x' = y' = 0$ whenever defined, and both residual factors are 1.

Suppose $m > T$. When T is odd, (159) gives $v_F(y) \geq \lfloor m/2 \rfloor \geq (T + 1)/2$, so $y' = 0$. When T is even, (141) applies to y . Thus the τ residual factor is always 1. For even m , the same linearization applies to x , proving the asserted pair $(1, 1)$.

Now suppose $m > T$ and m is odd. Put $c = m - T > 0$ and $w = u^{-1}$. Then $v_K(w) = c$ and $c + T = m$ is odd. Lemma 9.18(b) shows that

$$\delta_\chi := -\text{Tr}(w) \in \mathfrak{p}_F^{(m-1)/2} \setminus \mathfrak{p}_F^{(m+1)/2}$$

is a valid critical element for χ_F . We verify the required normalization. Since

$$v_F(\mathcal{N}\delta_\chi^2) = (T - m) + (m - 1) = T - 1,$$

the map

$$z \mapsto \psi_F(A\mathcal{N}\delta_\chi^2 \tilde{z})$$

is the nontrivial character induced by τ on U_F^{T-1}/U_F^T . Indeed, $T \geq 2$ and hence $T - 1 \geq \lfloor T/2 \rfloor$, so the stationary linearization of τ is valid on U_F^{T-1} ; nontriviality follows from the exact conductor T of τ .

Put

$$r = \frac{w}{\text{Tr}(w)}.$$

Then $\text{Tr}(r) = 1$ and, since $N(w) = \mathcal{N}^{-1}$,

$$N(r) = \frac{N(w)}{\text{Tr}(w)^2} = \frac{1}{\mathcal{N}\delta_\chi^2}, \quad N(r)^{-1} = \mathcal{N}\delta_\chi^2.$$

Moreover,

$$v_K(r) = v_K(w) - v_K(\text{Tr}(w)) = (m - T) - (m - 1) = 1 - T.$$

For every $z \in k$ and every lift $\tilde{z} \in \mathcal{O}_F$, the element $1 + (\tilde{z}/N(r))r$ is a unit of K : indeed,

$$v_K\left(\frac{\tilde{z}}{N(r)}r\right) \geq 2(T - 1) + (1 - T) = T - 1 > 0.$$

The quadratic norm identity therefore gives

$$\begin{aligned} N\left(1 + \frac{\tilde{z}}{N(r)}r\right) &= 1 + \frac{\tilde{z}}{N(r)}\text{Tr}(r) + \frac{\tilde{z}^2}{N(r)^2}N(r) \\ &= 1 + \mathcal{N}\delta_\chi^2(\tilde{z} + \tilde{z}^2). \end{aligned}$$

Since τ is trivial on norms, its stationary linearization gives

$$\psi_F\left(A\mathcal{N}\delta_\chi^2(\tilde{z} + \tilde{z}^2)\right) = 1 \quad (z \in k).$$

By (20), with $p = 2$ and $\lambda = 1$, the image of $z \mapsto z^2 + z$ is $\ker \text{Tr}_{k/\mathbf{F}_2}$. The displayed character is therefore a nontrivial character of the quotient $k/\ker \text{Tr}_{k/\mathbf{F}_2} \simeq \mathbf{F}_2$, hence is ψ_0 . Consequently

$$\psi_F(A\mathcal{N}\delta_\chi^2\tilde{z}) = \psi_0(z).$$

Hence $\delta_\chi = -\text{Tr}(w)$ gives the normalized polar character. Put $s = \text{Tr}_{K/F}(u)$. Since

$$\frac{s}{\mathcal{N}} = \text{Tr}(u^{-1}) = \text{Tr}(w)$$

and $x = -s/(1 + \mathcal{N})$, one has the identity

$$\frac{x}{\delta_\chi} = \frac{\mathcal{N}}{1 + \mathcal{N}} = 1 - \frac{1}{1 + \mathcal{N}}.$$

Here $v_F(\mathcal{N}) = T - m = -c$, so $v_F((1 + \mathcal{N})^{-1}) = c > 0$. Therefore the normalized correction class is

$$x' = \overline{x/\delta_\chi} = 1.$$

Equation (161) therefore gives the residual factor $h_\gamma(1)$, proving the case $m > T$, m odd.

If $m = T$ and T is even, the bound (159) gives $x, y \in \mathfrak{p}_F^{T/2}$, so (141) gives the pair $(1, 1)$.

Finally suppose $m = T$ and T is odd. Put $\rho = \bar{u}$, $s = \text{Tr}_{K/F}(u)$, and $d_\tau = (T - 1)/2$. Choose a lift $\tilde{\rho} \in \mathcal{O}_F^\times$ of ρ . Lemma 9.18(a) gives $v_F(2) = d_\tau$, so

$$\delta_\tau = 2, \quad \delta_\chi = 2/\tilde{\rho}$$

are critical elements for τ and χ_F , respectively. We verify the normalization of $\delta_\tau = 2$. Since T is odd and $m = T > 1$, one has $T \geq 3$. Hence $T - 1 \geq \lceil T/2 \rceil$, so the stationary linearization of τ is valid on $4\mathcal{O}_F$. Since $\delta_\tau = 2$ and $v_F(4) = T - 1$, the map

$$z \mapsto \psi_F(A\delta_\tau^2\tilde{z}) = \tau(1 + 4\tilde{z})$$

is the character induced by τ on U_F^{T-1}/U_F^T . It is nontrivial because τ has exact conductor T . For every $z \in k$, the element $1 + 2\tilde{z}$ lies in $F^\times \subset K^\times$, and hence

$$N(1 + 2\tilde{z}) = (1 + 2\tilde{z})^2 = 1 + 4(\tilde{z} + \tilde{z}^2).$$

The character τ is trivial on norms, so its stationary linearization gives

$$\psi_F\left(A\delta_\tau^2(\tilde{z} + \tilde{z}^2)\right) = 1.$$

By (20), with $p = 2$ and $\lambda = 1$, the image of $z \mapsto z^2 + z$ is $\ker \text{Tr}_{k/\mathbf{F}_2}$. The displayed character is nontrivial on $k/\ker \text{Tr}_{k/\mathbf{F}_2} \simeq \mathbf{F}_2$, so it is ψ_0 . Consequently

$$(162) \quad \psi_F(A\delta_\tau^2 \tilde{z}) = \psi_0(z) \quad (z \in k).$$

Since $\bar{\mathcal{N}} = \rho^2$, (162) gives

$$\psi_F(A\mathcal{N}\delta_\chi^2 \tilde{z}) = \psi_F\left(4A\frac{\mathcal{N}}{\bar{\rho}^2} \tilde{z}\right) = \psi_0(z).$$

Thus these choices give the normalized functions h_{γ_0} and h_γ already fixed in Proposition 9.20.

Since $u - \tilde{\rho} \in \mathfrak{p}_K$, the unit-trace congruence (144) gives

$$s - 2\tilde{\rho} = \text{Tr}(u - \tilde{\rho}) \in \mathfrak{p}_F^{d_\tau+1}.$$

Consequently

$$\overline{s/\delta_\tau} = \rho.$$

Combining $\overline{s/\delta_\tau} = \rho$, (143), and $\bar{\mathcal{N}} = \rho^2$, where $\rho = \bar{u}$, gives

$$(163) \quad x' = \overline{x/\delta_\chi} = \frac{\rho^2}{1 + \rho^2}, \quad y' = \overline{y/\delta_\tau} = \frac{\rho}{1 + \rho^2}.$$

Substituting (163) in (161) proves the remaining case of (160). $\square$

Proposition 9.24 (Quadratic stationary assembly). *Retain the minimal-conductor choice of χ_F . With X from Lemma 9.17, $\mathcal{R}_{\text{even}}$ from (158), and $\rho = \bar{u}$ when $m = T$, the error quotient satisfies*

$$(164) \quad \begin{aligned} \mathcal{E}_{K/F}(\chi_F) &= \mathcal{R}_{\text{even}} \psi_F(-AX) \\ &\times \begin{cases} 1, & m < T \text{ or } m \text{ even}, \\ h_\gamma(1), & m > T, \text{ } m \text{ odd}, \\ h_\gamma\left(\frac{\rho^2}{1 + \rho^2}\right) h_{\gamma_0}\left(\frac{\rho}{1 + \rho^2}\right), & m = T, \text{ } T \text{ odd}. \end{cases} \end{aligned}$$

Proof. Write $N = N_{K/F}$. Use (46), (11), and the following admissible denominators and stationary representatives. Rescaling a denominator and its numerator uses Lemma 4.3(b). For $m > T$, the pairs and their sources are

character	$(\gamma_\theta, \beta_\theta)$	source
χ_K	$(\gamma_F, \alpha\delta_h(\mathcal{N} - u))$	(80)
τ	$(\gamma_F/(\alpha\delta_h), 1)$	(82)
χ_F	$(\gamma_F, \alpha\delta_h\mathcal{N})$	(39), (140)
$\tau\chi_F$	$(\gamma_F, \alpha\delta_h(\mathcal{N} + 1))$	(80).

For $m \leq T$, they are

character	$(\gamma_\theta, \beta_\theta)$	source
χ_K	$(\Gamma_\tau/\xi_1, (\alpha/\xi_1)(\mathcal{N} - u))$	Proposition 6.10
τ	$(\Gamma_\tau/\alpha, 1)$	(86)
χ_F	$(\Gamma_\tau/\xi, (\alpha/\xi)\mathcal{N})$	(84), (86)
$\tau\chi_F$	$(\Gamma_\tau, \alpha(\mathcal{N} + 1))$	(87) ($j = 1$).

The factors contributed by the admissible elements are, in the high range,

$$\frac{\chi_K(\gamma_F)\tau(\gamma_F/(\alpha\delta_h))}{\chi_F(\gamma_F)(\tau\chi_F)(\gamma_F)} = \tau(\alpha\delta_h)^{-1},$$

and in the low range,

$$\frac{\chi_K(\Gamma_\tau/\xi_1)\tau(\Gamma_\tau/\alpha)}{\chi_F(\Gamma_\tau/\xi)(\tau\chi_F)(\Gamma_\tau)} = \tau(\alpha)^{-1} = 1.$$

Here $\chi_K = \chi_F \circ N$, $N(\xi_1) = \xi$, and α is a norm.

It remains to collect the factors $\theta(\beta)^{-1}$ from (45). In the high range they give

$$\begin{aligned} & \chi_K(\alpha\delta_h(\mathcal{N}-u))^{-1} \chi_F(\alpha\delta_h\mathcal{N})(\tau\chi_F)(\alpha\delta_h(\mathcal{N}+1)) \\ &= \tau(\alpha\delta_h) \chi_F\left(\frac{\mathcal{N}(\mathcal{N}+1)}{N(\mathcal{N}-u)}\right) \tau(\mathcal{N}+1) = \tau(\alpha\delta_h)H, \end{aligned}$$

where

$$H = \chi_F\left(\frac{\mathcal{N}(\mathcal{N}+1)}{N(\mathcal{N}-u)}\right) \tau(\mathcal{N}+1).$$

Indeed, $N(\alpha\delta_h) = (\alpha\delta_h)^2$, so the scalar factors in the two χ_F -values cancel that in the χ_K -value. The low-range contribution is

$$\chi_K\left(\frac{\alpha}{\xi_1}(\mathcal{N}-u)\right)^{-1} \chi_F\left(\frac{\alpha}{\xi}\mathcal{N}\right) (\tau\chi_F)(\alpha(\mathcal{N}+1)) = H,$$

because $N(\alpha/\xi_1) = \alpha^2/\xi$ and $\tau(\alpha) = 1$. Multiplication by the admissible-element quotients therefore leaves H in both ranges; in particular the high-range factor $\tau(\alpha\delta_h)$ cancels its inverse.

By (142),

$$\mathcal{N}-u = -u(1-\mathcal{N}/u), \quad N(\mathcal{N}-u) = \mathcal{N}N(1-\mathcal{N}/u) = \mathcal{N}(1+\mathcal{N})z_0.$$

Also $N(1+u) = (1+\mathcal{N})z_1$, and τ is trivial on norms. Consequently

$$H = \chi_F(z_0)^{-1} \tau(z_1)^{-1}.$$

Since $\psi_K = \psi_F \circ \text{Tr}_{K/F}$, the additive stationary values have signed sum

$$\text{Tr}_{K/F}(\mathcal{N}-u) + 1 - \mathcal{N} - (\mathcal{N}+1) = -\text{Tr}_{K/F}(u).$$

Together with H and the residual quotient (158), this gives

$$\mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{even}} \left[\psi_F(A \text{Tr}_{K/F}(u)) \chi_F(z_0) \tau(z_1) \right]^{-1}.$$

Substitute (160). The additive arguments combine as

$$-A(\text{Tr}_{K/F}(u) + \mathcal{N}(z_0 - 1) + (z_1 - 1)) = -AX$$

by the definition of X in Lemma 9.17. This proves (164). $\square$

9.7. The norm–trace term and final cancellation.

Proposition 9.25 (Depth and phase of the quadratic norm–trace term). *Retain the minimal-conductor choice of χ_F and X from Lemma 9.17. If $\text{char } F = 2$, then $X = 0$. If $\text{char } F = 0$ and either $m \neq T$ or T is even, then $X \in \mathfrak{p}_F^T$. With $\rho = \bar{u}$ when $m = T$, one has*

$$(165) \quad \psi_F(-AX) = \begin{cases} 1, & m \neq T \text{ or } T \text{ even,} \\ \psi_0\left(\frac{\rho}{1+\rho^2}\right), & m = T, \text{ } T \text{ odd.} \end{cases}$$

Proof. Equation (141) gives $\psi_F(A\mathfrak{p}_F^T) = 1$, since τ has conductor T . If $\text{char } F = 2$, then T is even: otherwise Lemma 9.18(a) would give $v_F(2) = (T-1)/2$, although $2 = 0$ in F . Thus the break $t = T-1$ is odd, and (143) gives $X = 0$. This proves the assertions in characteristic two.

Assume $\text{char } F = 0$. Put $e = v_F(2)$ and $s = \text{Tr}_{K/F}(u)$. Since $2 = \text{Tr}_{K/F}(1)$, the trace-ideal formula (17) gives

$$e \geq \left\lfloor \frac{T}{2} \right\rfloor, \quad v_F(s) \geq \left\lfloor \frac{a+T}{2} \right\rfloor.$$

If $a > 0$, then $1+\mathcal{N}$ is a unit, and (143) gives

$$v_F(X) \geq e + \left\lfloor \frac{a+T}{2} \right\rfloor \geq T.$$

If $a < 0$, put $c = -a > 0$. Since $v_F(1+\mathcal{N}) = -c$,

$$v_F(X) \geq e + \left\lfloor \frac{T-c}{2} \right\rfloor + c = e + \left\lfloor \frac{T+c}{2} \right\rfloor \geq T.$$

If $a = 0$ and T is even, $1 + \mathcal{N}$ is a unit and the bound is $e + T/2 \geq T$. Thus $X \in \mathfrak{p}_F^T$ and $\psi_F(-AX) = 1$ in all these cases.

It remains to take $m = T$ and T odd. Put $\rho = \bar{u}$, $C = \rho/(1 + \rho^2)$, and $y = z_1 - 1$. The critical element in (162) is $\delta_\tau = 2$, and (163) gives $y/2 \equiv C \pmod{\mathfrak{p}_F}$. By (143), $X = 2y$; since $v_F(4) = T - 1$, it follows that

$$X \equiv 4\tilde{C} \pmod{\mathfrak{p}_F^T}$$

for any lift $\tilde{C} \in \mathcal{O}_F$ of C . Using $\psi_F(\mathbf{A}\mathfrak{p}_F^T) = 1$ and (162), we obtain

$$\psi_F(-AX) = \psi_0(-C) = \psi_0(C),$$

because k has characteristic two. This proves (165). $\square$

Proposition 9.26 (Wild quadratic cancellation). *For every wildly ramified quadratic extension K/F of nonarchimedean local fields and every quasi-character χ_F of $F^\times$,*

$$\mathcal{E}_{K/F}(\chi_F) = 1.$$

Proof. By (15), choose χ_F of minimal conductor in its $S(K/F)$ -orbit. Conductors 0 and 1 were treated in Subsection 9.1, so assume $m_F(\chi_F) > 1$.

If $m < T$ or m is even, both the residual quotient in (158) and the additional factor in (164) are 1. If $m > T$ and m is odd, those factors are $h_\gamma(1)^{-1}$ and $h_\gamma(1)$, respectively, and cancel. Thus, whenever $m \neq T$ or T is even, (164) and (165) give $\mathcal{E}_{K/F}(\chi_F) = \psi_F(-AX) = 1$.

Suppose $m = T$ and T is odd. Put

$$r = \sqrt{\rho}, \quad A_\rho = \frac{\rho}{1 + \rho}, \quad B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

Equations (158), (164), and (165), with $h_\lambda(z) = \mathfrak{q}(z)\psi_0(\lambda z)$, give

$$\mathcal{E}_{K/F}(\chi_F) = \mathfrak{q}(A_\rho)\mathfrak{q}(B)\mathfrak{q}(C)\psi_0(\gamma_1 A_\rho + \gamma B + \gamma_0 C + C + A_\rho).$$

Since $B + C = A_\rho$, the refinement identity (145) gives

$$\mathfrak{q}(B)\mathfrak{q}(C) = \mathfrak{q}(A_\rho)\psi_0(BC), \quad BC = \frac{\rho^3}{1 + \rho^4}.$$

Substitution in the expression for $\mathcal{E}_{K/F}(\chi_F)$, using $\mathfrak{q}(A_\rho)^2 = \psi_0(A_\rho)$, gives $\mathcal{E}_{K/F}(\chi_F) = \psi_0(a_2)$ with

$$a_2 = \gamma_1 A_\rho + \gamma B + \gamma_0 C + C + \frac{\rho^3}{1 + \rho^4}.$$

Substituting the $m = T$ value from (149), $\gamma_1 = (\gamma_0 + \rho\gamma + r)/(1 + \rho)$ cancels the terms in γ and γ_0 and gives

$$a_2 = \frac{\rho^3}{(1 + \rho^2)^2} + \frac{\rho(1 + r)}{(1 + \rho)^2}.$$

The first term on the right is the square of $r^3/(1 + \rho^2)$. Frobenius invariance of ψ_0 therefore reduces its phase to that of

$$a_4 = \frac{r^3}{1 + \rho^2} + \frac{\rho(1 + r)}{(1 + \rho)^2} = \frac{\rho}{1 + \rho^2} = \frac{\rho}{(1 + \rho)^2}.$$

With $c = (1 + \rho)^{-1}$, $a_4 = c + c^2$, so $\psi_0(a_4) = 1$ by (62). $\square$

10. COMPLETION OF THE FIRST MAIN LEMMA

Proof of Theorem 1.1. Let F be a nonarchimedean local field and let K/F be cyclic of prime degree ℓ . If K/F is unramified, apply Proposition 7.1. If it is ramified and tame, the odd-degree and quadratic cases are Propositions 7.3 and 7.4.

Suppose that K/F is wild. Then $\ell = p$, the residue characteristic. By (15), replace χ_F by a member of minimal conductor in its $S(K/F)$ -orbit. Conductors 0 and 1 are treated in Subsection 9.1. For odd p , the stable range is Proposition 8.1, and the remaining non-stable range is Corollary 9.16. For $p = 2$, we have Proposition 9.26. These cases exhaust all cyclic extensions of prime degree. $\square$

REFERENCES

- [1] H. Davenport and H. Hasse, *Die Nullstellen der Kongruenzzetafunktionen in gewissen zyklischen Fällen*, J. Reine Angew. Math. **172** (1935), 151–182.
- [2] B. M. Dwork, *On the Root Number in the Functional Equation of the Artin–Weil L-Series*, Ph.D. thesis, Columbia University, 1954.
- [3] R. P. Langlands, *On the Functional Equation of the Artin L-functions*, Yale University preprint (1970), IAS electronic edition.
- [4] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics **67**, Springer, 1979.

RESEARCH INSTITUTE FOR MATHEMATICAL SCIENCES, KYOTO UNIVERSITY, KYOTO 606–8502, JAPAN

Email address: `fueda@kurims.kyoto-u.ac.jp`